Supplementary material: Complex Hamiltonian learning and KAM-preserving magnetic control of a catheter model

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Abstract

This supplement gives the distributed force accounting, complete learning-error estimates, nonlinear normal forms, invariant-torus proofs and extended numerical experiments supporting the main article. It also supplies the full magnetic-feedback lift, energy-based clearance proofs and explicit local KAM measure calculation. Numerical observations and infinite-time conclusions retain their respective domains and assumptions. External theorems are attributed at their point of application; model-specific derivations and proofs are included here.

S1 Distributed mechanics and the unified magnetic model

S1.1 Rod mechanics and a variationally consistent reduction

Following the spatial rod formulation (Simo 1985; Antman 2005), let $s \in [0, L]$ denote reference arclength, $r(s,t) \in \mathbb{R}^3$ the centreline and $D(s,t) \in SO(3)$ its director frame. Define

$$\Gamma = D^{\mathsf{T}} r_s, \quad D_s = D\widehat{\kappa}, \quad D_t = D\widehat{\omega}_b. \tag{S1}$$

The hat map is defined by $\widehat{a}b = a \times b$. The extensible, unshearable Kirchhoff specialization has $\Gamma = (0, 0, 1+e)^{\mathsf{T}}$; inextensibility additionally sets $e = 0$. Transverse internal forces are then constraint reactions. They must not be set to zero by substituting zero shear into a shearable constitutive law. In spatial components the balances are

$$\rho A r_{tt} = n_s + f_{\mathrm{mag}} + f_{\mathrm{h}} + f_{\mathrm{c}} + \rho A g, \tag{S2}$$

$$\partial_t (D\mathcal{J}_b \omega_b) = m_s + r_s \times n + \tau_{\mathrm{ext}}. \tag{S3}$$

For constant body inertia, the left side of the second equation is $D(\mathcal{J}_b \dot{\omega}_b + \omega_b \times \mathcal{J}_b \omega_b)$. With stored energy density $W(\Gamma, \kappa)$, unconstrained components of $D^{\mathsf{T}} n$ and $D^{\mathsf{T}} m$ are its strain derivatives. A fixed base specifies position and orientation; a free tip specifies the appropriate force and moment resultants, including any concentrated magnetic load.

For a body-fixed magnetic moment density $\mu_b(s)$ and a static applied field $B_0(r)$, neglecting magnetic interactions between distinct material elements gives the applied-field potential

$$V_{\mathrm{mag}} = -\int_0^L (D\mu_b) \cdot B_0(r)\, ds. \tag{S4}$$

Its translational and rotational variations give $f_{\mathrm{mag}} = \nabla_r (\mu \cdot B_0)$ at fixed orientation and $\tau_{\mathrm{mag}} = \mu \times B_0$, where $\mu = D\mu_b$. Thus static magnetic loading belongs to the conservative energy. Field modulation supplies explicit time-dependent work or an input port. The permanent-dipole load is the leading term of the Maxwell-stress resultant derived below; it is counted once in the force balance.

For a complete localized magnetic element, the magnetostatic force can be evaluated on a surrounding vacuum control surface Σ:

$$F_{\mathrm{mag}} = \int_\Sigma T_M n_\Sigma \, \mathrm{d}A, \qquad T_M = \mu_0^{-1}(BB^{\mathsf{T}} - \tfrac{1}{2}|B|^2 I). \tag{S5}$$

Here B is the total field, including the element's self-field. The associated moment integrates the moment arm crossed with $T_M n_\Sigma$. To recover the permanent-dipole approximation, represent the element by its complete localized steady current $j(\xi)$, including bound currents, with $\xi = x - r$. The prescribed magnetization is unaffected by the applied field B_{ext}, whose sources lie outside Σ. Current closure gives $\nabla \cdot j = 0$ and no flux through the boundary. The identity $\nabla \cdot T_M = j \times B$ and symmetry of T_M convert

the stress resultants into Lorentz force and moment. The self-field has zero net force and moment, as seen by taking its stress surface to infinity; the applied-field-only stress also has zero resultant in this source-free volume. The remaining interaction is

$$\begin{aligned} m_{\mathrm{d}} &= \tfrac{1}{2}\int \xi \times j(\xi)\ \mathrm{d}V, \\ F_{\mathrm{mag}} &= \int j(\xi) \times B_{\mathrm{ext}}(r+\xi)\ \mathrm{d}V \\ &= \nabla_r(m_{\mathrm{d}} \cdot B_{\mathrm{ext}}(r)) + R_F, \\ \tau_{\mathrm{mag}} &= \int \xi \times [j(\xi) \times B_{\mathrm{ext}}(r+\xi)]\ \mathrm{d}V \\ &= m_{\mathrm{d}} \times B_{\mathrm{ext}}(r) + R_\tau. \end{aligned} \tag{S6}$$

The gradient holds the moment orientation fixed. Integration by parts gives $\int j\ \mathrm{d}V = 0$ and $\int \xi_i j_j\ \mathrm{d}V = \epsilon_{ijk}(m_{\mathrm{d}})_k$. Taylor expansion of B_{ext} and $\nabla \cdot B_{\mathrm{ext}} = 0$ then give (S6). For $J_2 = \int |\xi|^2 |j(\xi)|\ \mathrm{d}V$, the remainders obey $|R_F| \le \frac{1}{2} J_2 \sup \|D^2 B_{\mathrm{ext}}\|$ and $|R_\tau| \le J_2 \sup \|D B_{\mathrm{ext}}\|$, with operator norms taken over the element and its Taylor segments. Surface-current contributions are included in these integrals when present. Neglecting the higher multipoles recovers the permanent-dipole force and torque used in magnetic robotics (Abbott et al. 2020, Section 4.1, Eqs. (12) and (16)).

A modified Morison-type local closure, stated per unit reference length, is

$$f_{\mathrm{h}} = m_f a_f + m_a(a_f - r_{tt}) - c_1 v_r - c_2 |v_r| v_r, \tag{S7}$$

$$v_r = r_t - v_f, \qquad c_1, c_2 \ge 0. \tag{S8}$$

Here $m_f = \rho_f A_f$ and m_a are displaced-fluid and added mass per unit length; a_f is the prescribed fluid acceleration. This is a reduced closure requiring calibration, not a solved Navier–Stokes interaction. In still fluid its drag power is $-c_1|r_t|^2 - c_2|r_t|^3$. In moving fluid it is dissipative relative to the fluid velocity and can do positive work on the rod. A constant added mass may be moved to the inertial side exactly once.

For a stationary wall, define penetration $\delta(r)$ positive inside the wall and $n_c = \nabla\delta$. One nonadhesive penalty model is

$$V_c = \frac{k_c}{m+1}[\delta]_+^{m+1}, \tag{S9}$$

$$f_c = -k_c[\delta]_+^m n_c - c_c \mathbf{1}_{\{\delta>0\}}[\dot{\delta}]_+ n_c. \tag{S10}$$

The elastic term is included through V_c and the second term dissipates power during compression. Friction can be added as a dissipative port. This switching law is used for force accounting outside the contact-free KAM domain, not as an analytic perturbation through an impact.

Choose a configuration map $(r(s,q), D(s,q))$ that satisfies the essential boundary conditions and the rod constraints. A linear centreline ansatz alone is insufficient to specify rotational inertia. Define

$$r_i = \partial_{q_i} r, \qquad a_i = \mathrm{vee}(D^{\mathsf{T}} \partial_{q_i} D). \tag{S11}$$

Then $\omega_b = \sum_i a_i \dot{q}_i$ and the reduced kinetic energy is

$$T_N = \tfrac{1}{2}\dot{q}^{\mathsf{T}} M(q) \dot{q}, \tag{S12}$$

$$M_{ij}(q) = \int_0^L \left(\rho A r_i \cdot r_j + a_i^{\mathsf{T}} \mathcal{J}_b a_j\right) ds. \tag{S13}$$

An added-fluid-mass contribution may be included when its consistent kinetic metric and associated inertial terms are specified. Otherwise hydrodynamic inertia remains in the generalized force balance and is not also included in M. Galerkin projection of the virtual-work identity gives

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + \nabla V_N(q) = Q_{\mathrm{nc}}, \tag{S14}$$

$$Q_i = \int_0^L (f \cdot r_i + \tau \cdot D a_i) ds + Q_i^{\partial}. \tag{S15}$$

Here $C\dot{q}$ denotes the Christoffel contribution of M, and V_N contains elastic, static magnetic and retained gravitational potentials. Microgravity reduces the last contribution; it does not eliminate fluid drag or establish a particular blood-flow profile. Relative-flow drag, moving-base work and contact losses are retained in Q_{nc}. A stationary-fluid quadratic drag, for example, has nonpositive power; a moving fluid may inject energy. Contact switching is excluded from the analytic neighbourhood used for KAM analysis.

For $M(q) \succ 0$, the Legendre transform yields

$$p = M(q)\dot{q}, \qquad H_N(q,p) = \tfrac{1}{2} p^{\mathsf{T}} M(q)^{-1} p + V_N(q). \tag{S16}$$

In particular, $\dot{p}_i = -\frac{1}{2} p^{\mathsf{T}} \partial_{q_i}(M^{-1}) p - \partial_{q_i} V_N + (Q_{\mathrm{nc}})_i$. The configuration dependence of inertia is retained. Potential forces appear either in V_N or in Q_{nc}, never both.

S1.2 Complex energy–dissipation representation with an explicit decoder

Energy and Rayleigh dissipation have different physical units. Let E_0 and t_0 be reference energy and time, and choose diagonal position and momentum scales Q_0, P_0 with

$Q_0^\mathsf{T} P_0 = E_0 t_0 I$. Set $q = Q_0 x$, $p = P_0 y$, $\tau = t/t_0$ and

$$h(x, y) = H_N(Q_0 x, P_0 y)/E_0, \tag{S17}$$

$$d(x, v) = \frac{t_0}{E_0} \mathcal{D}_N(Q_0 x, Q_0 v/t_0). \tag{S18}$$

Here $v = dx/d\tau$ and $\mathcal{D}_N$ is a Rayleigh dissipation potential. The two entries of $h + id$ are now dimensionless. A complex network provides coupled features and two physically decoded real readouts,

$$\mathcal{H}_\theta(x, y, v) = h_\theta(x, y) + i d_\theta(x, v), \tag{S19}$$

$$h_\theta = h_{\text{prior}} + h_{\text{res},\theta}. \tag{S20}$$

One admissible dissipative head is

$$d_\theta(x, v) = \tfrac{1}{2} v^\mathsf{T} D_\theta(x) v, \qquad D_\theta(x) = L_\theta(x) L_\theta(x)^\mathsf{T} \succeq 0. \tag{S21}$$

More general heads may be used if $d_\theta(x, \cdot)$ is differentiable, convex and minimized at zero. The decoder is

$$v = \partial_y h_\theta, \qquad x' = v, \tag{S22}$$

$$y' = -\partial_x h_\theta - \partial_v d_\theta(x, v) + B(x)u. \tag{S23}$$

The derivative of d_θ is taken with respect to its independent velocity argument before substituting $v = \partial_y h_\theta$. It is not a total derivative through this substitution.

Proposition S1. Power-consistent complex representation. *For a differentiable h_θ, the decoded system satisfies*

$$\frac{dh_\theta}{d\tau} = -v^\mathsf{T} \partial_v d_\theta(x, v) + u^\mathsf{T} B(x)^\mathsf{T} v + \partial_\tau h_\theta. \tag{S24}$$

For the quadratic head, the first term equals $-v^\mathsf{T} D_\theta(x) v \le 0$. For the convex head it is also nonpositive.

Proof. Apply the chain rule to h_θ and substitute (S23). The two canonical terms cancel. Convexity and the minimum at zero imply $v^\mathsf{T} \partial_v d_\theta(x, v) \ge d_\theta(x, v) - d_\theta(x, 0) \ge 0$.

With $z = (x + iy)/\sqrt{2}$, the canonical convention is $\{z_j, \bar{z}_k\} = -i\delta_{jk}$ and

$$z' = -i\partial_{\bar{z}} h_\theta - \frac{i}{\sqrt{2}} \partial_v d_\theta + \frac{i}{\sqrt{2}} B(x)u. \tag{S25}$$

This is a unified complex representation of conservative and nonconservative dynamics with a specified physical decoder. The rule $z' = -i\partial_{\bar{z}}(h_\theta + i d_\theta)$ would generally produce a different vector field and has no automatic dissipativity guarantee. Real analyticity in (x, y), and a controlled analytic extension to a complex neighbourhood, are the relevant requirements for analytic KAM. Holomorphicity in z alone is not required of a nonconstant real energy $h_\theta(z, \bar{z})$. In particular, ReLU or modReLU features require a separate smooth approximation or a theorem of appropriate regularity; their ordinary piecewise-smooth implementation is not analytic.

Table S1. Declared parameters of the unified magnetic cantilever. These are illustrative mechanical parameters, not measurements from a clinical or spaceflight device.

Quantity	Symbol	Value
Length	L	$0.08\,\mathrm{m}$
Circular radius	r_c	$0.0006\,\mathrm{m}$
Young's modulus	E	$5.0 \times 10^6\,\mathrm{Pa}$
Density	ρ	$1100\,\mathrm{kg\,m^{-3}}$
Tip dipole	μ	$2.0 \times 10^{-5}\,\mathrm{A\,m^2}$
Axial bias field	B_0	$0.03\,\mathrm{T}$
Mass damping rate	α	$0.55558524\,\mathrm{s^{-1}}$
Transverse field amplitude	B_a	$0.006\,\mathrm{T}$
Drive frequency	Ω_d	$9.4449490\,\mathrm{rad\,s^{-1}}$
Ramp time	t_r	$0.25\,\mathrm{s}$
Uniform Hermite elements	N_e	32

The dimensional specialization below uses explicit SI units. The learning and KAM statements use dimensionless canonical variables, with their energy and time scales declared before fitting.

S1.3 One clamped–free magnetic beam throughout the computational study

We instantiate the rod reduction as a straight, planar, small-slope Euler–Bernoulli cantilever with a concentrated magnetic dipole at its free tip. The base position and slope are fixed. The elastic part is linear in curvature; the full trigonometric magnetic potential is retained. These choices define a reproducible mechanical specialization of the spatial rod, rather than a catheter-specific material identification. Table S1 states all dimensional parameters. Gravity, imposed fluid flow and field gradient are zero in this specialization. Still-fluid/structural losses are represented by a mass-proportional Rayleigh term. The more general force accounting above remains applicable when those terms are present.

Write $w(s, t)$ for transverse displacement and $\vartheta(t) = w_s(L, t)$ for the small-slope tip rotation. A uniform field $(B_0, B_y(t))$ has zero translational dipole force and the tip moment

$$M_{\text{tip}} = \mu\{B_y(t) \cos\vartheta - B_0 \sin\vartheta\}. \tag{S26}$$

This is the natural moment boundary load. It does not impose a second kinematic constraint at the free tip. The magnetic potential is $-\mu(B_0 \cos\vartheta + B_y \sin\vartheta)$, so that (S26) follows

by variation. Keeping this potential exact does not remove the small-slope approximation in the beam kinematics.

Let $q = (w_1, \vartheta_1, \ldots, w_{N_e}, \vartheta_{N_e})^{\mathsf{T}}$ contain the unconstrained cubic-Hermite nodal coordinates, e_ϑ select its final rotation, and M, K be the consistent mass and bending stiffness matrices. The physical equations and static-bias energy are

$$M\ddot{q} + \alpha M\dot{q} + Kq = e_\vartheta M_{\mathrm{tip}}, \tag{S27}$$

$$H_0(q,p) = \tfrac{1}{2}p^{\mathsf{T}}M^{-1}p + \tfrac{1}{2}q^{\mathsf{T}}Kq + E_0\{1 - \cos(e_\vartheta^{\mathsf{T}}q)\}, \tag{S28}$$

$$E_0 = \mu B_0 = 6 \times 10^{-7}\,\mathrm{J}, \qquad p = M\dot{q}. \tag{S29}$$

There is no axial end restraint and no artificial stretching quartic. The nonlinearity used in learning and in the KAM calculation is the magnetic nonlinearity of this same cantilever.

The static-field linearization is $K_b = K + E_0 e_\vartheta e_\vartheta^{\mathsf{T}}$. Choose $\Phi^{\mathsf{T}}M\Phi = I$ and $\Phi^{\mathsf{T}}K_b\Phi = \mathrm{diag}(\omega_j^2)$, with positive tip-slope entries fixing modal signs. For n retained modes, $q = \Phi_n a$, and the modal momentum is $\pi = \dot{a}$. Set $\omega_0 = \omega_1$, $\tau = \omega_0 t$, $\nu_j = \omega_j/\omega_0$, and

$$a_j = \sqrt{\frac{E_0}{\omega_0\omega_j}}X_j, \qquad \pi_j = \sqrt{\frac{E_0\omega_j}{\omega_0}}Y_j. \tag{S30}$$

The product of each position–momentum scale is E_0/ω_0; division of energy by E_0 and rescaling time therefore give the standard dimensionless canonical equations. Define

$$S = \Phi_n\,\mathrm{diag}\sqrt{\frac{E_0}{\omega_0\omega_j}}, \quad d = S^{\mathsf{T}}e_\vartheta, \quad \ell(s)^{\mathsf{T}} = N(s)S, \tag{S31}$$

where $N(s)$ is the Hermite displacement row. Thus the full centreline and tip slope are recovered in SI units as $w(s) = \ell(s)^{\mathsf{T}}X$ and $\vartheta = d^{\mathsf{T}}X$. The single dimensionless Hamiltonian used below is

$$h_*(X,Y) = h_2(X,Y) + R_{\mathrm{mag}}(X), \tag{S32}$$

$$h_2 = \tfrac{1}{2}\sum_{j=1}^{n}\nu_j(X_j^2 + Y_j^2), \tag{S33}$$

$$R_{\mathrm{mag}} = 1 - \cos(d^{\mathsf{T}}X) - \tfrac{1}{2}(d^{\mathsf{T}}X)^2. \tag{S34}$$

The subtracted quadratic avoids counting the static magnetic stiffness twice. For the two-mode model, $(\omega_1, \omega_2) = (11.41728728, 70.24049294)\,\mathrm{rad\,s^{-1}}$. All learned models use this fixed chart; states are not independently renormalized after fitting.

With $u = B_y/B_0$, $\gamma = \alpha/\omega_0$ and $\mathsf{N} = \mathrm{diag}(\nu_j)$, the physical port decoder becomes

$$X' = \partial_Y h, \tag{S35}$$

$$Y' = -\partial_X h - \gamma\mathsf{N}^{-1}\partial_Y h + u\,d\cos(d^{\mathsf{T}}X). \tag{S36}$$

For $h = h_*$, the damping term is exactly $-\gamma Y$. The Rayleigh head is $\mathcal{D}(v) = \frac{1}{2}\gamma v^{\mathsf{T}}\mathsf{N}^{-1}v$. For a learned energy, retaining the velocity-based form in (S36) preserves nonnegative dissipation. Using $-\gamma Y$ after learning a momentum-dependent residual would, in general, lose that guarantee. The corresponding power identity is

$$h' = -\gamma(\partial_Y h)^{\mathsf{T}}\mathsf{N}^{-1}\partial_Y h + u\cos(d^{\mathsf{T}}X)d^{\mathsf{T}}\partial_Y h. \tag{S37}$$

Alternatively, including $-u\sin(d^{\mathsf{T}}X)$ in the time-dependent energy moves the input power to $-u'\sin(d^{\mathsf{T}}X)$. These are equivalent bookkeeping conventions.

S1.4 Independent COMSOL realization of the same discretization

The model was also constructed and solved in COMSOL Multiphysics 6.3.0.290 through LiveLink for MATLAB. Its two-dimensional Beam interface uses the Euler–Bernoulli formulation, 32 uniform elements, a consistent mass matrix, a fixed base, and the nonlinear tip moment (S26). Geometric nonlinearity is disabled to match the declared small-slope mechanical model. The zero axial displacement subspace decouples, leaving 64 transverse displacement/rotation degrees of freedom.

After selecting transverse degrees of freedom by their field names and coordinates, the independently assembled COMSOL and Python mass and elastic stiffness matrices have relative Frobenius differences 7.76×10^{-15} and 3.05×10^{-14}, respectively. The first six modal frequencies agree to a maximum relative difference of 1.88×10^{-9}. These checks establish agreement of the stated discretizations; they do not establish the material validity of the chosen parameters. The native model, assembly exports, parameter values and time-domain verification records accompany the manuscript.

S1.5 A global result for the unforced physical model

Proposition S2. Unforced global stability of the full FE target. *For the full clamped–free Hermite model (S28), set $B_y = 0$. With the declared parameters, the potential is globally strongly convex and the origin is its unique equilibrium. If $\alpha > 0$, the origin is globally asymptotically*

stable. If $\alpha = 0$, every finite-energy trajectory remains in a bounded energy sublevel and the origin is Lyapunov stable. These are global statements for the declared small-slope mathematical model; they do not establish its physical validity at large deflections or a global property of arbitrary learned quartic residuals.

Proof. Write $b = e_\vartheta$ and $U(q) = q^\mathsf{T} K q/2 + E_0\{1 - \cos(b^\mathsf{T} q)\}$. The clamped conditions give $|w_s(L)|^2 \le L\int_0^L |w_{ss}|^2\, ds$, including for every Hermite finite-element function. Equivalently, $bb^\mathsf{T} \preceq [L/(EI)]K$; equality of the corresponding tip compliance follows from the static quadratic bending shape. Thus, for every q,

$$D^2U(q) = K + E_0\cos(b^\mathsf{T} q)bb^\mathsf{T} \succeq (1-\kappa)K \succeq 0.905K, \tag{S38}$$

$$\kappa = \frac{E_0 L}{EI} = \frac{8}{27\pi} < 0.095. \tag{S39}$$

In particular, $\nabla U = 0$ only at $q = 0$. Both M and K are positive definite, and

$$H_0 \ge \tfrac{1}{2}p^\mathsf{T} M^{-1} p + \tfrac{1}{2}q^\mathsf{T} K q.$$

Its sublevels are compact. Along the unforced dynamics,

$$\dot H_0 = -\alpha p^\mathsf{T} M^{-1} p \le 0.$$

This proves boundedness, global continuation and Lyapunov stability. For $\alpha > 0$, the largest invariant set with $\dot H_0 = 0$ has $p = 0$ and $\nabla U(q) = 0$, hence consists only of the origin. LaSalle's invariance principle on each compact sublevel gives global asymptotic stability.

S2 Complex Hamiltonian learning

S2.1 A fixed canonical chart

For a positive definite constant mass M and reference stiffness K_{ref}, select U such that $U^\mathsf{T} M U = I$ and $U^\mathsf{T} K_{\mathrm{ref}} U = \Omega^2$, where $\Omega = \mathrm{diag}(\omega_1, \dots, \omega_n)$. In the magnetic specialization, $K_{\mathrm{ref}} = K_b$ includes the static-field stiffness. The transformation

$$q = U\Omega^{-1/2}Q, \qquad p = MU\Omega^{1/2}P \tag{S40}$$

is symplectic: its configuration and momentum blocks satisfy $S_q^\mathsf{T} S_p = I$. Consequently $H_2 = \frac{1}{2}\sum_j \omega_j(Q_j^2 + P_j^2)$. The computational variables in (S30) obey $(Q,P) = \sqrt{E_0/\omega_0}(X,Y)$, together with the stated energy and time normalization. That fixed dimensionless chart is used for training, derivative errors and normal forms. Canonical normalization requires the position and momentum scalings to preserve the symplectic form. Optional training weights change only the loss metric.

Write $x = (Q,P)$ and use one complex normalization throughout:

$$z_j = \frac{Q_j + iP_j}{\sqrt{2}}, \quad I_j = z_j\bar z_j, \qquad \{z_j, \bar z_k\} = -i\delta_{jk}. \tag{S41}$$

Wirtinger calculus gives $\partial_{\bar z_j} = (\partial_{Q_j} + i\partial_{P_j})/\sqrt{2}$ and therefore $\dot z = -i\partial_{\bar z} H$ for real H. This is a change of representation of a real symplectic system. Holomorphicity in z alone would exclude most real energy functions, including $H_2 = \sum_j \omega_j z_j \bar z_j$. The analyticity required by analytic KAM instead means a holomorphic extension of the real variables (Q,P) to a complex neighbourhood, in which z and $\bar z$ are independent linear combinations.

S2.2 Architecture and objective

A shared complex feature map $F_\theta(z, \bar z, b)$, with magnetic configuration b, drives real readouts for H_θ and a dissipation factor L_θ. The latter defines $\mathcal{D}_\theta(q,v) = \frac{1}{2}\left\|L_\theta(q)^\mathsf{T} v\right\|^2$. The physical decoder in Section S1 determines the vector field. An affine complex layer has the real block realization

$$(A + iB)(a + ib) = (Aa - Bb) + i(Ba + Ab). \tag{S42}$$

Thus complex weights impose a specific coupling, rather than furnishing an extra physical state. Smooth split activations $\sigma(a + ib) = \tanh(a) + i\tanh(b)$ are real analytic on bounded real domains and extend to a sufficiently small pole-free complex neighbourhood when weights are bounded. Their use does not justify assuming an arbitrarily wide analytic strip.

For observed dimensionless states and velocities, a suitable objective is

$$\widehat{\mathcal{L}} = \frac{1}{N}\sum_{i=1}^{N}\left\|\widehat{\dot x}_i - f_\theta(x_i, u_i)\right\|^p + \lambda_E\widehat{\mathcal{L}}_E + \lambda_R\mathcal{R}(\theta). \tag{S43}$$

Energy observations, if available, fix a gauge through $\widehat{\mathcal{L}}_E$; vector-field training alone identifies H only up to a constant. To infer conservative gradient error from the full decoder loss, errors in dissipation and inputs must first be bounded or measured and included in the label-error budget. They cannot be absorbed into a Hamiltonian residual merely because their magnitudes are small.

The reproducible experiments specialize this architecture to a finite analytic complex feature network. The exact

magnetic feature R_{mag} in (S34) is represented by complex exponentials and a quadratic correction, with a learned real readout. Quartic features complement its degree-four Taylor coefficient; this keeps the full cosine target in the hypothesis class while avoiding a duplicate quartic direction. Products of complex affine forms and their conjugates generate all real quadratic and quartic monomials. They can be implemented by a finite network with polynomial activation, using $uv = ((u+v)^2 - (u-v)^2)/4$. A real readout combines the full magnetic analytic feature with the polynomial residual; conjugate features enforce the reality of the energy. The prescribed features exclude a spurious cubic term and, for the gray-box model, preserve the known static-field quadratic part exactly. Fitting the output coefficients from energy derivatives is linear least squares. The experiment therefore tests an explicitly specified finite-feature CHNN. The magnetic coefficient and polynomial residual readouts are estimated by linear least squares within this finite-feature architecture.

S2.3 What complex approximation theory establishes

Zhang et al. (2022, Theorem 1) establish universal approximation with the specified complex-reaction activation. Their Theorem 2 gives a separate L^2 comparison with real networks under a particular distribution and resource budget. These results have distinct hypotheses and approximation criteria. Such a theorem supplies neither higher derivative accuracy nor a universal advantage for magnetic catheter energies. In particular, piecewise reaction activations cannot be inserted unchanged into a smooth KAM proof.

There is nevertheless a direct smooth approximation route. For input $(Q+iP)/\sqrt{2}$, first-layer weights $\sqrt{2}(A_Q - iA_P)$ and real bias b give preactivation $(A_QQ + A_PP + b) + i(A_QP - A_PQ)$. Split activations and subsequent real weights keep the real channel independent of the imaginary channel, so a real-part output realizes standard real networks. For bounded, nonconstant C^m activations such as $\tanh$, Hornik (1991, Theorem 3) gives uniform approximation through derivative order m of C^m targets on compact sets, with network width allowed to grow. Alternatively, increasing the polynomial degree and network size realizes successive truncations of a convergent analytic power series, with uniform derivative control on compact polydiscs inside its convergence domain. The fixed quartic feature class used in the experiments instead requires the quantitative coefficient certificate developed below.

S3 From nonzero training loss to KAM persistence

S3.1 A statistical bridge in the norm required by KAM theory

All Hamiltonians below are real-valued scalar functions, including when their parameters and hidden representations are complex. The realification of a complex parameter vector has dimension P. Let $\mathcal{D} = \mathbb{T}^n \times B$, where B is a bounded ball in action space, and write $d = 2n$. Lebesgue measure on $\mathcal{D}$ is denoted by dx. The following result completes the derivative-norm interface needed to combine the learning argument of Chen et al. (2022) with a finite-differentiability KAM theorem. It does not infer derivative smallness from function values alone.

Theorem S3. Uniform learning-to-C^ℓ bound. *Fix $d < p < \infty$, an integer $\ell \geq 1$, and an integer $s > \ell + d/p$. Let $X_1, \dots, X_N$ be independent samples with density $f_\mu \geq m_0 > 0$ on $\mathcal{D}$. Let $H_\star$ be the target Hamiltonian and $\{H_\vartheta : \vartheta \in \Theta\}$ the hypothesis class, with $\Theta \subset [-B_\Theta, B_\Theta]^P$. Assume that for all $x \in \mathcal{D}$ and $\vartheta, \eta \in \Theta$,*

$$\|\nabla H_\vartheta(x) - \nabla H_\star(x)\|_p \leq Q, \tag{S44}$$

$$\|\nabla H_\vartheta(x) - \nabla H_\eta(x)\|_p \leq L_g \|\vartheta - \eta\|_\infty. \tag{S45}$$

After fixing the additive gauge by

$$e_\vartheta = H_\vartheta - H_\star - \frac{1}{|\mathcal{D}|}\int_{\mathcal{D}} (H_\vartheta - H_\star)\, dx,$$

$$\sup_{\vartheta\in\Theta} \|e_\vartheta\|_{W^{s,p}(\mathcal{D})} \leq M_s, \tag{S46}$$

suppose that M_s is a valid uniform bound. Define

$$\widehat{R}_N(\vartheta) = \frac{1}{N}\sum_{i=1}^{N} \|\nabla H_\vartheta(X_i) - \nabla H_\star(X_i)\|_p^p, \tag{S47}$$

$$L_{\mathrm{par}} = pQ^{p-1}L_g, \qquad L_{\max} = Q^p. \tag{S48}$$

For any fixed $0 < a \leq 2B_\Theta$ and $0 < \delta < 1$, put

$$\Gamma_N(a,\delta) = 2L_{\mathrm{par}}a + \frac{L_{\max}}{\sqrt{2N}}\sqrt{P\log(1 + 2B_\Theta/a) + \log(1/\delta)}, \tag{S49}$$

$$S_N(\vartheta) = \widehat{R}_N(\vartheta) + \Gamma_N(a,\delta), \tag{S50}$$

$$G_N(\vartheta) = C_{\mathrm{P}}\, m_0^{-1/p} S_N(\vartheta)^{1/p}, \tag{S51}$$

$$\alpha = \frac{\ell - 1 + d/p}{s-1} \in (0,1). \tag{S52}$$

Here C_{P} is a Poincare constant chosen so that a mean-zero function satisfies $\|e\|_{W^{1,p}} \leq C_{\mathrm{P}}\|\nabla e\|_{L^p}$. With probability at

least $1-\delta$*, simultaneously for every* $\vartheta \in \Theta$*,*

$$\|e_\vartheta\|_{C^\ell(\overline{\mathcal{D}})} \le E_N(\vartheta) := C_{\mathrm{GN}}\left[G_N(\vartheta) + M_s^\alpha G_N(\vartheta)^{1-\alpha}\right]. \tag{S53}$$

C_{GN} *depends only on the domain and* (d, p, s, ℓ)*. We use* $\|e\|_{C^\ell} = \max_{|a|\le\ell} \sup |D^a e|$*; equivalent conventions only change the displayed constants. The event is uniform over the hypothesis class and therefore includes parameters selected using these same training data.*

Proof. The loss $r_\vartheta(x) = \|\nabla H_\vartheta(x) - \nabla H_\star(x)\|_p^p$ lies in $[0, L_{\max}]$ and is L_{par}-Lipschitz in the ℓ^∞ parameter metric. Partitioning the parameter cube into cells of side length a and choosing a representative from each nonempty cell gives an a-net with at most $(1 + 2B_\Theta/a)^P$ points. The one-sided Hoeffding inequality and a union bound control $R(\eta) - \widehat{R}_N(\eta)$ at every net point by the square-root term in Γ_N, where $R(\eta) = \int r_\eta \, d\mu$. The population and empirical losses each change by at most $L_{\mathrm{par}} a$ between a parameter and its net representative. Consequently $R(\vartheta) \le S_N(\vartheta)$ uniformly.

The density lower bound gives $\|\nabla e_\vartheta\|_{L^p(dx)} \le m_0^{-1/p} S_N(\vartheta)^{1/p}$. The gauge in (S46) and Poincare's inequality then yield $\|e_\vartheta\|_{W^{1,p}} \le G_N(\vartheta)$. Apply the Gagliardo–Nirenberg inequality to $g = \nabla e_\vartheta$, with derivative orders $j = \ell - 1$, $m = s - 1$, and both endpoint Lebesgue exponents equal to p. Its interpolation identity is

$$0 = \frac{j}{d} + \alpha\left(\frac{1}{p} - \frac{m}{d}\right) + \frac{1-\alpha}{p},$$

which gives the stated α. The strict inequality $s > \ell + d/p$ ensures $\alpha < 1$ and avoids the critical Sobolev endpoint. Extension from the smooth bounded action domain, together with periodic charts in the angular variables, gives

$$\|e\|_{C^\ell} \le C_{\mathrm{GN}}\left(\|e\|_{W^{1,p}} + \|e\|_{W^{1,p}}^{1-\alpha}\|e\|_{W^{s,p}}^{\alpha}\right).$$

Lower-order derivatives are included by the inhomogeneous term and the $W^{1,p} \hookrightarrow C^0$ embedding, which uses $p > d$. Substitution proves (S53).

For instance, one may select an integer $\ell > 2(\tau + 1)$, $p > 2n$, and $s = \ell + 1$. This makes every exponent explicit, although the resulting interpolation bound can be conservative. Bounds on M_s, Q, L_g follow from bounded parameters and derivatives of smooth activations on a compact domain; the corresponding bounds for $H_\star$ must also be justified. Merely declaring the functions smooth does not provide these quantitative bounds. Alternatively, a loss containing all spatial derivatives of the vector field through order ℓ controls $W^{\ell+1,p}$ directly, and Sobolev embedding then gives a C^ℓ error with exponent $1/p$, without interpolation.

Corollary S4. KAM transfer for the target and the trained Hamiltonian. *Let* $n \ge 2$*,* $\tau > n - 1$ *and an integer* $\ell > 2(\tau + 1)$ *be fixed. Suppose that, in one specified canonical action–angle chart,*

$$H_\star = h_0(I) + P_\star(I, \varphi), \qquad H_\vartheta = h_0(I) + P_\vartheta(I, \varphi).$$

Assume $h_0 \in C^{\ell+2}(\overline{B})$ *with finite norm, and* $\nabla h_0 : B \to \Omega$ *is a diffeomorphism with bounded inverse in* C^ℓ*. Fix* $\gamma > 0$ *and set*

$$\Omega_{\gamma,\tau} = \left\{ \omega \in \Omega : \begin{array}{l} |k \cdot \omega| \ge \gamma |k|_1^{-\tau} \ (0 \ne k \in \mathbb{Z}^n), \\ \mathrm{dist}(\omega, \partial\Omega) \ge \gamma \end{array} \right\}. \tag{S54}$$

Let $c_{\mathrm{K}} > 0$ *be the constant in Theorem A of Bounemoura (2020) for this domain and integrable Hamiltonian. If a certified bound* $\beta_\vartheta \ge \|P_\vartheta - \overline{P}_\vartheta\|_{C^\ell}$ *satisfies*

$$\beta_\vartheta + E_N(\vartheta) < c_{\mathrm{K}} \gamma^2, \tag{S55}$$

then, on the event in Theorem S3, both $H_\star$ *and* H_ϑ *possess invariant Lagrangian tori for every frequency in* $\Omega_{\gamma,\tau}$*. The flow on each torus is* C^1*-conjugate to the linear flow with that frequency. For a piecewise smooth frequency-domain boundary, the complement of each family has measure at most* $C\gamma$*. In particular, each family has positive measure when* γ *is small enough.*

Proof. Write $\overline{P}$ for the spatial mean over $\mathcal{D}$. The gauge-fixed difference satisfies

$$P_\star - \overline{P}_\star = P_\vartheta - \overline{P}_\vartheta - e_\vartheta.$$

Thus the two perturbation norms are at most β_ϑ and $\beta_\vartheta + E_N(\vartheta)$, respectively. Constants added to a Hamiltonian do not change its vector field. Apply the cited finite-smoothness KAM theorem separately to the two Hamiltonians with the same h_0, domain, and Diophantine set. The theorem supplies tori of class $C^{\tau+1}$ with Lipschitz dependence as invariant graphs on frequency. The two systems need not have identical invariant embeddings.

The certified budget includes the physical perturbation from h_0, any normal-form remainder, and the learning error. A small learning error alone does not establish that the physical system is close to an integrable one. Theorem A in the published 2020 version of Bounemoura (2020) gives the measure bound $C\gamma$; the $C\gamma^2$ measure bound printed in the 2018 preprint should not be used.

S3.2 A coefficient certificate for a finite analytic CHNN

The finite analytic readout used in the experiments yields a stronger certificate that avoids a dimension-dependent interpolation loss. Let $\psi_1, \dots, \psi_m$ be real-valued basis functions on the real phase domain. Complex features may be converted into this basis by taking their real and imaginary parts and collecting equal monomials. The certificate applies to the effective coefficients of the actual trained Hamiltonian, irrespective of how they are parameterized inside the network.

Proposition S5. Deterministic global certificate from sampled gradients. *Suppose*

$$H_\vartheta = H_{\text{prior}} + \sum_{j=1}^{m} c_{\vartheta,j}\psi_j,$$
$$H_\star = H_{\text{prior}} + \sum_{j=1}^{m} c_{\star,j}\psi_j + R.$$

The basis contains no constant gauge direction. Let

$$G_N = \frac{1}{N}\sum_{i=1}^{N} D\psi(X_i)^\top D\psi(X_i),$$
$$\lambda_N = \lambda_{\min}(G_N) > 0,$$

where the jth column of $D\psi$ is $\nabla\psi_j$. For gradient observations $Y_i = \nabla H_\star(X_i) + \xi_i$, define

$$r_N = \left(\frac{1}{N}\sum_i \|\nabla H_\vartheta(X_i) - Y_i\|_2^2\right)^{1/2},$$
$$t_N = \left(\frac{1}{N}\sum_i \|\nabla R(X_i) + \xi_i\|_2^2\right)^{1/2}.$$

Then

$$\|c_\vartheta - c_\star\|_2 \le \frac{r_N + t_N}{\sqrt{\lambda_N}}. \tag{S56}$$

For any function norm $\|\cdot\|_{\mathcal{X}}$ satisfying the triangle inequality, put $K_{\mathcal{X}} = (\sum_j \|\psi_j\|_{\mathcal{X}}^2)^{1/2}$. If $\|R\|_{\mathcal{X}} \le \varepsilon_R$, then

$$\|H_\vartheta - H_\star\|_{\mathcal{X}} \le K_{\mathcal{X}}\frac{r_N + t_N}{\sqrt{\lambda_N}} + \varepsilon_R. \tag{S57}$$

The norm may be C^ℓ on a real compact domain or the supremum norm on a prescribed complex neighborhood where every basis function and R is holomorphic. No independent sampling assumption is needed.

Proof. Let $\Delta c = c_\vartheta - c_\star$. The empirical norm of $D\psi\Delta c$ is $(\Delta c^\top G_N \Delta c)^{1/2}$. Since $D\psi\Delta c = (\nabla H_\vartheta - Y) + (\nabla R + \xi)$ at every sample, the triangle inequality gives $\sqrt{\lambda_N}\|\Delta c\|_2 \le r_N + t_N$. Finally, $\|\sum_j \Delta c_j\psi_j - R\|_{\mathcal{X}} \le K_{\mathcal{X}}\|\Delta c\|_2 + \varepsilon_R$ by Cauchy–Schwarz.

For raw monomials $\psi_\nu(x) = x^\nu$, an explicit C^ℓ bound on $[-r, r]^d$ is obtained from

$$\|\psi_\nu\|_{C^\ell} \le \max_{|a|\le\ell,\, a\le\nu} \frac{\nu!}{(\nu-a)!}\, r^{|\nu|-|a|}.$$

The analytic sup norm on $\{|x_j| \le r\} \subset \mathbb{C}^d$ is $r^{|\nu|}$. For all 35 degree-four monomials in four variables, $K_{\text{an}} \le \sqrt{35}\, r^4$. If a canonical chart $x = \Phi(I, \varphi)$ maps an action–angle complex domain into this polydisc, the same analytic bound holds after composition. For a C^ℓ certificate after composition, the derivative bounds of Φ must instead be included by the chain rule. Unmodeled physical terms require the tail bounds t_N and ε_R. Their approximation-tail contribution is zero only when the target lies exactly in the declared class. The full-cosine feature class below has this property for its specified reduced target; it does not remove truncation error relative to the full beam.

The statistical result requires independent state samples with the stated coverage density. Consecutive states on a trajectory do not satisfy this assumption automatically. Accordingly, the trajectory-based experiments below use the deterministic Gram certificate, which requires no sampling independence. Their implementation divides both squared residuals and the Gram matrix by $4N$ rather than N; the coefficient bound is unchanged provided the noise norm uses that same scalar-observation normalization.

S4 Nonlinear stability of the learned magnetic model

S4.1 Normal form of the same magnetic cantilever

In the dimensionless chart of Section S1.3, write

$$h_\star = h_2 + g(d^\top X),$$
$$h_2 = \frac{1}{2}\sum_{j=1}^{n} \nu_j(X_j^2 + Y_j^2), \tag{S58}$$
$$g(z) = 1 - \cos z - \frac{z^2}{2}.$$

The magnetic term in this expression is the complete analytic function. In particular, its fourth-order part and its

untruncated remainder are

$$h_{4,\star} = -\frac{(d^\top X)^4}{24},$$
$$T_{6,\star} = 1 - \cos(d^\top X) - \frac{(d^\top X)^2}{2} + \frac{(d^\top X)^4}{24}.$$

Proposition S6. Magnetic twist and analytic tail. *Let $I_j = (X_j^2 + Y_j^2)/2$, and let $A_\star$ denote the Hessian with respect to I of the angular average of $h_{4,\star}$. With $v = (d_1^2, \dots, d_n^2)^\top$,*

$$(A_\star)_{jj} = -\frac{d_j^4}{8},$$
$$(A_\star)_{ij} = -\frac{d_i^2 d_j^2}{4} \quad (i \neq j), \tag{S59}$$
$$A_\star = \frac{\operatorname{diag}(d_j^4) - 2vv^\top}{8}.$$

and

$$\det A_\star = (1-2n)8^{-n} \prod_{j=1}^{n} d_j^4. \tag{S60}$$

Thus every retained mode with nonzero tip-rotation participation contributes to a nonsingular twist. For $n \geq 2$, the twist has one negative and $n-1$ positive eigenvalues; positive definiteness is not required here. If $D = \sum_j |d_j|$ and $\mathbb{D}_r^{2n}$ is a complex Cartesian polydisc, then

$$\|T_{6,\star}\|_r \leq \frac{(Dr)^6}{720} \cosh(Dr). \tag{S61}$$

Proof. The angular averages are $\langle X_j^4 \rangle = 3I_j^2/2$ and $\langle X_i^2 X_j^2 \rangle = I_i I_j$. All terms with an odd exponent in some X_j have zero average. Substitution gives (S59). Factoring $\operatorname{diag}(d_j^2)$ on both sides gives a matrix congruent to $(I_n - 2\mathbf{1}\mathbf{1}^\top)/8$. Its eigenvalues are $1/8$ with multiplicity $n-1$ and $(1-2n)/8$, which proves the determinant and inertia statements. For the analytic tail, expand the cosine and use $(6+2k)! \geq 6!(2k)!$ and $|d^\top X| \leq Dr$.

For the two-mode, 32-element model, the magnetic-bias modes give $\nu \simeq (1, 6.152117504)$ and $d \simeq (0.22567077, 0.32179881)$. Consequently $\det A_\star \simeq -1.303708 \times 10^{-6}$ and $\sigma_{\min}(A_\star) \simeq 5.8064 \times 10^{-4}$. These quantities belong to the same cantilever model used for learning and shape reconstruction.

S4.2 Complete learned Hamiltonian and transfer of twist

The fitted family is

$$h_\eta = h_2 + \frac{b}{s_4} g(d^\top X) + \sum_{j=1}^{34} c_j \psi_j(X, Y), \tag{S62}$$

where $\eta = (b, c_1, \dots, c_{34})$, $a = b/s_4$, $s_4 = \|q_\star\|_2$ is the fixed quartic coefficient normalizer and the ψ_j span the coefficient-space complement of the target quartic direction. The class contains the complete target at $\eta_\star = (s_4, 0, \dots, 0)$; no cosine truncation error is assigned to the training labels. The polynomial part is represented by matched real or complex features. All bounds use the actually exported feature matrix, so numerical orthogonality of its columns is not a proof assumption.

Let $F(z) = \big(g(d^\top X)/s_4, \psi_1(z), \dots, \psi_{34}(z)\big)$, and put $D(z) = [\nabla F_0(z), \dots, \nabla F_{34}(z)] \in \mathbb{R}^{4\times 35}$. With $G_N = (4N)^{-1} \sum_i D(z_i)^\top D(z_i)$, $\lambda_{\min}(G_N) \geq \underline{\lambda} > 0$, empirical derivative root-mean-square loss r_N (averaged over all $4N$ scalar observations), and a certified derivative-label error with the same root-mean-square normalization bounded by ϵ_N, the deterministic estimate is

$$\|\eta - \eta_\star\|_2 \leq b_N := \frac{r_N + \epsilon_N}{\sqrt{\underline{\lambda}}}. \tag{S63}$$

This estimate follows from the smallest singular value of the stacked derivative design matrix and the triangle inequality.

The effective quartic coefficient vector is $q_\eta = (b/s_4)q_\star + Uc$, where U is the exported matrix of polynomial features. Let $\mathcal{A}q$ be the symmetric twist matrix obtained by angular averaging. Then the directly computable bound

$$\|A_\eta - A_\star\|_2 \leq \|\mathcal{A}[q_\star/s_4\, U]\|_{2\to F}\, b_N + \delta_{A,\mathrm{dec}} < \sigma_{\min}(A_\star) \tag{S64}$$

is sufficient for nonsingularity of the trained twist. The subscript F denotes the Frobenius norm on symmetric matrices. In the two-mode raw quartic basis, the previously derived bound $\|\mathcal{A}\Delta q\|_2 \leq \sqrt{19}\|\Delta q\|_2$ is also valid. A direct rational or interval check of A_η can be sharper than the sufficient loss-based bound. Here $\delta_{A,\mathrm{dec}}$ bounds the exported decoder's rounding error relative to its exact coefficient map; it is zero for exact arithmetic and is explicitly included in the numerical certificate. Specifically, the exported effective-quartic coefficient error δ_{dec} gives $\delta_{A,\mathrm{dec}} = \sqrt{19}\delta_{\mathrm{dec}}$.

Define the finite-feature analytic bound by

$$B_F(r)^2 = \left(\frac{(Dr)^4}{24 s_4}\cosh(Dr)\right)^2 + r^8 \sum_{j=1}^{34} \|\psi_j\|^2_{\mathrm{coeff},1}.$$

The same coefficient estimate gives a complete analytic perturbation bound,

$$\|h_\eta - h_\star\|_r \le b_N B_F(r) + \delta_{h,\mathrm{dec}}(r). \tag{S65}$$

This controls the entire learned function on a complex domain, including every higher-order magnetic term.

Suppose $\alpha_4 = \min_{0<|k|_1\le 4} |k\cdot\nu| > 0$. The exact quartic Lie transformation associated with the full effective $h_{4,\eta}$ gives

$$h_\eta \circ \Phi_\eta = \nu\cdot I + \tfrac{1}{2} I^\top A_\eta I + \mathcal{R}_{6,\eta}.$$

If C_4, C_Z, C_χ are the Cartesian coefficient norms of its quartic part, angular average and homological generator, then

$$\|\mathcal{R}_{6,\eta}\|_r \le 512 C_\chi (C_4 + C_Z) r^6 + \frac{|a|(2Dr)^6}{720}\cosh(2Dr). \tag{S66}$$

This holds when $32 C_\chi r^2 < 1$. The first term is the exact quartic Lie remainder; the second bounds $aT_{6,\star}\circ\Phi_\eta$ because the time-one flow maps the radius-r polydisc into the radius-$2r$ polydisc. This identity retains the full cosine rather than applying a theorem to an isolated quartic approximation.

With nonsingular A_η and fourth-order nonresonance, the ordinary Kolmogorov theorem applies on sufficiently small positive-action boxes (Pöschel 2001, Theorems A–B, arXiv:0908.2234v1). Indeed, their action width is $O(r^2)$, their integrable Hessian is A_η, and (S66) is $O(r^6)$. Choosing a Diophantine constant proportional to r^2 makes the analytic KAM smallness quotient $O(r^2)$; reducing its proportionality constant first leaves a positive-measure set of admissible frequencies. This establishes local KAM families for the complete target and trained cantilever Hamiltonians. The finite-amplitude result below is verified separately by an a posteriori theorem and is not inferred from this asymptotic statement.

S4.3 Validated invariant tori at finite physical amplitude

The local normal-form argument can be supplemented by a finite-amplitude certificate for the same complete Hamiltonian. We use the a posteriori theorem of Haro and Luque (2019); the equation and table numbering below refer to Theorem 3.1 and Appendix B, Tables 1–3, of its archived preprint, arXiv:1711.05515v1. The computation includes the analytic interpolation error and uses ball arithmetic, rather than identifying a small residual at finitely many sample points with an analytic norm.

The mathematical model certified here is specified exactly by the exported binary coefficients. In particular,

$$\nu = \left(1, \frac{6926668524826847}{1125899906842624}\right), \quad d = \left(\frac{4065323231059343}{18014398509481984}, \frac{2898505962069547}{9007199254740992}\right). \tag{S67}$$

The target uses $a = 1$ and zero quartic residual in (S62). Each trained case uses the exact dyadic value of its exported cosine multiplier and every exported quartic coefficient. This convention identifies the function being certified without hiding floating-point parameter uncertainty. It establishes a result for the exported two-mode reduction; independent finite-element and COMSOL comparisons do not themselves prove an infinite-dimensional KAM theorem for the continuum beam.

Theorem S7. Finite-amplitude tori of the target and trained models. *For the target Hamiltonian* (S58)*, with parameters* (S67)*, for each of the three primary exported GC Hamiltonians with training seeds* 17*,* 43*, and* 91*, and for the additional seed-*17 *fit at noise amplitude* 10^{-8}*, and for the GC model fitted to independent COMSOL position–velocity trajectories, there is a real analytic, embedded Lagrangian torus* $K_\infty : \mathbb{T}^2 \to \mathbb{R}^4$ *satisfying the invariance equation below, with the displayed frequency for the first five models:*

$$X_h(K_\infty) = DK_\infty \varpi, \quad \varpi = \frac{w_1}{2\pi}(1,\zeta), \quad w_1 = \frac{49999589341}{50000000000}. \tag{S68}$$

where, with $\phi = (1+\sqrt{5})/2$*,*

$$\zeta = \frac{1876278\phi + 592877}{304979\phi + 96369}. \tag{S69}$$

For the COMSOL-fitted model, the frequency is instead specified by

$$w_1 = \frac{99999143141}{100000000000}, \quad \zeta = \frac{974993\phi + 427310}{158480\phi + 69457}. \tag{S70}$$

It is chosen near that actual model's quartic normal-form frequency. Angles have period one; ϖ is measured in cycles per unit dimensionless time. Each torus lies within 3×10^{-41}, in the supremum norm on the complex strip of width $1/80$, of its computed finite Fourier polynomial. The reconstructed beam has nonzero millimetre-scale motion: the certified tip-displacement witness and the bound valid for every point of the centerline are given in Table S2.

Proof. We give the analytic estimates and specify the finite checks that close them. Vector norms are maximum norms. Following the source convention, analytic matrix norms are $\max_i\sum_j\sup_\theta|M_{ij}(\theta)|$, with componentwise suprema on $\mathbb{T}^2_\rho=\{\theta:|\operatorname{Im}\theta|_\infty<\rho\}$. The phase-space domain is $\mathbb{D}^4_{3/10}$.

Frequency and the cohomological equation. The continued fraction of ζ has prefix $[6;6,1,1,2,1,23,1,13,6,3]$ followed by an infinite tail of ones. For the COMSOL-fitted model the prefix is $[6;6,1,1,2,1,20,1,1,4,1,1,3,2]$, also followed by ones. All partial quotients in either case are at most 23. The bounded-partial-quotient inequality gives $|q\zeta-p|\ge1/(25|q|)$ for $q\neq0$. As $w_1>9/10$ and $2\pi<7$, this implies

$$|k\cdot\varpi|\ge\frac{\gamma}{|k|_1},\qquad\gamma=\frac{9}{1750},\qquad k\in\mathbb{Z}^2\setminus\{0\}.\tag{S71}$$

The case $k_2=0$ follows directly from $\varpi_1>9/70$. We use the weaker exponent $\tau_K=3$ in the a posteriori theorem. For a zero-mean analytic function v, Fourier division by $2\pi ik\cdot\varpi$ and the count $\#\{k:|k|_1=j\}=4j$ give, with $q=e^{-2\pi\delta}$,

$$\begin{aligned}\|\mathcal{R}_\varpi v\|_{\rho-\delta}&\le\frac{2}{\pi\gamma}\frac{q(1+q)}{(1-q)^3}\|v\|_\rho\\&\le\frac{e^{4\pi\rho}}{2\pi^4\gamma\delta^3}\|v\|_\rho<\frac{\|v\|_\rho}{\gamma\delta^3}.\end{aligned}$$

These inequalities hold for $0<\delta<\rho=1/40$, so the cohomological constant $c_R=1$ is justified directly. Reconstructing the continued-fraction convergent matrix verifies the stated frequency and Diophantine bound.

Global bounds and theorem constants. Write $h=h_2+ag(d^\top X)+p_4$, with quartic coefficient norm $C_4=\sum_\alpha|(p_4)_\alpha|$. The exact exported coefficients satisfy $|a|\le2$, $C_4<1/100$, $\|d\|_1<11/20$, and $\max\nu_j<7$. The full cosine and its derivatives then give, on $\mathbb{D}^4_{3/10}$,

$$\begin{aligned}c_{X_h,0}&=3,\quad&c_{X_h,1}&=8,\\c_{X_h^\top,1}&=32,\quad&c_{X_h,2}&=1.\end{aligned}\tag{S72}$$

The third quantity bounds the derivative of the *row-vector map* $X_h^\top$. It bounds the sum of the absolute Jacobian entries and is not the matrix norm of $(DX_h)^\top$. For example, its linear part is $2(\nu_1+\nu_2)<28$; its remaining bound is $2\|d\|_1^2(\cosh(3\|d\|_1/10)-1)+12C_4(3/10)^2<1$. For the second derivative, the row-sum tensor estimate $2\|d\|_1^3\sinh(3\|d\|_1/10)+24C_4(3/10)<1$ suffices. All geometric tensors are constant canonical tensors with norm one and zero derivatives.

Let $J=\left(\begin{smallmatrix}0&I\\-I&0\end{smallmatrix}\right)$, $L=DK$, $B=(L^\top L)^{-1}$, and $N=-JLB$. In the source convention the symplectic matrix is $\Omega=-J$. The torsion used in the theorem is therefore

$$T=-N^\top J\big(DX_h(K)N-D_\varpi N\big).\tag{S73}$$

There are no additional first integrals ($n=d=2$, canonical Case III). Choose $\rho=1/40$, $\rho_\infty=1/80$, $a_1=a_2=2$, $a_3=12$, $\delta=\rho/12$, and the iteration parameter $\nu_{\mathrm{it}}=1$. The bounds used with strict margins are

$$\begin{aligned}\|DK\|_\rho&<1<\sigma_K=2,\\\|DK^\top\|_\rho&<\tfrac32<\sigma_{K^\top}=2,\\\|B\|_\rho&<3<\sigma_B=4,\end{aligned}$$

$$\|\langle T\rangle^{-1}\|<10^5<\sigma_T=2\times10^5,\qquad\|K\|_\rho<\tfrac15.$$

The last inequality leaves a phase-space distance margin $1/10$. Substitution in Tables 1–3 gives exact rational constants with the following decimal approximations:

$$\begin{aligned}C_1&\simeq4.843521623\times10^{46},\\C_2&\simeq4.940319879\times10^{21},\\\frac{\gamma^4\rho^{12}}{C_1}&>8.6086\times10^{-76}.\end{aligned}\tag{S74}$$

The rational recurrence in those tables is evaluated using rational or ball arithmetic; rounded decimal values are used only for display. The source's C_{sym} follows from the Case III block identity in Remark 4.7, immediately after Eq. (4.39): $C_{\mathrm{sym}}=\max\{1,\sigma_B^2\}C_{\Omega_L}=192$, using $\sigma_B=4$ and $C_{\Omega_L}=12$. This uses C_{Ω_L} from the geometric identity in place of the $C_{\widehat{\Omega}_L}$ printed in the corresponding row of Table 1.

Analytic residual and torsion checks. For each case the approximate embedding K is an exact dyadic finite Fourier polynomial supported on $|k_j|\le63$. Conjugate symmetry is checked exactly. Its derivative and wide-strip norms are bounded by $\sum_k|K_k|e^{2\pi s|k|_1}$ and the corresponding differentiated sums. To bound B, compare K with $K_0=(\frac{1}{10}\cos2\pi\theta_1,\frac{1}{10}\cos2\pi\theta_2,-\frac{1}{10}\sin2\pi\theta_1,-\frac{1}{10}\sin2\pi\theta_2)$. Put $e_L=\|D(K-K_0)\|$ and $e_{L^\top}=\|D(K-K_0)^\top\|$. As

$DK_0^\top DK_0 = (2\pi/10)^2 I$ exactly, the checked inequality

$$\|DK_0^\top\| e_L + e_{L^\top}\|DK_0\| + e_{L^\top} e_L < (2\pi/10)^2$$

provides a Neumann-series inverse bound for the analytic Gram matrix.

The residual $E = X_h(K) - DK\varpi$ is evaluated with 512-bit Arb balls on a 256×256 grid, including ball sine evaluations and all quartic terms. Let $\mathcal{I}_m E$ be its trigonometric interpolant, $\widehat{\rho} = 1/4$, and $M \geq \|E\|_{\widehat{\rho}}$, computed directly from the finite Fourier polynomial and the entire Hamiltonian. Fourier decay, the two-dimensional union bound for the omitted modes, and the alias sum give

$$\begin{aligned} \|E\|_\rho &\leq \|\mathcal{I}_m E\|_\rho + E_{\text{tail}}, \\ E_{\text{tail}} &= 8M \frac{1+q}{(1-q)^2} e^{-\pi(\widehat{\rho}-\rho)m}. \end{aligned} \tag{S75}$$

Here $q = e^{-2\pi(\widehat{\rho}-\rho)}$ and $m = 256$. Both the omitted Fourier tail and the aliased coefficients are included in the factor 8. The larger strip is used only to estimate the composition residual; the entire full-cosine Hamiltonian is defined there even when $K(\mathbb{T}^2_{\widehat{\rho}})$ leaves the smaller phase-space domain used by the KAM iteration.

For the mean torsion, the same Gram estimate on the strip $7/100$ proves $\|N\|, \|N^\top\| < 6$. Since $\|\varpi\|_1 < 2$, Cauchy's inequality on the strip $6/100$ gives $\|D_\varpi N\| < 1200$, and hence $\|T\| < 6(8 \cdot 6 + 1200) < 10^4$. An Arb trapezoidal average on 128^2 nodes is corrected by

$$10^4 \left[\left(\frac{1+z}{1-z} \right)^2 - 1 \right], \qquad z = e^{-2\pi(6/100)128}.$$

The derivative of B in this evaluation is calculated by differentiating its inverse identity, so no differentiated interpolation of N is used. Inverting the resulting interval 2×2 matrix proves the stated bound on $\langle T \rangle^{-1}$.

The resulting bounds in Table S2 satisfy

$$\frac{C_1 \|E\|_\rho}{\gamma^4 \rho^{12}} < 1. \tag{S76}$$

The cited theorem supplies the invariant Lagrangian torus and $\|K_\infty - K\|_{\rho_\infty} \leq C_2 \|E\|_\rho / (\gamma^2 \rho^6) < 3 \times 10^{-41}$. For completeness, embeddedness can also be checked quantitatively: writing $X_j - iY_j = 0.1 e^{2\pi i \theta_j} + \xi_j$, the verified C^0 and C^1 differences from K_0 make the phase map a degree-one map with derivative less than $1/2$ from the identity. It is a diffeomorphism. The same inequality holds after adding the KAM correction and its Cauchy derivative bound on a smaller strip.

Physical displacement. The exact exported shape matrix maps X to the Hermite nodal displacements and slopes. On an element of length ℓ, the cubic Bernstein control values are w_0, $w_0 + \ell\vartheta_0/3$, $w_1 - \ell\vartheta_1/3$, and w_1. For a control row b, its value on the invariant torus is bounded by $\sum_{j=1}^2 |b_j| (\sum_k |(X_j)_k| + 3 \times 10^{-41})$. The convex-hull property therefore bounds every centerline point, every phase, and all time. A ball evaluation at phase $(0, 0)$ gives the separate positive lower witness in the table. These bounds use the same exported shape map as the learning and beam reconstruction.

The additional low-noise fit connects the two certificates on one actual learned Hamiltonian. Its nonzero derivative MSE is $3.3866649 \times 10^{-17}$. The outward Gram lower bound 6.0167×10^{-8}, together with the label and decoder error budgets, gives a twist-error upper bound 2.81119×10^{-4}, smaller than the target singular-value lower bound 3.03081×10^{-4}. The fifth row of the table uses these same fitted coefficients. Thus the nonzero-loss twist certificate and the finite-amplitude KAM certificate concern the same fitted function; the latter additionally uses the full analytic invariance residual, as required by the theorem.

The sixth row certifies the actual GC function obtained from four independent COMSOL position–velocity trajectories by integral matching. The fit uses state increments and feature integrals, without reconstructed acceleration labels. Its exported cosine multiplier is $a = 1.022324882191665$; every exported quartic term is included in the analytic verification, using the fitted coefficients without re-estimation. The frequency in (S70) admits the same conservative $\gamma = 9/1750$, so the global bounds and theorem constants above remain unchanged. In this case the checked inverse averaged torsion is below 94762, within the required 10^5 margin.

This result certifies one specified nonresonant torus in each Hamiltonian at a useful physical amplitude. It does not attach a measured phase-space volume to these individual tori. Positive-measure families near the elliptic equilibrium follow from the separate normal-form argument above; the finite-amplitude theorem and the energy-based safety theorem have distinct hypotheses and conclusions.

S4.4 *Positive-measure families around the certified tori*

The nondegeneracy retained by the a posteriori iteration gives a further conclusion at the same finite-amplitude motion.

Corollary S8. KAM stability of the certified finite-amplitude tori. *Each of the six tori in Theorem S7, and the additional*

Table S2. Finite-amplitude certificates for the complete target and actual trained Hamiltonians. Displayed bounds are rounded outward. The lower witness is tip displacement at one phase; the upper bound covers the entire reconstructed centerline at all phases.

Model	$\|E\|_\rho$ upper	KAM quotient upper	Tip lower (mm)	Centerline upper (mm)
Target	$2.7310\,10^{-77}$	0.031724	1.8977905	1.8977964
GC17	$2.7372\,10^{-77}$	0.031796	1.8927460	1.8927517
GC43	$2.7538\,10^{-77}$	0.031988	1.8837514	1.8837573
GC91	$2.7285\,10^{-77}$	0.031694	1.9077295	1.9077354
GC17 (10^{-8} noise)	$2.7311\,10^{-77}$	0.031724	1.8977399	1.8977457
GC (COMSOL states)	$2.7330\,10^{-77}$	0.031746	1.8977374	1.8977433

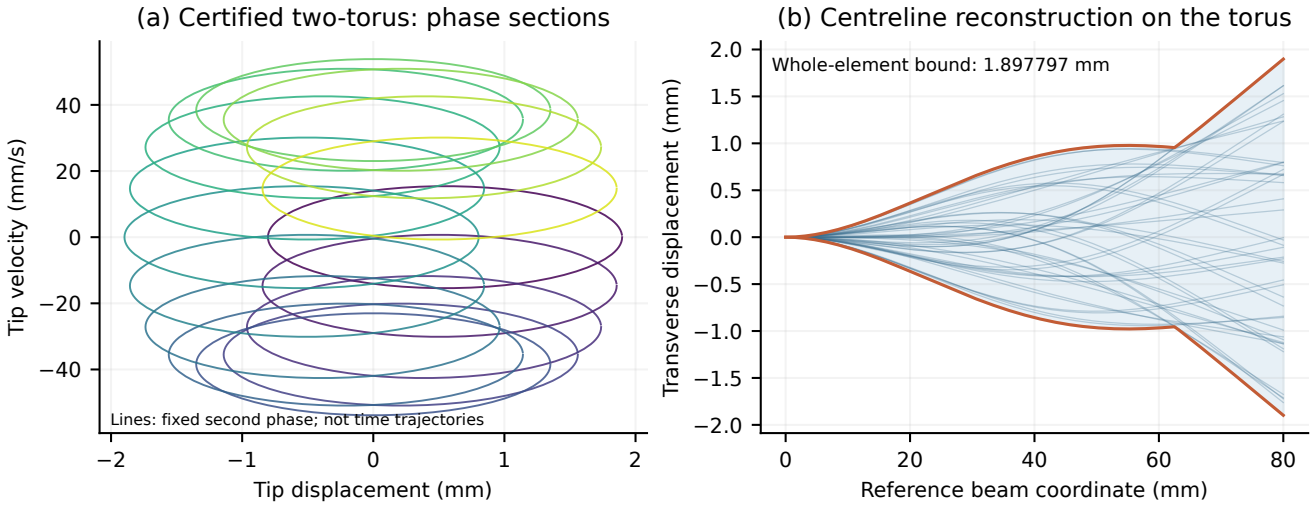


Figure S1. The certified target torus in the same SI beam coordinates. (a) Tip displacement and velocity along sampled sections with the second phase fixed; these curves are not time trajectories. (b) Sampled centerlines and a plotted phase envelope. The separate whole-element Bernstein/Fourier certificate bounds every centerline point on the corrected invariant torus by 1.897797 mm. Sampling illustrates the geometry; the analytic certificate supplies the all-phase bound.

integral-COMSOL torus in Section S4.5, is accumulated by positive-measure families of analytic Lagrangian invariant tori. In analytic symplectic tubular coordinates (θ, I) centered on that torus, their relative Lebesgue measure in $\mathbb{T}^2 \times \{|I|_\infty < r\}$ tends to one as $r \to 0$. This is an asymptotic statement about neighborhoods of the certified millimetre-amplitude torus; it does not specify a numerical tube thickness or a percentage in a preassigned experimental domain. In particular, for any prescribed fraction $p < 1$, a sufficiently thin tube has relative invariant-torus measure greater than p. The present statement is existential in the tube radius.

Proof. The inequalities controlling the inverse averaged torsion in the Haro–Luque iteration are uniform at every step: see their Eqs. (5.65)–(5.67) and the summation in Section 5.3. The strict margin used in (S74) therefore gives

$$\|\langle T_\infty \rangle^{-1}\| \le 2 \times 10^5.$$

This is nondegeneracy of the final invariant object, not only of its numerical approximation.

An analytic Lagrangian neighborhood chart Ψ can be chosen with $\Psi(\theta, 0) = K_\infty(\theta)$ and first derivative $D\Psi(\theta, 0) = [L_\infty, N_\infty]$. One construction starts with $K_\infty(\theta) + N_\infty(\theta)I$ and uses the relative analytic Moser correction. The forms agree on the zero section; the relative primitive and its correcting vector field are $O(|I|^2)$, so the correction preserves this first jet. In these coordinates invariance implies

$$h \circ \Psi = e + \varpi \cdot I + \tfrac{1}{2} I^\top A(\theta) I + O(|I|^3). \tag{S77}$$

The automatic-reducibility identity of Haro–Luque Section 2.4 is

$$DX_h(K_\infty)P - D_\varpi P = P \begin{pmatrix} 0 & T_\infty \\ 0 & 0 \end{pmatrix},$$

where $P = [L_\infty, N_\infty]$. Comparing this with the linearized vector field of (S77) gives $A = T_\infty$. More generally, changing the symplectic normal frame to $N_\infty + L_\infty S(\theta)$, with S symmetric, changes this block to $T_\infty - D_\varpi S$, whose average is the same. Hence $\overline{A} = \langle A \rangle$ is nonsingular independently of this choice of normal frame.

The quadratic normal form is justified without a formal-series convergence assumption. Solve the matrix

homological equation $D_\varpi S = A - \overline{A}$ with zero average by

$$S_k = \frac{A_k}{2\pi i k \cdot \varpi} \quad (k \neq 0), \qquad S_0 = 0.$$

Equation (S71) and analyticity make this series convergent on every strictly smaller angle strip; the cohomological bound established above controls it there. The exact time-one Hamiltonian flow of $\chi_2(\theta, I) = \frac{1}{2} I^\top S(\theta) I$ exists on a sufficiently thin tube and fixes $I = 0$. With the pullback convention $L_\chi F = \{F, \chi\}$, it changes the quadratic coefficient to $\overline{A}$ and leaves an analytic $O(|I|^3)$ remainder. The resulting normal form satisfies the hypotheses of Bounemoura (2016, Theorem 6, analytic case): the frequency is nonresonant, the averaged quadratic form is nonsingular, and the full remainder is analytic. That theorem gives the claimed density-one families. Symplectic coordinate changes preserve volume, and the constant factor relating the dimensionless chart to SI canonical coordinates cancels in relative measure.

This corollary establishes the local abundance of invariant tori around the certified motion. Section S4.6 defines their relative measure on a specified phase-space domain. For the integral-COMSOL fit, Appendix S11 supplements the local result with an explicit symplectic chart, a complete analytic remainder and a conservative quantitative bound on a fixed tube. The tube width and the resulting measure bound are reported together to distinguish this rigorous local estimate from the wider numerical sampling domain.

S4.5 A validated COMSOL state-to-KAM learning certificate

We add a controlled identification experiment for the same complete two-mode magnetic Hamiltonian. COMSOL independently integrates its four state equations for 64 initial conditions, with 17 state observations on each dimensionless interval $[0, T]$, $T \simeq 0.04$. This experiment is explicitly a two-mode modal ODE experiment; the earlier 64-DOF finite-element trajectories remain a separate validation. Only positions and velocities enter the new training labels. Neither acceleration reconstruction nor a solver tolerance is used as an error certificate.

For each observed orbit, fit a degree-eight polynomial curve $c_\ell : [0, T] \to \mathbb{R}^4$ to the observed states. Its coefficients are frozen as exact dyadic rationals, with its constant and linear coefficients corrected exactly so that both endpoints equal the exported observations z_ℓ^-, z_ℓ^+. This construction uses no evaluation of the physical acceleration. Let $Lz = X_{h_2}(z)$ and let $V(z) = [J\nabla F_0(z), \dots, J\nabla F_{34}(z)]$ be the nonlinear feature field of the same learned family. Define

$$\begin{aligned} D_\ell &= \frac{1}{T} \int_0^T V(c_\ell(\tau))\, d\tau, \\ y_\ell &= \frac{z_\ell^+ - z_\ell^-}{T} - \frac{1}{T} \int_0^T L c_\ell(\tau)\, d\tau. \end{aligned} \tag{S78}$$

The integral design and labels are therefore functions of the observed state curves, not instantaneous force-balance labels.

Proposition S9. Integral loss, dynamic defect and coefficient error. *Let $\eta_\star = (s_4, 0, \dots, 0)$ represent the declared full-cosine target, and let m be the number of observed curves. Put*

$$\begin{aligned} e_\ell &= \frac{1}{T} \int_0^T \big(\dot{c}_\ell - X_{h_\star}(c_\ell)\big)\, d\tau, \\ G_I &= \frac{1}{4m} \sum_{\ell=1}^{m} D_\ell^\top D_\ell, \\ r_I^2 &= \frac{1}{4m} \sum_{\ell=1}^{m} \|D_\ell \widehat{\eta} - y_\ell\|_2^2. \end{aligned} \tag{S79}$$

If $\lambda_{\min}(G_I) \geq \underline{\lambda} > 0$ and $(4m)^{-1} \sum_\ell \|e_\ell\|_2^2 \leq \epsilon_I^2$, then

$$\|\widehat{\eta} - \eta_\star\|_2 \leq b_I := \frac{r_I + \epsilon_I}{\sqrt{\underline{\lambda}}}. \tag{S80}$$

The curves need not be exact solutions of the physical differential equation. Their explicitly verified dynamic defect supplies the required error budget.

Proof. Exact endpoint matching and the fundamental theorem of calculus give $y_\ell - D_\ell \eta_\star = e_\ell$. The triangle inequality in the stacked, $1/(4m)$-normalized Euclidean norm gives $\|D(\widehat{\eta} - \eta_\star)\|_2 / \sqrt{4m} \leq r_I + \epsilon_I$. The Gram lower bound proves (S80).

The defect is independently evaluated against the explicitly declared physical target. Thus the proposition does not infer arbitrary unknown model error from a small training loss. It certifies consistency with this target using additional, verified physical information.

Validated integrals and numerical error. All polynomial feature integrals are evaluated by exact polynomial integration with Arb ball coefficients. For $\vartheta = d^\top (c_\ell)_X$, the magnetic feature uses the odd Taylor terms of $\sin\vartheta - \vartheta$ through degree 19, together with

$$\begin{aligned} S_{19}(\vartheta) &= \sum_{\substack{3 \leq j \leq 19 \\ j \text{ odd}}} \frac{(-1)^{(j-1)/2} \vartheta^j}{j!}, \\ |\sin\vartheta - \vartheta - S_{19}(\vartheta)| &\leq \frac{R^{21} \cosh R}{21!}, \quad |\vartheta| \leq R. \end{aligned} \tag{S81}$$

Table S3. Outward bounds for the independent COMSOL integral-learning experiment. The MSE has a strictly positive certified lower bound.

Quantity	Certified bound
Training MSE	$[3.6030, 3.6031]\,10^{-23}$
r_I	$\le 6.0026\,10^{-12}$
ϵ_I	$\le 6.8835\,10^{-12}$
$\lambda_{\min}(G_I)$	$\ge 2.9302\,10^{-7}$
$\|\widehat{\eta} - \eta_\star\|_2$	$\le 2.3805\,10^{-8}$
Twist error	$\le 1.0377\,10^{-7}$
$\sigma_{\min}(A_\star)$	$\ge 4.9032\,10^{-4}$
$\|h_{\widehat{\eta}} - h_\star\|_{0.3}$	$\le 1.3310\,10^{-9}$

The computed maximum remainder is below 4.8752×10^{-43}. The remainder is included in every affected feature integral and dynamic defect; the certified function remains the complete cosine model. The exact identity $y - D\eta_\star = e$ is also checked componentwise by independent ball evaluations.

The interval Gram inverse is computed at 256-bit precision. Since the exact Gram matrix is real symmetric and positive semidefinite, a validated inverse makes it positive definite and gives $\underline{\lambda} = 1/\|G_I^{-1}\|_\infty$. Decoder rounding is retained when the real readout coefficients are exported as a cosine multiplier and a quartic polynomial. With $B = [q_\star/s_4\ U]$, this gives

$$\begin{aligned} \|A_{\widehat{\eta}} - A_\star\|_2 &\le \sqrt{19}\big(\|B\|_2 b_I + \delta_{\mathrm{dec}}\big), \\ \|h_{\widehat{\eta}} - h_\star\|_r &\le b_I B_F(r) + \delta_{h,\mathrm{dec}}(r). \end{aligned} \tag{S82}$$

Here $\|B\|_2 \le \|B^\top B\|_\infty^{1/2}$ is verified from the exported feature matrix; exact numerical orthogonality is unnecessary. The analytic bound controls every derivative on a smaller polydisc by Cauchy's estimate:

$$\begin{aligned} &\|\partial^\alpha (h_{\widehat{\eta}} - h_\star)\|_{r'} \\ &\quad \le \alpha!\,(r - r')^{-|\alpha|}\,\|h_{\widehat{\eta}} - h_\star\|_r. \end{aligned}$$

The target is unchanged. The singular-value lower bound in Table S3 is a sharper bound for the same two-by-two symmetric matrix: $\sigma_{\min}(A_\star) \ge |\det A_\star|/\|A_\star\|_\infty$. The previously reported, smaller lower bound remains valid. The strict twist margin, fixed quadratic part and complete analytic tail therefore give local positive-measure KAM families for this actual fitted Hamiltonian by the normal-form argument above.

An independent bound on trajectory error. The observation curves deviate from every exported sample by at most 5.2538×10^{-14} and satisfy

$$\sup_\tau \|\dot{c}_\ell - X_{h_\star}(c_\ell)\|_2 \le 1.0973 \times 10^{-8}.$$

The linear modal field L is skew symmetric in these balanced coordinates. On the real box $|z|_\infty < 0.3$, the nonlinear part has Euclidean Lipschitz constant

$$\|d\|_2^2 \frac{(0.3\|d\|_1)^2}{2} < 0.002084 < 0.01.$$

For the exact physical solution z_ℓ with $z_\ell(0) = c_\ell(0)$, variation of constants and Gronwall's inequality therefore give

$$\begin{aligned} &\sup_{0 \le \tau \le T} \|z_\ell(\tau) - c_\ell(\tau)\|_2 \\ &\quad \le T e^{0.01T} \|\dot{c}_\ell - X_{h_\star}(c_\ell)\|_{\infty,2} \\ &\quad < 4.3907 \times 10^{-10}. \end{aligned}$$

The verified curve bound $|c_\ell|_\infty < 0.2000$ closes the bootstrap inside the radius-0.3 box. This is an a posteriori bound on the independently observed trajectories, not an interpretation of COMSOL's requested time-stepping tolerances.

The same fitted function at millimetre amplitude. The frozen network has cosine multiplier $a = 0.9999996888354415$. Its complete exported coefficients are also used in a separate, 512-bit a posteriori torus verification with the frequency in Eqs. (S68)–(S69) and the theorem constants of Theorem S7. The verified analytic invariance residual is at most 2.7310×10^{-77}, the KAM smallness quotient is at most 0.031724, and the inverse averaged torsion is below 80526. The invariant torus lies within 2.0894×10^{-41} of its exact Fourier candidate. The tip-displacement witness is at least 1.8977904 mm, and the whole reconstructed centerline is bounded by 1.8977962 mm for every phase and all time. Thus the nonzero integral-loss certificate and the finite-amplitude KAM certificate concern exactly the same independently trained function. The first proves coefficient, analytic and twist control; the second checks the additional invariance-residual condition needed at the stated finite amplitude.

The full-FE trajectory fit has an independent a posteriori torus certificate, but its integral observation matrix yields a coefficient bound above 0.025 before accounting for continuous-quadrature and model errors. This bound does not satisfy the coefficient condition used here. The independently observed two-mode trajectories provide the defect bounds needed for the complete integral-learning theorem; model-reduction errors remain part of the separate full-FE evaluation.

S4.6 Phase-space measure of the invariant family

The positive-measure family established above motivates a domain-specific measure of quasiperiodic initial states. A quantitative fraction requires a specified volume measure and initial-state domain. Following the measure formulation of Biasco and Chierchia (2018); Chierchia and Koudjinan (2021), let $U \subset \mathbb{R}^4$ be a bounded domain of positive Liouville volume for the identified two-degree-of-freedom Hamiltonian h, and let $\mathcal{K}_h$ denote the union of its analytic Lagrangian invariant tori carrying Diophantine quasiperiodic motion. Define

$$f_{\mathrm{KAM}}(U) = \frac{\mu_L(U \cap \mathcal{K}_h)}{\mu_L(U)}, \qquad \mu_L = \mathrm{d}X_1\,\mathrm{d}X_2\,\mathrm{d}Y_1\,\mathrm{d}Y_2. \tag{S83}$$

An explicitly certified subfamily provides a lower bound for this fraction. The complement of that subfamily can contain other regular motion; it is not thereby a chaotic set. The sampling confidence in a learning bound such as that of Chen et al. (2022) is a probability over training data, whereas (S83) measures initial states of one fixed fitted Hamiltonian.

For a symplectic chart Ξ and an action box D, set $U = \Xi(\mathbb{T}^2_{2\pi} \times D)$, where $\mathbb{T}^2_{2\pi} = (\mathbb{R}/2\pi\mathbb{Z})^2$. Then $\mu_L(U) = (2\pi)^2\,\mathrm{area}(D)$. The period-one angles used by the earlier torus certificate are changed to radians together with the inverse action rescaling, preserving the symplectic form. The constant volume factor in the SI canonical chart also cancels from the relative fraction. The radius and location of D must accompany every reported percentage.

Where the assumptions of Theorem S10 hold, the invariant-family lift to the 64-mode closed loop carries this low-dimensional parameter measure forward. It does not turn it into a positive fraction of the full 128-dimensional Lebesgue volume. Accordingly, the measure estimate in Appendix S11 and the numerical sampling experiment in Section S5.6 refer to the conservative low-mode Hamiltonian realized by the magnetic feedback, with their respective domains specified separately.

The explicit constants of Chierchia and Koudjinan (2021) yield a computable local measure bound after construction of a symplectic tube and estimation of the full analytic remainder. Appendix S11 gives that construction for the integral-COMSOL Hamiltonian. Its sufficient estimates certify an extremely thin transverse domain; the central motion remains at millimetre amplitude, but the resulting width does not quantify an experimentally resolvable initial-state tolerance. To examine a broader range of amplitudes, Section S5.6 samples a prescribed action domain and evaluates finite-time quasiperiodicity with independent integration and averaging checks.

S5 Reproducible learning and independent COMSOL validation

All experiments in this section use the clamped-free beam, static-field eigenvectors, complete magnetic cosine, and SI parameters defined above. The two-mode learning problem and its higher-order validation models therefore share the same mechanics. The reference energy is

$$h_\star(X,Y) = \frac{1}{2}\sum_{j=1}^{2}\nu_j(X_j^2+Y_j^2) + g(d^TX), \qquad g(s) = 1-\cos s - \frac{1}{2}s^2. \tag{S84}$$

Here $\nu = (1, 6.152117504)$ and $d = (0.225670773, 0.321798806)^T$; one dimensionless energy unit is 6.0×10^{-7} J and $\tau = 11.417287283t$. The tip displacement in metres is $0.013506054X_1 + 0.005470926X_2$. These mappings are used for every reported displacement, slope, and energy error, including the independent COMSOL comparison.

S5.1 Finite-feature identification and matched controls

The CHNN uses $z_j = (X_j + \mathrm{i}Y_j)/\sqrt{2}$, conjugate polynomial features, and a complex exponential feature for the full cosine. Its real energy readout is

$$h_\vartheta = h_2 + a\,g(d^TX) + m_4(X,Y)^TUc, \tag{S85}$$

where m_4 lists the 35 degree-four real monomials in (X_1, X_2, Y_1, Y_2). If v_4 is the coefficient vector of $-(d^TX)^4/24$, the 34 columns of U span its orthogonal complement. The feature $g/\|v_4\|_2$ and these complementary quartics give 35 fitted real parameters, avoid duplicating the leading cosine term, and retain the complete analytic cosine tail. The target corresponds to $a = 1, c = 0$ and is therefore in the feature class. The stored matrix U, rather than idealized exact orthogonality, is used in the certificate computations.

We compare real (R) and complex (C) implementations, each with either the fixed mechanical quadratic part (G) or a fitted general quadratic part (B). Thus GR and GC have 35 fitted real parameters, whereas BR and BC have 45. The two representations span the same real energy class, receive identical data, and use the same regularizer expressed in common real feature coordinates.

Both implementations evaluate their respective real or complex features. Because the representations are related by an invertible transformation, this comparison tests their numerical implementation within a common hypothesis class.

For each seed in $\{17, 43, 91\}$, we generate 12 conservative trajectories of 4 s from the two-mode Python Galerkin model, sampled at 129 times. Initial modal radii are drawn uniformly from $[0.03, 0.20]$ and $[0.03, 0.16]$, with independent uniform phases. The largest training tip slope is 0.0891 rad. Each component of the dimensionless vector-field label receives independent uniform noise in $[-10^{-6}, 10^{-6}]$; states are noiseless. Linear readout fitting minimizes derivative mean-square error with a 10^{-14} quadratic regularizer. Eight independent trajectories, generated with a separate seed and integrated for 16 s at 1601 output times, are held out in full. The reference solver is DOP853 with relative and absolute tolerances 2×10^{-10} and 2×10^{-12}. Tightening both tolerances by a factor of 100 changes the tip of the first held-out trajectory by at most 1.92×10^{-8} mm.

Table S4 and Fig. S2 show the benefit of fixing the known quadratic mechanics in this experiment, while matched real and complex fits agree to numerical precision. GC identifies $a = 1.00326 \pm 0.00154$ with nonzero training residuals. The energy error evaluates the true energy along the predicted and reference states; it is not an energy-conservation surrogate for trajectory accuracy. The small fitting errors reflect exact representation, accurate synthetic derivatives, and deliberately controlled label noise. They do not establish corresponding physical measurement accuracy.

We also continue one independent initial condition for 200 s, or 363 first-mode linear periods, using the three primary GC fits. The seed-level tip RMSE ranges from 1.38×10^{-4} to 4.03×10^{-4} mm; the largest modal phase error is 4.19×10^{-4} rad. The relative drift of each learned model's own energy remains below 5.11×10^{-9} with conservative integration at relative and absolute tolerances 2×10^{-11} and 2×10^{-13}. Figure S3 shows its Poincaré section, phase error, and numerical energy drift. These finite-time diagnostics assess the fitted dynamics; they do not estimate the measure of KAM tori or replace the analytic certificate.

S5.2 A nonzero-loss certificate for the fitted magnetic energy

The feature derivatives make the empirical Gram matrix and the complete quartic Taylor map computable. We repeat the same three fits at noise amplitudes 10^{-8} and 10^{-9}, in addition to the 10^{-6} experiments. Outward-rounded interval calculations enclose the derivatives of the analytic sine feature at the stored states, the Gram inverse residual, the observed loss, clean-label rounding, and the conversion from readout weights to the exported energy coefficients. The degree-17 sine series is enclosed with its degree-19 remainder.

For the target magnetic quartic, $A_{jj} = -d_j^4/8$ and $A_{12} = -d_1^2 d_2^2/4$. This twist is indefinite and nonsingular. Exact rational arithmetic on the stored dyadic d gives the conservative lower bound $\sigma_{\min}(A_\star) > 3.03081 \times 10^{-4}$. If G is the derivative Gram matrix, B the quartic coefficient map, ℓ the noisy derivative MSE, and η_{eff} the enclosed noise and arithmetic budget, the verified sufficient inequality is

$$\sqrt{19}\left(\|B\|_2 \frac{\sqrt{\ell} + \eta_{\text{eff}}}{\sqrt{\lambda_{\min}(G)}} + \delta_{\text{dec}}\right) < \sigma_{\min}(A_\star), \qquad \text{(S86)}$$

where δ_{dec} encloses readout decoding roundoff. At $\eta = 10^{-8}$, Table S5 verifies this inequality with strictly positive losses for all seeds. Thus the data-dependent error bound itself transfers nondegeneracy to the actual learned Hamiltonians. The local KAM conclusions additionally use the nonresonance and analytic normal-form hypotheses proved for this same model. This table alone does not certify a finite experimental oscillation radius. For the primary 10^{-6} fits, this sufficient error bound is too loose; their directly computed twist singular values are nevertheless between 5.78×10^{-4} and 5.93×10^{-4}.

S5.3 Learning dissipation and opening the magnetic input port

The imaginary readout is the nonnegative Rayleigh potential $\mathcal{D}_\gamma(v) = \gamma \sum_j v_j^2/(2\nu_j)$, with $v = h_{\vartheta,Y}$. For each seed, eight additional damped, unforced 5 s trajectories provide noisy derivative labels. Nonnegative least squares fits the single γ using the already fitted conservative gradient. In particular, the decoder retains the learned momentum-dependent quartics:

$$\begin{aligned} X' &= h_{\vartheta,Y}, \\ Y' &= -h_{\vartheta,X} - \gamma\, \mathrm{diag}(\nu^{-1}) h_{\vartheta,Y} \\ &\quad + u\, d \cos(d^T X). \end{aligned} \qquad \text{(S87)}$$

The fitted dimensional mass-drag coefficients are 0.555585710, 0.555587102, and 0.555584024 s^{-1}, compared with 0.555585237 s^{-1} for the generator. On six independent driven trajectories with amplitudes 4–8 mT and frequency ratios 0.65–1.20 relative to the biased first

Table S4. Conservative held-out errors, mean ± sample standard deviation over three training seeds. Each seed is evaluated on all eight independent 16 s trajectories. These are errors within the declared simulated model.

Model	Tip RMSE (mm)	Slope RMSE (mrad)	Energy RMSE (nJ)
BR	$(2.28\pm0.75)10^{-5}$	$(5.80\pm1.20)10^{-4}$	$(3.21\pm0.51)10^{-6}$
BC	$(2.28\pm0.75)10^{-5}$	$(5.80\pm1.20)10^{-4}$	$(3.21\pm0.51)10^{-6}$
GR	$(1.20\pm0.71)10^{-5}$	$(3.90\pm1.63)10^{-4}$	$(3.14\pm0.71)10^{-6}$
GC	$(1.20\pm0.71)10^{-5}$	$(3.90\pm1.63)10^{-4}$	$(3.14\pm0.71)10^{-6}$

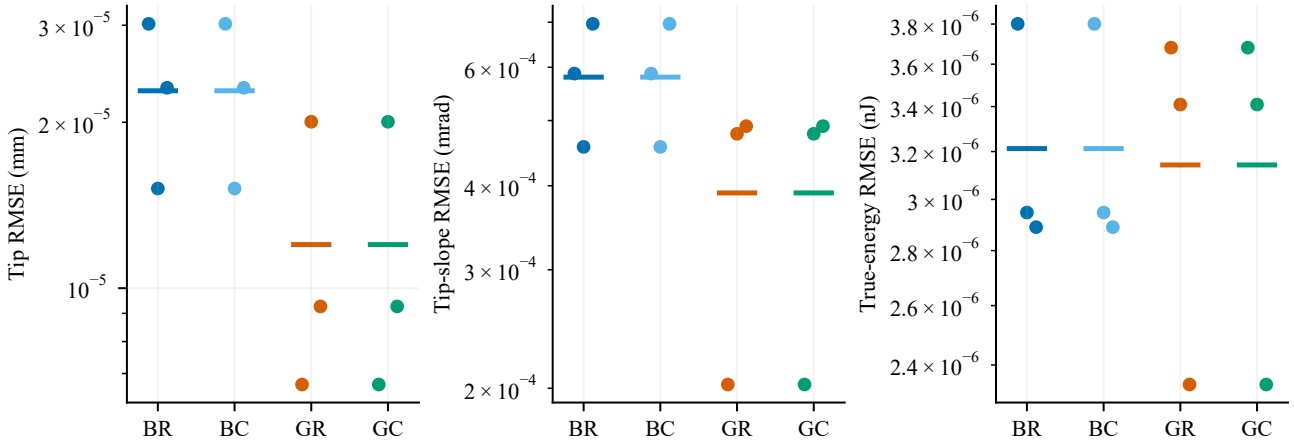


Figure S2. Identical full-cosine beam data and real parameter budgets for the four learning controls. Points are individual seeds and horizontal marks are means. The quadratic mechanics prior changes the result; the matched real/complex representation changes it only at numerical precision.

frequency, the three seed-level tip RMSEs are 8.83×10^{-5}, 1.37×10^{-4}, and 6.40×10^{-5} mm. Closing the learned drag port increases the tip RMSE to approximately 3.01 mm. The integrated work balance is

$$h_\vartheta(\tau)-h_\vartheta(0)=\int_0^\tau u\cos(d^TX)d^Th_{\vartheta,Y}\,ds - \int_0^\tau \gamma\sum_j h_{\vartheta,Y_j}^2/\nu_j\,ds. \tag{S88}$$

Its largest numerical residual across these cases is 4.41×10^{-17} J. These driven tests assess the port model, rather than autonomous KAM persistence. Some stress cases exceed the channel clearance and are free-motion continuations; safe operating trajectories are evaluated separately with the physical energy bound and input governor.

S5.4 Independent COMSOL assembly and nonlinear time integration

COMSOL Multiphysics 6.3 independently assembles the same 32-element clamped-free beam and solves its nonlinear magnetic-tip response. The relative Frobenius differences between the exported and Python consistent-mass matrices are 7.75×10^{-15} for mass and 3.05×10^{-14} for bending stiffness. The nonlinear held-out COMSOL trajectory starts from rest, uses the same mass drag, and applies

$$B_y(t)=0.006\,[1-\exp(-(t/0.25)^2)]\sin(\Omega t)\quad \mathrm{T} \tag{S89}$$

with $\Omega=9.4449490256$ rad/s for 8 s, relative tolerance 10^{-7}, and 2001 output samples. The exported nodal displacement, rotation, velocity, and angular velocity are evaluated directly, with no fitting to this driven trajectory. Its peak tip displacement is 3.09032 mm. A separate COMSOL solve with relative tolerance 10^{-8} changes the tip by 4.59×10^{-5} mm RMSE and reduces its discrepancy against the full Python FE solution to 1.03×10^{-5} mm RMSE. The original held-out dataset is used for the primary comparison below.

The nested modal comparison in Table S6 separates reduction error from learning error. Against the exact two-mode response, GC contributes only 1.81×10^{-6}–3.48×10^{-6} mm tip RMSE for this input. Against COMSOL, both have approximately 0.0156 mm RMSE. The two-mode maximum slope discrepancy is 3.69 mrad; its true stored-energy RMSE is 0.132 nJ. These discrepancies decrease with modal enrichment. Figure S5 shows the response and both error scales. The primary training data were generated by the two-mode Python Galerkin model; the COMSOL time history is an independent held-out validation dataset. Neither dataset is a physical hardware measurement.

S5.5 COMSOL-derived training data and an independent integral check

We generate a separate conservative COMSOL dataset comprising four 4 s training trajectories and two independent

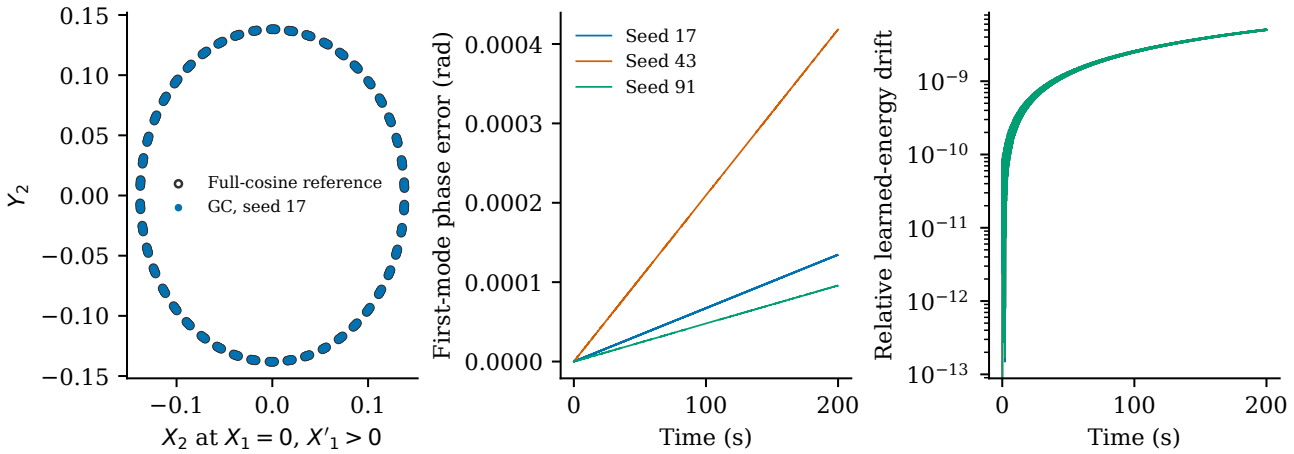


Figure S3. Conservative 200 s continuation of an independent initial state. Left: 364 upward crossings of $X_1 = 0$ for the reference and GC17. Middle: unwrapped first-mode phase errors of the three learned energies. Right: each learned energy's relative integration drift. All ports are closed for these conservative diagnostics.

Table S5. Enclosed loss-to-twist checks at noise amplitude 10^{-8}. All twist-error upper bounds are below 3.03081×10^{-4}.

Seed	Noisy MSE	Gram eigenvalue lower bound	Twist-error upper bound
17	3.387×10^{-17}	6.0167×10^{-8}	2.8112×10^{-4}
43	3.275×10^{-17}	2.2734×10^{-7}	1.4374×10^{-4}
91	3.283×10^{-17}	2.0968×10^{-7}	1.4973×10^{-4}

8 s test trajectories, sampled every 0.004 s. Initial displacements and velocities excite both retained modes. The remaining 62 modes receive the static response to the nonlinear magnetic residual and its compatible first time derivative; all 64 modes remain active during integration. This initial tail condensation is not assumed to be an exact nonlinear invariant manifold. Native solver-vector checks confirm that the exported initial displacement and velocity are correct; small changes introduced by consistent initialization are handled by starting each predictor from the actual exported state.

The derivative-label construction requires care. At the stored output times, the exported second time derivatives differentiate a time interpolant and show a substantial lag relative to the velocity evolution. For example, the first trajectory has a dimensionless modal equation-residual RMS of 4.58×10^{-2} when those derivatives are used directly. We therefore reconstruct instantaneous acceleration from the exported COMSOL mass and stiffness matrices:

$$\ddot{q} = M_{\mathrm{C}}^{-1} \left[-K_{\mathrm{C}} q - \mu B_0 \sin(\theta_{\mathrm{tip}}) e_\theta \right]. \tag{S90}$$

The training states and velocities therefore come from the COMSOL time solver, while the acceleration labels use known physical force balance. This is surrogate identification for the declared mechanical model with force-balance labels. The unresolved-mode force is included: its largest component is 3.36×10^{-9} in dimensionless units for the checked training trajectory, and the remaining reconstructed equation-residual RMS is 1.81×10^{-10}.

These COMSOL-based models are fitted separately from the primary Python-trained GC models. Three seeds vary the additional componentwise label noise, bounded by 10^{-6}, while using the same four COMSOL trajectories. The real/complex and prior/no-prior controls use the same parameter budgets as before.

An independent check uses only the COMSOL displacement and velocity samples. A nine-point closed Newton–Cotes rule integrates the vector-field features on nonoverlapping 0.032 s windows, and the observed state increments provide the regression targets. No acceleration or force-balance label enters this integral fit. With the initial solver settings (relative tolerance 10^{-6} and scaled absolute tolerance 10^{-5}), the integral residual is too large to resolve the weak magnetic correction reliably: on the first trajectory, its RMS is 2.16 times the nonlinear magnetic signal. An unconstrained 35-feature integral fit then yields $a = 1.437$, illustrating that a good fit to a numerically dispersed trajectory need not recover the physical magnetic coefficient. A stricter COMSOL solve, with relative tolerance 10^{-8} and scaled absolute tolerance 10^{-7}, reduces that ratio to 0.112 on the same trajectory. This independent data-quality check resolves time-integration error that the force-balance loss alone does not detect.

All six trajectories were then integrated with these stricter tolerances. The refined runs use generalized-alpha

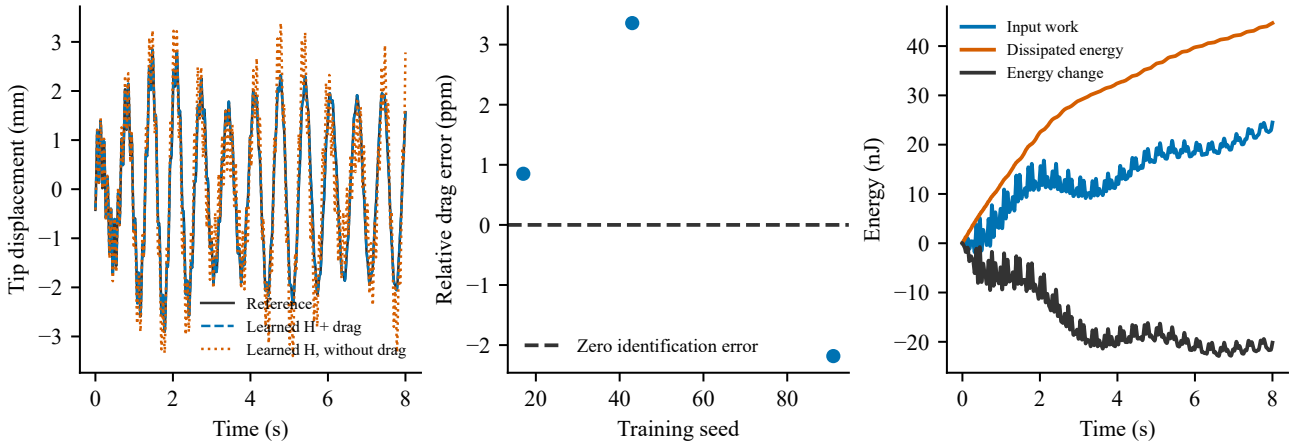


Figure S4. The same learned conservative energy with an identified nonnegative drag readout and magnetic input port. Left: an independent driven trajectory and the drag ablation. Middle: signed relative errors in the drag coefficient. Right: supplied work, dissipated energy, and stored-energy change for that trajectory.

Table S6. Independent nonlinear COMSOL trajectory comparison. Modal models use the common static-field chart. GC includes learned drag; its range covers the three primary training seeds.

Predictor	Tip RMSE (mm)	Maximum tip error (mm)
2-mode exact cosine	0.0156168	0.0226548
4-mode exact cosine	0.00407422	0.00592596
8-mode exact cosine	0.00102484	0.00152782
64-mode Python FE	0.00005614	0.00011835
2-mode learned GC	0.0156171–0.0156177	0.0226545–0.0226549

integration with high-frequency spectral radius zero and backward-Euler consistent initialization. The four refined training trajectories provide 500 integral windows, or 2000 scalar equations, for the 35 real readout parameters. The resulting GC model has $a = 1.0223248822$ and a nonzero averaged-derivative integral RMS residual of 4.0524×10^{-7}. The corresponding residuals of the quadratic prior alone and the physical full-cosine target are 3.9099×10^{-6} and 6.6454×10^{-7}; the nonlinear magnetic signal RMS is 3.6601×10^{-6}. The fitted quartic twist has minimum singular value 5.1166×10^{-4}. The conservative weights were frozen before evaluating the two independent refined test trajectories.

Table S7 and Figure S6 report the held-out errors. The integral fit partly absorbs the remaining time-solver dispersion: its agreement with COMSOL is better than that of the exact closed two-mode target, despite a 2.23% error in the cosine multiplier. These errors quantify identification against simulated observations. The requested solver tolerances do not supply a rigorous defect budget for applying the loss-based certificate to these integral residuals. The separate two-mode short-trajectory experiment in Section S4.5 supplies a verified integral defect budget and the corresponding loss-based certificate for its own frozen fitted function.

For comparison, the separately fitted models using the known-force labels in (S90) recover $a = 1.0000690$ without added label noise. With the three added-noise seeds, the prior-informed GC models give 0.25347–$0.25940\,\mu$m pooled tip RMSE against the same refined COMSOL tests; their additional error relative to the closed two-mode trajectories is 0.00364–$0.00771\,\mu$m. Matched real and complex features again give coincident results to numerical precision.

Finally, the position–velocity fit is used in the independent a posteriori proof of Section S4.3. Its complete cosine and all fitted quartic coefficients yield an analytic residual below 2.7330×10^{-77} and a theorem smallness quantity below 0.031746. The resulting exact invariant torus has a tip-displacement witness above 1.8977374 mm and a whole-centreline upper bound below 1.8977433 mm. This certificate concerns the actual fitted Hamiltonian and does not require its coefficients to equal the physical target. The same frozen conservative readout is paired with a nonnegative drag coefficient refitted on the existing Python damped-beam trajectories, $\gamma = 0.0486617702$ for seed 17. Its independent driven COMSOL tip RMSE is 0.015622 mm. Section S7.4 then uses this same energy, full gradient and Hessian, and drag readout to generate the magnetic tracking input.

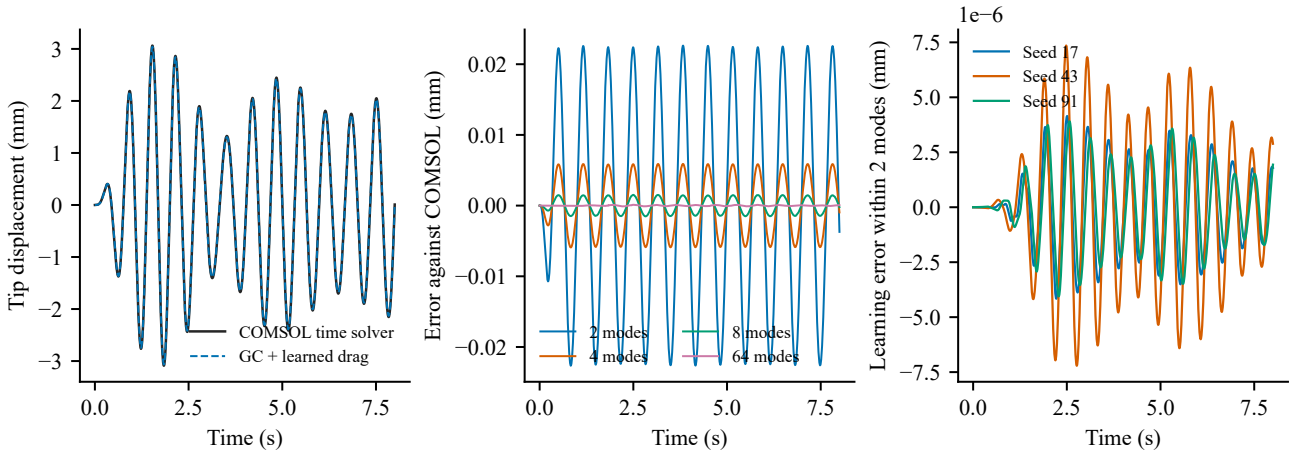


Figure S5. Independent nonlinear COMSOL time integration. Left: COMSOL and the learned two-mode GC response with learned drag. Middle: errors of the nested exact-cosine modal models against COMSOL. Right: the additional GC identification error within the two-mode model. Reduction error is the larger contribution in this validation.

Table S7. Independent refined COMSOL tests of the GC model identified only from displacement and velocity increments. Predictions start from each actual exported initial state. Energy errors use the physical stored-energy function evaluated along the learned trajectory.

Test	Tip RMSE (μm)	Max. tip error (μm)	Slope RMSE (mrad)	Energy RMSE (nJ)	Exact 2-mode tip RMSE (μm)
1	0.14836	0.39240	0.008677	0.0002067	0.24172
2	0.05140	0.17435	0.002411	0.0002253	0.27459

S5.6 Finite-time fraction in an operating action annulus

We examine the phase-space fraction defined in Section S4.6 on a domain spanning millimetre-scale oscillations. The Hamiltonian is identified from the two-mode COMSOL position–velocity data in Section S4.5. In its canonical coordinates, set

$$\begin{aligned} I_j &= \tfrac{1}{2}(X_j^2 + Y_j^2), \quad j = 1, 2, \\ \phi_j &= \operatorname{atan2}(-Y_j, X_j). \end{aligned} \tag{S91}$$

and prescribe, before classifying any orbit,

$$\mathcal{D}_{\mathrm{op}} = [0.0032, 0.0072]^2 \times \mathbb{T}^2_{2\pi}. \tag{S92}$$

Both harmonic modal radii therefore lie in $[0.08, 0.12]$, surrounding the radius 0.1 of the previously certified $1.898\,\mathrm{mm}$ torus. The corresponding harmonic tip envelope $|\ell_1|\sqrt{2I_1} + |\ell_2|\sqrt{2I_2}$ ranges from 1.518 to $2.277\,\mathrm{mm}$. The canonical polar Jacobian has absolute determinant one, so independent uniform actions and phases sample normalized four-dimensional Liouville volume:

$$\mu_L(\mathcal{D}_{\mathrm{op}}) = (2\pi)^2(0.004)^2 = 6.3165468 \times 10^{-4}. \tag{S93}$$

We draw $N = 1024$ independent points with PCG64 seed 20261002; the domain, initial points, observables, and decision thresholds are specified before classification. These points are independent of the training observations and the certified-torus phase grid.

Following weighted Birkhoff averaging (Das et al. 2017; Duignan and Meiss 2023), we compare consecutive windows using the smooth bump $b(s) = \exp[-1/(s(1-s))]$ for $0 < s < 1$, with $b = 0$ otherwise. The normalized weighted mean is

$$W_T(F; z) = \frac{\int_0^T b(t/T) F(\Phi_h^t z)\, dt}{T \int_0^1 b(s)\, ds}, \tag{S94}$$

where Φ_h^t denotes the flow of the fitted Hamiltonian h. The computations use normalized sampled sums of this weight; integration and quadrature refinements are reported below. Eight observables are used: $X_j/0.1$, $Y_j/0.1$, $I_j/0.005$, $X_1X_2/0.01$, and $Y_1Y_2/0.01$. We also average the polar angular velocities

$$\dot{\phi}_j = \frac{X_j \partial_{X_j} h + Y_j \partial_{Y_j} h}{X_j^2 + Y_j^2}. \tag{S95}$$

For each $T \in \{512, 1024, 2048, 4096\}$, let $e_F(T)$ be the largest absolute difference of the eight weighted means on $[0, T]$ and $[T, 2T]$, and let $e_\omega(T)$ be the largest relative difference of the two weighted frequencies. A candidate passes the principal test when $e_F, e_\omega \le 10^{-8}$ at both $T = 2048$ and 4096. It must also keep both sampled actions above 3.2×10^{-4}, have sampled relative energy drift below 10^{-8},

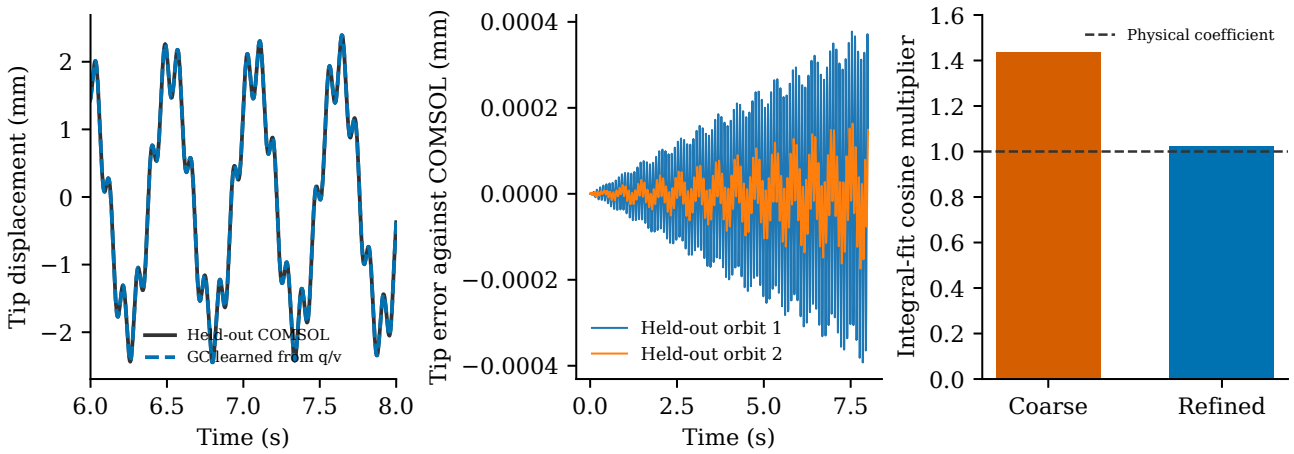


Figure S6. Identification using full-FE COMSOL displacement and velocity outputs only. Left: independent test response. Middle: prediction errors for both held-out trajectories. Right: the fitted cosine multiplier under the two declared COMSOL tolerance settings; the physical value is one. The refined coefficients are held fixed during the subsequent evaluations.

and satisfy the finite-order screen

$$\min_{\substack{0<|k|_1\le 64\\ k\in\mathbb{Z}^2 \text{ primitive}}} \frac{|k\cdot\overline{\omega}|}{|k|_1\|\overline{\omega}\|_\infty} > 10^{-7}, \tag{S96}$$

where $\overline{\omega}$ uses the two longest consecutive windows. The total simulated time is $8192/\omega_0 = 717.508\,\mathrm{s}$. Full cosine and momentum-dependent quartic terms are retained.

Step refinement resolves integration error in the diagnostics. Fixed-step eighth-order DOP853 at $\Delta\tau = 1/16$ produces relative energy drift up to 3.94×10^{-7} and fails the numerical validity test. At $\Delta\tau = 1/32$, all points pass the principal eight-digit criterion and the energy test, but the consecutive-window averages fail the ten-digit criterion. Refining all 1024 trajectories to $\Delta\tau = 1/64$, with the same initial points and thresholds, resolves the ten-digit criterion. The largest relative energy drift then falls to 6.11×10^{-12}; $e_F(4096) \le 4.49\times10^{-12}$ and $e_\omega(4096) \le 3.63\times10^{-14}$. Halving the step again for the preselected first 64 points changes no six-, eight-, or ten-digit labels, with maximum relative frequency change 2.40×10^{-14}. Halving the averaging output interval on identical saved trajectories changes the weighted means by at most 3.82×10^{-14}. An independent adaptive solver, phase points from the certified torus's Fourier approximation, an exactly resonant oscillator, and regular/chaotic two-wave examples provide additional checks. In particular, the resonant oscillator passes the averaging test but is rejected by (S96).

The refined ensemble gives $1024/1024$ candidates at each of the six-, eight-, and ten-digit thresholds. For the specified finite-time classifier C,

$$\widehat{p}_C = \frac{1}{1024}\sum_{i=1}^{1024} \mathbf{1}_C(z_i) = 1, \qquad \mathrm{CI}_{95\%}(p_C) = [0.996404, 1]. \tag{S97}$$

using the two-sided exact Clopper–Pearson interval. The smallest normalized finite-order detuning is 2.113×10^{-4}; the largest sampled tip displacement is $2.269\,\mathrm{mm}$. Figure S7 shows the fixed-domain sample and the integration and threshold checks. This interval quantifies only random initial-condition sampling for C. It does not bound classification error, unresolved high-order resonance, sticky chaos, or physical model error. Thus (S97) is an empirical finite-time candidate fraction in the operating domain. Its 99.6404% lower confidence endpoint concerns the classifier probability, whereas Appendix S11 establishes a rigorous KAM-volume bound on a separate, narrower analytic domain.

S5.7 State-only identification beyond the nominal hypothesis class

The nominal full-cosine experiment places its target inside the chosen feature class. We additionally test unknown material and magnetic parameters, state noise, and two omitted physical effects. All training observations in this experiment are generated by the full 32-element, 64-DOF Python FE plant; the predictors retain the same four nominal static-field modes. Let $e = E/E_{\mathrm{ref}}$ and $m = \mu/\mu_{\mathrm{ref}}$. The local applied field is

$$B_x = B_0 - G(x-L), \qquad B_y = B_y(t) + Gy. \tag{S98}$$

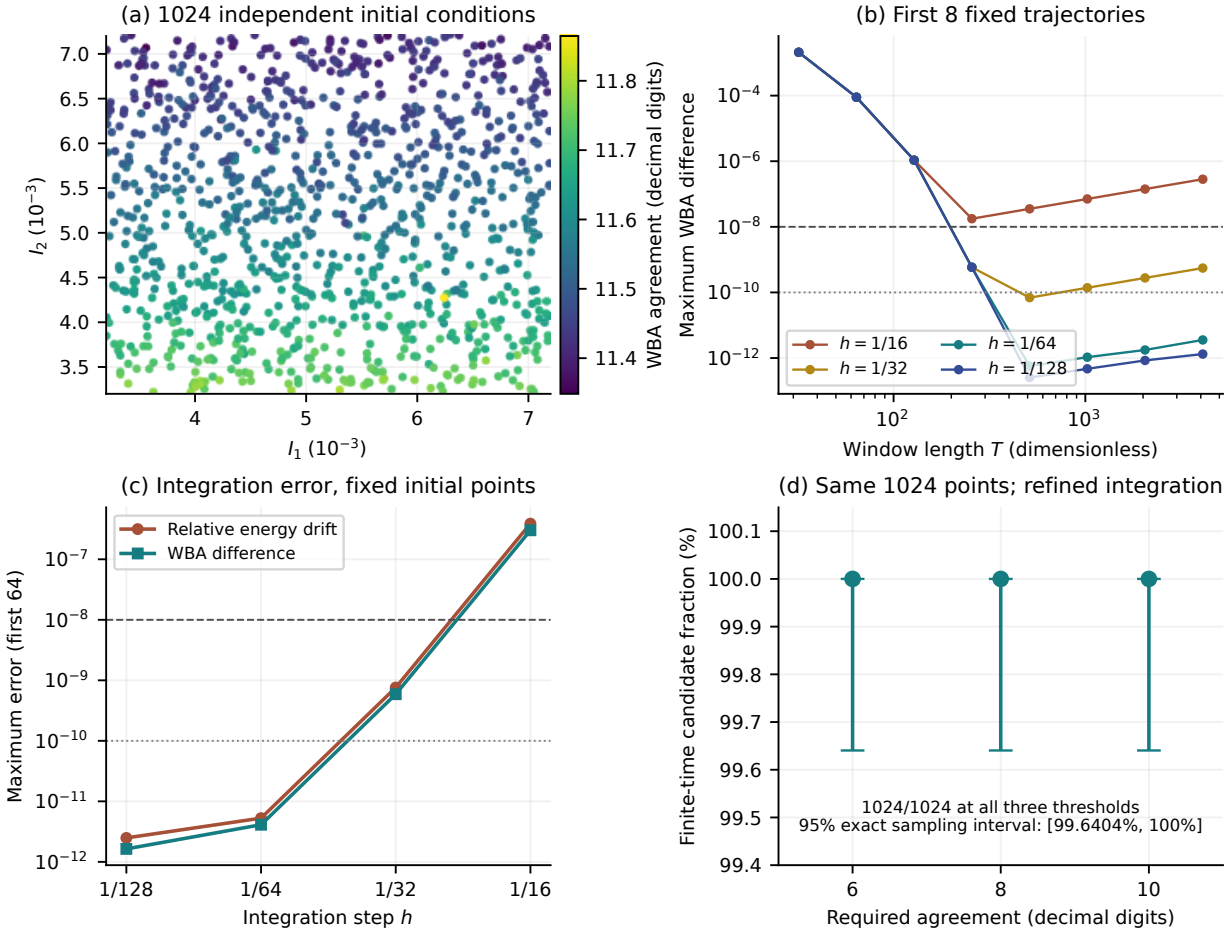


Figure S7. Finite-time quasiperiodicity diagnostics for the fitted Hamiltonian. (a) Uniform action sampling; angles are independently uniform. (b) Consecutive-window agreement on the first eight fixed trajectories; the long-window floor reflects integration error and roundoff. (c) Step refinement on the first 64 fixed initial points identifies the coarse integration error. (d) Candidate fractions and exact sampling intervals at three thresholds, using the same 1024 points with refined integration. Candidate status describes finite-time agreement of the specified observables.

The field is curl-free and divergence-free. Under the declared linear-beam kinematics, $x_{\mathrm{tip}} = L$, so its additional potential is $-\mu G w(L)\sin\vartheta$. It supplies both the transverse tip force $\mu G\sin\vartheta$ and the corresponding moment $\mu G w(L)\cos\vartheta$. The axial force $-\mu G\cos\vartheta$ is balanced by the axial constraint reaction and does no virtual work in this specialization. We also add the still-fluid Morison drag density $-c_2|w_t|w_t$, with dissipation potential $c_2\int_0^L |w_t|^3 ds/3$.

In the fixed dimensionless chart, the conservative energy is

$$h = \tfrac{1}{2}Y^{\mathsf{T}}\,\mathrm{diag}(\nu)Y + \tfrac{e}{2}X^{\mathsf{T}}A_{\mathrm{el}}X + m(1-\cos\vartheta) - g\frac{G_{\mathrm{ref}}}{B_0}(\ell^{\mathsf{T}}X)\sin\vartheta, \qquad \vartheta = d^{\mathsf{T}}X, \tag{S99}$$

where $A_{\mathrm{el}} = \mathrm{diag}(\nu) - dd^{\mathsf{T}}$, $G_{\mathrm{ref}} = 0.5$ T/m, and $g = mG/G_{\mathrm{ref}}$. The five fitted readouts are $(e, m, g, \alpha/\alpha_{\mathrm{ref}}, c_2/c_{2,\mathrm{ref}})$, with $c_{2,\mathrm{ref}} = 0.02$ kg/m^2. The elastic and magnetic strengths and both dissipative coefficients are constrained to be nonnegative; the gradient may have either sign. The kinetic metric is known. Complex features evaluate the polynomial and exponential terms through $z = (X + iY)/\sqrt{2}$ and its conjugate.

Each of three parameter settings provides six independent 3 s training trajectories and three 4 s tests with different initial conditions and multisine inputs. The two changed settings use respectively $(e, m, \alpha/\alpha_{\mathrm{ref}}, G, c_2) = (0.85, 1.20, 1.40, 0.5, 0.02)$ and $(1.20, 0.85, 0.70, -0.4, 0.012)$ in SI units where dimensional. The first four modes are excited, and the remaining initial coordinates are statically condensed with zero initial velocity. This construction does not assume an invariant modal manifold. Samples are separated by 1 ms. On overlapping 32 ms windows, multiplication by $\psi = \sin^4(\pi s)$ and integration by parts gives

$$-\int \psi' Y\, d\tau = \int \psi F_\theta(X, Y, u)\, d\tau. \tag{S100}$$

Table S8. Mean tip RMSE over three independent full-FE tests in μm. The noisy entries average three independent training-noise seeds; the clean fit is counted once. The last control has a freely learned quadratic potential.

Case	Noise	Omitted	Expanded	Free H_2
Nominal	0	2.415	2.415	2.417
Nominal	0.5%	2.833	2.807	2.996
Soft/$+G$	0	182.988	9.829	7.258
Soft/$+G$	0.5%	183.028	9.757	7.216
Stiff/$-G$	0	166.496	2.121	1.978
Stiff/$-G$	0.5%	166.916	2.841	2.808

The endpoints of ψ vanish. Fitting therefore uses only positions, velocities and the applied field; neither accelerations nor reconstructed force-balance labels enter the regression.

We compare the expanded CHNN with a three-readout model that re-identifies elasticity, magnetic strength and linear drag but omits the gradient and quadratic drag. A further control freely learns the ten entries of the symmetric quadratic potential while retaining the analytic magnetic features and dissipative heads, giving fourteen readouts. Thus every comparison uses four modes. Matched real and complex realizations agree in effective coefficients to 1.31×10^{-11} or better. The experiment does not attribute an accuracy advantage to complex arithmetic.

Independent Gaussian noise is added to the observed modal positions and velocities at 0.1% and 0.5% of each coordinate's training standard deviation, with dimensionless floors 10^{-4} for X and 10^{-5} for Y. The modal projections are assumed observable; this test does not validate a particular sensor layout. At 0.5%, the reconstructed training-tip noise RMS is 3.80, 5.72 and 3.33 μm for the three settings. No noise is added to future reference labels. Figure S8 and Table S8 show that adding the missing physics resolves the main model-class error. The free-quadratic model can absorb reduction error and is slightly more accurate in the two changed settings; a universal advantage for the physical prior is not inferred. For the soft setting, the expanded clean fit identifies $e = 0.849522$, $m = 1.199860$, $G = 0.500272$ T/m, $\alpha = 0.753909$ s^{-1} and $c_2 = 0.020832$ kg/m^2. The last two values differ from their full-plant values by approximately 3.1% and 4.2%. Thus accurate state prediction is not presented as exact parameter recovery. A four-mode predictor supplied with the true soft-setting parameters itself has 10.894 μm mean tip RMSE, showing the retained influence of reduction error.

An independent COMSOL solve tests the changed soft-rod setting from rest for 4 s, including the same gradient force, magnetic moment and distributed quadratic drag. Against the separate full-FE Python solve, the tip RMSE is 0.000189 μm and the maximum difference is 0.000623 μm; the input-field difference is below 2.73×10^{-17} T. The learning coefficients were frozen before this COMSOL test and are not refitted. The omitted-physics, expanded and free-quadratic predictors obtain respectively 190.055, 9.644 and 6.946 μm tip RMSE. This comparison checks an independently implemented altered mechanical model, rather than only another trajectory of the original exact-in-class target.

S5.8 Causal adaptation to slowly changing unknown parameters

To test time variation separately from recalibration between settings, we generate a 29 s full-FE trajectory with a smooth drift. Define $s = \mathrm{clip}_{[0,1]}((t - 4\,\mathrm{s})/(24\,\mathrm{s}))$ and $a = (1 - \cos \pi s)/2$. The hidden plant has $e = 0.85 + 0.20a$, $m = 1.20 - 0.20a$ and $\alpha/\alpha_{\mathrm{ref}} = 1.40 - 0.40a$, while $G = 0.5$ T/m and $c_2 = 0.02$ kg/m^2 remain fixed. Multisine excitation continues throughout. At cutoffs $8, 12, 16, 20, 24, 28$ s, each estimator receives only the preceding 4 s of positions, velocities and known input. It then predicts the next 1 s from the currently observed state. The coefficients are frozen before future state labels are accessed.

A constant sliding-window fit and a locally affine coefficient fit use the same weak identity (S100); the latter adds features multiplied by $(t - t_{\mathrm{cut}})/(4\,\mathrm{s})$. Linear constraints preserve the nonnegative readouts over both the fitting interval and the forecast interval. The known future input is supplied to all predictors, but future states and true parameter values are withheld. Across the six forecasts, the mean next-second tip RMSE is 387.483 μm for the initial fit frozen at 4 s, 59.217 μm for causal constant refitting and 9.867 μm for causal affine adaptation. The affine errors range from 2.701 to 17.295 μm. Constant refitting is better at the final cutoff, where the physical drift stops; this limitation of extrapolating a local trend is retained. These are six forecasts on one declared drift trajectory, not independent patient experiments or a general online-learning theorem.

For this nonautonomous plant, the physical energy ledger includes $E_0[\dot{e} X^{\mathsf{T}} A_{\mathrm{el}} X/2 + \dot{m}(1 - \cos\vartheta) - \dot{g}(G_{\mathrm{ref}}/B_0)(\ell^{\mathsf{T}} X)\sin\vartheta]$ in addition to magnetic input power and fluid losses. Its maximum numerical residual is 6.06×10^{-14} J. The stationary conservative KAM certificates are not transferred to these driven, dissipative, time-dependent trajectories.

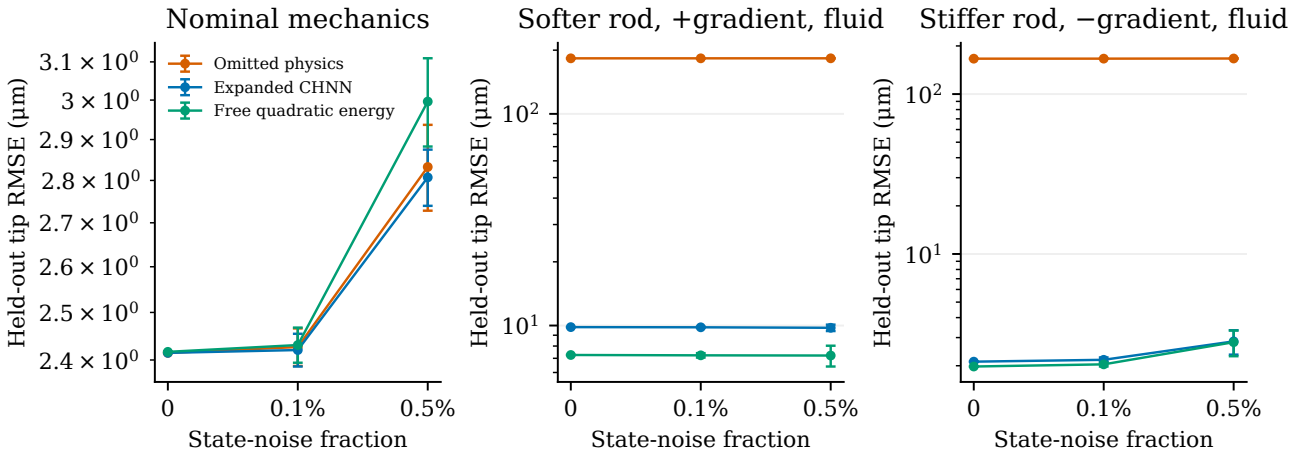


Figure S8. State noise and physics outside the nominal hypothesis class. Each point averages three independent held-out trajectories. Bars show the sample standard deviation across three training-noise seeds; the clean fit is counted once. The higher-error model still re-identifies its retained parameters.

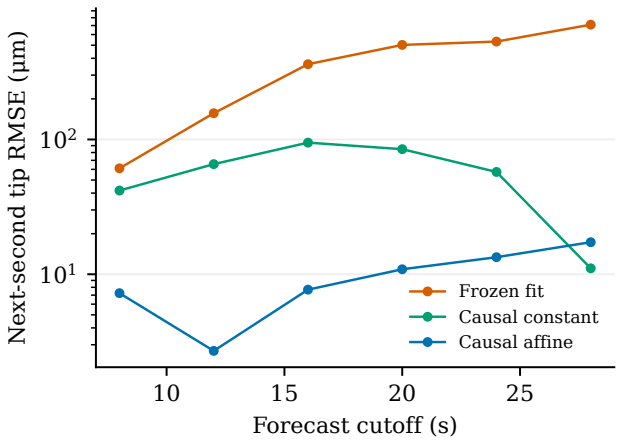


Figure S9. One-second forecasts made using only past data under slow material and magnetic drift. The frozen, constant and affine predictors share the same current observed state and prescribed future input.

S5.9 Forced-response stability and clearance boundaries

To distinguish resonance amplification from a dynamical loss of stability, we additionally scan the same nominal beam under a prescribed transverse field. This experiment concerns attracting responses of a driven, damped system; it does not measure destruction of the conservative KAM torus. The physical parameters, frozen static-field modes and mass-drag coefficient $\alpha = 0.5555852368\,\mathrm{s}^{-1}$ are unchanged. No field-gradient input, learned-model substitution, governor or contact force is used. In physical time, the retained modal equations are

$$\begin{aligned}\dot{X}_j &= \omega_j Y_j,\\ \dot{Y}_j &= -\omega_j X_j - \alpha Y_j\\ &\quad + \omega_0 d_j[\vartheta - \sin\vartheta + u(t)\cos\vartheta],\end{aligned} \tag{S101}$$

where $\vartheta = d^{\mathsf{T}}X$, $u = B_y/B_0$, $B_0 = 30$ mT and $\omega_0 = 11.417287283\,\mathrm{rad\,s}^{-1}$.

Fixed protocol and stability diagnostics. Independent runs use $B_y = A[1-\exp(-(t/(0.25\,\mathrm{s}))^2)]\sin(r\omega_0 t)$. The amplitude set is $A \in \{1,2,4,6,10,15,20\}$ mT. The frequency set contains the 0.02 grid over $[0.70, 1.30]$ and the 0.005 grid over $[0.94, 1.06]$, giving 49 distinct ratios. For each pair, the initial low state is either zero or $\pm(0.18, 0.04, 0, 0)$, giving 1029 independent integrations. Increasing- and decreasing-frequency continuations add 686 runs. Each continuation changes frequency at the zero-field phase and inherits the previous endpoint; the ramp is applied only at its first frequency. Each run discards 128 forcing periods and measures the next 64, for a total duration of 81.28–150.95 s. Fixed-step DOP853 uses 256 steps per forcing period in the two-mode grid.

We integrate the analytic variational equation simultaneously, renormalize one tangent vector every period, and average its logarithmic growth over the last 64 periods. These conditional finite-time Lyapunov exponents hold the prescribed forcing phase fixed. We also integrate the complete 4×4 fundamental matrix over a further period with 512 steps, and calculate the Floquet multipliers of every converged two-mode orbit. The nontrivial Jacobian block is

$$D_X\dot{Y} = -\operatorname{diag}(\omega_j) + \omega_0[1-\cos\vartheta - u\sin\vartheta]dd^{\mathsf{T}}. \tag{S102}$$

The identity $\det\mathcal{M} = \exp(-2\alpha T)$ for its monodromy matrix supplies an independent numerical check.

All 1715 runs converge to forcing-periodic responses. The largest single-period canonical residual is 6.65×10^{-11}; the maximum endpoint differences among the three initial states and between the two continuation directions are

2.31×10^{-11} and 4.01×10^{-11}, respectively. Conditional exponents in the independent grid range from -0.28046 to $-0.27044\,\mathrm{s}^{-1}$. Floquet spectral radii range from 0.803809 to 0.889055, with the determinant identity accurate to 1.41×10^{-14}. Fifteen cases at amplitudes 2, 6 and 20 mT and frequency ratios 0.8, 0.95, 1, 1.05 and 1.2 retain periodic recurrence and negative conditional exponents when both the integration time and steps per period are doubled, and separately with four and eight modes. The latter use 1024 and 4096 steps per period. Time refinement changes the sampled two-mode steady peak by at most 3.81×10^{-7} mm. An independently written adaptive DOP853 implementation also agrees in canonical state to 3.13×10^{-13} on three twelve-period checks.

These are numerical local-stability and basin checks at specified points, not a proof over a continuous parameter set. Of the 1029 independent runs, 550 keep their sampled whole-beam slope bound below 0.1 rad throughout startup and measurement. We use this value as a conservative reporting screen for the small-slope model, not as a calibrated physical validity threshold. Large-amplitude rows are retained in the data and marked in Figure S10, but do not establish physical device performance. No resolved multistability or chaos is detected in this scan; neither a chaos threshold nor conservative-torus breakup is inferred from it.

Full-model resonance and clearance contours. To resolve the operational boundary without high-mode truncation, we separately calculate periodic responses using all 64 frozen FE modes. For $f = \vartheta - \sin\vartheta + u\cos\vartheta$ with Fourier coefficients f_k and $\Omega = r\omega_0$, the exact linear modal response is

$$X_{j,k} = \frac{\omega_0 \omega_j d_j f_k}{\omega_j^2 - (k\Omega)^2 + \mathrm{i}\alpha k\Omega}. \tag{S103}$$

Iterating this response with the nonlinear force gives a numerical periodic solution; its full nonlinear ODE residual is checked after reconstructing $\vartheta = d^{\mathsf{T}}X$. Amplitude root-finding then locates a 4 mm peak-displacement crossing at 31 of 35 frequency ratios; the remaining four ratios, 1.225–1.30, have no such crossing below 20 mT on the calculated branch. Cubic-Hermite Bernstein bounds cover every centreline point at each sampled phase. At every reported crossing these spatial upper bounds equal the actual tip peak to 8.9×10^{-16} mm, so the contour also locates a tip reaching 4 mm in the numerical periodic solution.

The steady contour has a numerical minimum near $r = 0.999245$ ($\Omega = 11.408664\,\mathrm{rad\,s}^{-1}$), at $A = 1.915911$ mT and a spatial slope bound of 0.06750 rad. Thus a 20 mT actuator limit alone cannot ensure clearance near resonance.

Table S9. Numerical 4 mm crossing amplitudes in mT. Steady crossings use all 64 modes. Startup crossings use eight modes, zero initial state, the stated 0.25 s ramp, and 192 forcing periods. They are protocol-dependent response contours, not certified safe-field limits.

Ω/ω_0	Steady	Startup
0.8	13.6362	8.6513
0.9	7.5005	5.1882
1.0	1.9168	1.9169
1.1	8.7909	5.7662
1.2	18.5936	10.4455

At five representative crossings, increasing the Fourier phase grid from 1024 to 4096 changes the peak by at most $0.00526\,\mu$m. Comparisons with 2, 4, 8, 16 and 32 modes are retained: at these same inputs the two-mode peak differs from the full FE peak by up to $67.73\,\mu$m, whereas the eight-mode discrepancy is below $4.18\,\mu$m. Independent full-FE Radau integration from the computed periodic states at $r = 0.8, 1, 1.2$ agrees in tip position to 1.11×10^{-12} m over one period.

Startup overshoot imposes a different contour. The eight-mode startup roots in Table S9 use 512 observation phases per cycle; doubling time, integration resolution and observation resolution at $r = 0.8, 1, 1.2$ changes the peak by at most $0.0678\,\mu$m. Full-FE integration from rest over the first 32 periods, combined with the separately computed periodic response, changes the checked peak by at most $2.44\,\mu$m. At $r = 0.8$, the approximately 8.6513 mT request has a 2.5411 mm full-FE steady amplitude but reaches 4.0020 mm during startup. This is a numerical overshoot example with no wall-contact force. The continuous-time input-work governor of Section S7.1 remains the proved clearance mechanism; the response contours explain why both frequency and startup must enter a practical operating choice.

S6 Preserving KAM dynamics through magnetic actuation

S6.1 *Magnetic feedback that preserves the learned KAM dynamics*

A spatial magnetic-field gradient supplies a tip-force input that, together with the transverse field, preserves the certified conservative dynamics despite physical damping. Both inputs act through the magnetic dipole on the same finite-element beam.

Let a_j be the physical mass-normalized modal positions of the complete 64-mode FE model, with velocities v_j,

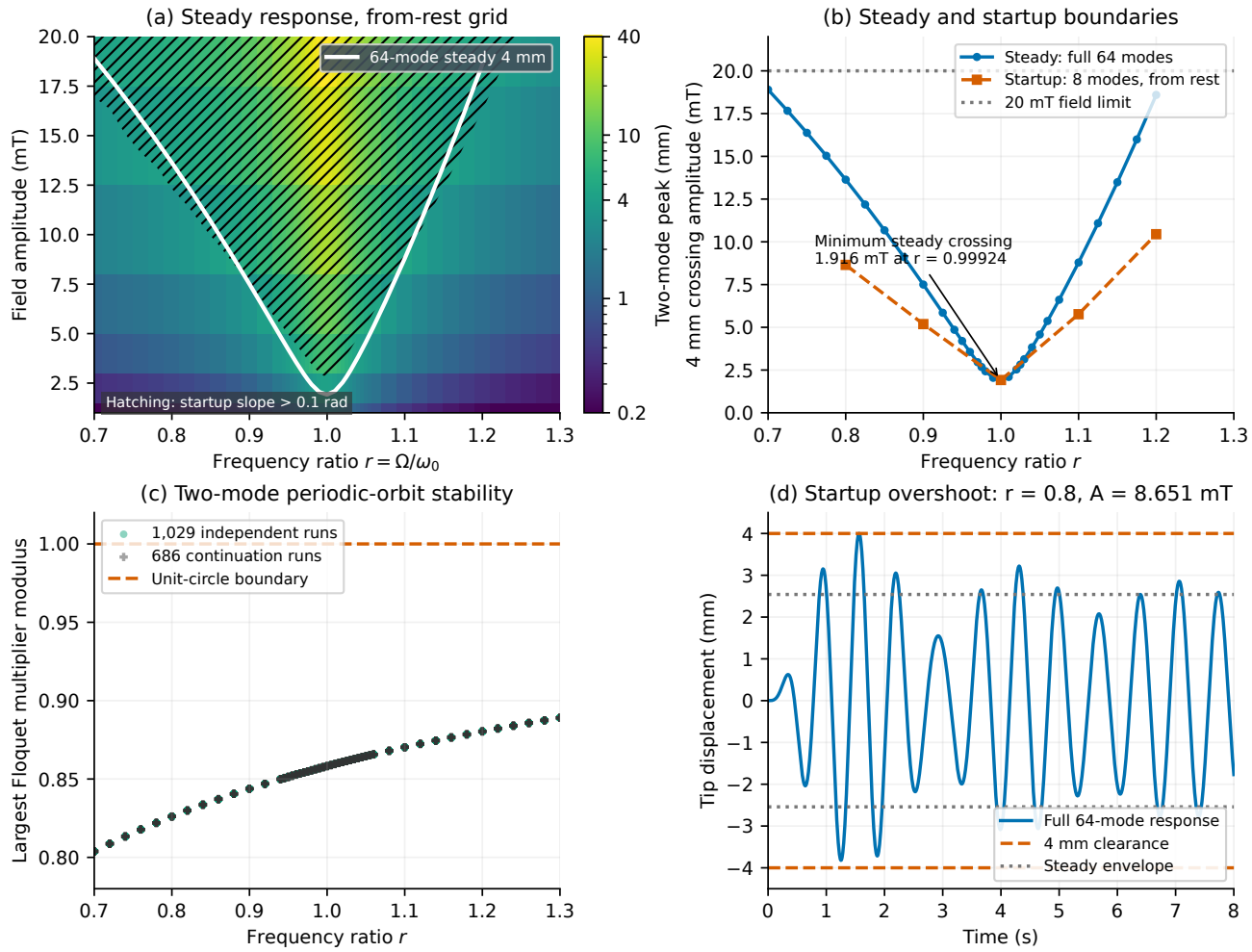


Figure S10. Driven-response and clearance study of the same magnetic beam. (a) Two-mode steady peak on the from-rest grid; hatching marks sampled startup slopes above the 0.1 rad reporting screen. The white line is the independently computed full-FE steady 4 mm contour. (b) Full-FE steady and eight-mode startup contours under the fixed ramp. (c) Floquet multipliers of the two-mode periodic responses from all independent and continuation runs. (d) Full-FE startup overshoot near the eight-mode numerical crossing. Stability of an attracting forced periodic response is distinct from conservative KAM-torus persistence.

frequencies ω_j, tip-displacement participation b_j, and tip-rotation participation t_j. The numerical theorem uses the exported modal coefficients and Hermite shape map as the exact definition of the computational model. The diagonalization residual relative to the assembled nodal matrices is quantified below. Write $\vartheta = \sum_j t_j a_j$ and $E_0 = \mu B_0$. With transverse tip force F and transverse field B_y, the modal equations are

$$\begin{aligned} \ddot{a}_j + \alpha \dot{a}_j + \omega_j^2 a_j &= b_j F + t_j U, \\ U &= E_0(\vartheta - \sin\vartheta) + \mu B_y \cos\vartheta. \end{aligned} \tag{S104}$$

The frequencies already include the linear stiffness from B_0. For the first two modes define

$$A_{\mathrm{mag}} = \begin{pmatrix} b_1 & t_1 \\ b_2 & t_2 \end{pmatrix}. \tag{S105}$$

An interval enclosure of its determinant excludes zero.

Let $h(X, Y)$ be the fitted Hamiltonian used by the KAM certificate, including momentum-dependent quartics. Set $a_i = S_i X_i$ and recover Y from

$$h_Y(X, Y) = \mathrm{diag}(S_i)^{-1} v_{1:2}/\omega_0. \tag{S106}$$

This implicit velocity-to-momentum map cannot be replaced by $Y_i = v_i/(S_i\omega_0\nu_i)$ when the learned momentum quartics are retained. The verified lower bound on h_{YY} below makes the map locally invertible and injective in Y on the stated real box. Define

$$a^{\mathrm{des}}_{i,tt} = S_i\omega_0^2\big(h_{YX}h_Y - h_{YY}h_X\big)_i, \tag{S107}$$

$$\begin{pmatrix} F \\ U \end{pmatrix} = A_{\mathrm{mag}}^{-1}\left(a^{\mathrm{des}}_{1:2,tt} + \alpha v_{1:2} + \mathrm{diag}(\omega_1^2, \omega_2^2) a_{1:2}\right). \tag{S108}$$

Consequently $\dot{X} = \omega_0 h_Y$ and $\dot{Y} = -\omega_0 h_X$ exactly where the implicit inverse is defined. The proof follows by differentiating (S106): the specified acceleration cancels the h_{YX} term and invertibility of h_{YY} gives the second canonical equation.

The physical commands realizing (S108) are

$$B_y = \frac{U - E_0(\vartheta - \sin\vartheta)}{\mu\cos\vartheta}, \qquad G = \frac{F}{\mu}\begin{pmatrix} -\sin\vartheta & \cos\vartheta \\ \cos\vartheta & \sin\vartheta \end{pmatrix}. \tag{S109}$$

Use $B(r) = B_{\mathrm{tip}} + G(r - r_{\mathrm{tip}})$ locally, with $B_{\mathrm{tip}} = (B_0, B_y)$. The matrix G is symmetric and traceless, so this local planar field satisfies the source-free curl and divergence constraints. For dipole $m = \mu(\cos\vartheta, \sin\vartheta)$ it gives $Gm = (0, F)$, while the tip torque is $\mu(B_y\cos\vartheta - B_0\sin\vartheta)$. Moreover $\|G\|_2 = |F|/\mu$. The applied mechanical power is $F\dot{w}(L) + \mu B_y\cos\vartheta\,\dot{\vartheta}$, in addition to the static magnetic energy already included in the Hamiltonian. The construction assumes these local field and gradient commands are available and the required modal and tip states are observed.

S6.2 Lifting a certified torus to the complete controlled FE model

Theorem S10. Invariant-torus lift under physical magnetic feedback. *Suppose a real analytic embedded two-torus K_∞ of h has frequency $\Omega \in \mathbb{R}^2$ in radians per physical second. Suppose h_{YY} is positive definite on a neighbourhood of this torus with convex Y-fibres for each X, A_{mag} is invertible, $\alpha > 0$, and the complete lifted solution has $|\vartheta| < \pi/2$. Then the full FE closed loop (S104)–(S109) has an embedded invariant two-torus projecting to K_∞. For a fixed low-mode orbit on this torus, each high-mode solution converges exponentially to a unique quasiperiodic response, provided the feedback remains in its stated field and state domain.*

Proof. The first two canonical modes follow h exactly. Hence F and U in (S108) depend only on their torus phase, even though the physical field command also uses the complete tip angle. For every $j \geq 3$, the remaining dynamics are the stable linear equation (S104), forced by analytic quasiperiodic functions. If F_k, U_k denote their Fourier coefficients in the convention $e^{ik\cdot\phi}$, the unique response graph is

$$a^*_{j,k} = \frac{b_j F_k + t_j U_k}{\omega_j^2 - (k\cdot\Omega)^2 + i\alpha(k\cdot\Omega)}. \tag{S110}$$

For real $k\cdot\Omega$, this denominator never vanishes: at zero frequency its real part is $\omega_j^2 > 0$, and at nonzero frequency its imaginary part is nonzero. Its inverse is uniformly bounded over real frequencies. Exponential Fourier decay is therefore preserved on a possibly smaller analytic strip. The velocity graph is obtained by multiplication by $ik\cdot\Omega$ and is analytic on a smaller strip. Positive definiteness on each convex Y-fibre makes $Y \mapsto h_Y(X, Y)$ strictly monotone and injective. Thus the physical low-mode positions and velocities preserve the injective canonical embedding. Differences in the high modes obey the homogeneous damped equations and decay exponentially.

The lifted closed loop combines Hamiltonian low-mode motion with contracting high-mode fibres. The contraction leaves each initial low-mode torus invariant; it does not select a unique torus or make the full controlled system Hamiltonian. Each lifted two-torus has zero volume in the full 128-dimensional phase space. The KAM-family measure therefore refers to the conservative low-mode coordinates.

All-phase response and startup bounds. We bound both the response graph and the startup with all high modal positions and velocities initially zero. Write $h = h_2 + R$, and let the certified torus satisfy $\|K_\infty - K_0\|_\infty \leq \epsilon_K$, where $K_0 = 0.1(\cos\phi_1, \cos\phi_2, -\sin\phi_1, -\sin\phi_2)$. The first two components of the force demand can be written as

$$r_i = S_i\{\omega_0^2[(R_{YX}h_Y)_i - \nu_i R_{X_i} - (R_{YY}h_X)_i] + \alpha\omega_0 h_{Y_i} + (\omega_i^2 - \omega_0^2\nu_i^2)X_i\}, \tag{S111}$$

This identity cancels the large linear restoring terms before bounding. The linear part on K_0 consists of just the two known physical frequencies. Each is bounded using its actual damped denominator in (S110). The remaining terms are bounded on $|X_i|, |Y_i| \leq 0.1 + \epsilon_K$ using every fitted quartic coefficient and the complete trigonometric remainder bounds $|\sin z - z| \leq |z|^3/6$ and $|1 - \cos z| \leq z^2/2$ for real z.

For $\beta = \alpha/2$ and $\omega_{d,j} = \sqrt{\omega_j^2 - \beta^2} > 0$, the impulse response $g_j(t) = e^{-\beta t}\sin(\omega_{d,j}t)/\omega_{d,j}$ obeys

$$\|g_j\|_{L^1(0,\infty)} \leq \frac{2}{\alpha\omega_{d,j}}, \qquad \|g_j'\|_{L^1(0,\infty)} \leq \frac{2}{\alpha} + \frac{1}{\omega_{d,j}}. \tag{S112}$$

These estimates include the entire nonlinear forcing remainder and its Fourier tail. Let P_j, V_j be the resulting all-phase bounds on the graph's position and velocity. Starting high mode j at zero adds at most

$$T_j = P_j + \frac{V_j + \beta P_j}{\omega_{d,j}} \tag{S113}$$

to its position envelope. This follows from the exact homogeneous solution initialized by $(-a_j^*(0), -\dot{a}_j^*(0))$. The bounds sum all 62 high modes and use the cubic Bernstein control rows of every Hermite element to convert these modal envelopes into bounds on every centreline point and every phase.

For the two-mode COMSOL integral fit in Section S4.5, which satisfies the nonzero-loss and finite-amplitude KAM conditions, 256-bit Arb arithmetic encloses $h_{YY} \succeq 0.9999999983I$ on the real 0.3 box. The complete response graph has centreline amplitude below 1.922696 mm. Including the zero-high-mode startup yields an all-time bound below 1.970263 mm, strictly inside the 4 mm net corridor. The latter envelope gives $|\vartheta| < 0.155261$ and $\cos\vartheta > 0.987971$, which closes the feedback-domain bootstrap. The required transverse field is below 0.921665 mT and the gradient spectral norm below $0.02572538\,\mathrm{T\,m^{-1}}$ at every phase and throughout that startup. These bounds validate feasibility under a 20 mT transverse field limit and a $1\,\mathrm{T\,m^{-1}}$ gradient limit; they are derived bounds, not the extrema of a simulated time history. Applying the same computation to the fit trained on complete COMSOL FE trajectories in Section S5.5 gives graph and zero-high-state startup bounds of 1.937308 mm and 2.014078 mm, respectively.

The theorem applies to initial low states on the exact certified torus. The numerical Fourier initial state approximates it within the certified KAM correction, and finite-time integration tests the feedback implementation. The theorem uses the exported modal FE coefficients and fitted weights; independent COMSOL comparisons assess the corresponding nodal mechanical implementation. To make the modal definition explicit, if Φ is the stored full shape basis, M, K_b are the independently assembled nodal matrices and $D_\omega = \mathrm{diag}(\omega_j)$, outward Arb products give

$$\|\Phi^{\mathsf{T}} M \Phi - I\|_\infty < 1.304685 \times 10^{-14}, \quad \text{(S114)}$$

$$\|D_\omega^{-1}\Phi^{\mathsf{T}} K_b \Phi D_\omega^{-1} - I\|_\infty < 3.000518 \times 10^{-9}. \quad \text{(S115)}$$

These are relative kinetic- and stiffness-energy form bounds. They quantify the relationship between the frozen modal model and the original assembled FE representation; they are not used to assert an exact conjugacy or an unproved persistence result for perturbed matrices. The two-mode shape map used for identification differs from the reassembled map by at most 1.258×10^{-14} m, and the reference frequency ω_0 is unchanged.

S6.3 Full-model implementation and frequency preservation

The feedback is integrated on all 64 modes for 12 s, using the two-mode COMSOL integral fit certified in Section S4.5. The initial low state is the phase-zero Fourier candidate and all higher modal positions and velocities are zero. A separate high-accuracy integration of the four-dimensional learned Hamiltonian supplies an implementation reference. The plant is actuated through the physical tip force and moment. The maximum canonical state difference is 1.54×10^{-10}, the relative learned-energy change is below 2.64×10^{-9}, and the full physical work-balance residual is below 6.06×10^{-17} J. The sampled field and gradient peaks are 0.890325 mT and 0.0256844 T/m. Figure S11 compares this trajectory with the same initial state under passive damping, for which the learned energy decreases by 99.8726%. The numerical trajectory is an implementation check; the all-time envelopes come from the preceding interval proof. An independent implementation integrates the low Hamiltonian separately and propagates each of the 62 high damped oscillators by its exact exponential response to piecewise-linear force demands. Force-grid steps of 1, 0.5 and 0.25 ms give maximum tip differences from the full-state integration of 1.073, 0.277 and 0.106 nm, respectively. Every high-mode homogeneous transient is retained. This check separates time-step error from modal truncation and supports the full-state implementation. The resulting differences measure numerical convergence; the rigorous bounds are supplied by the interval proof.

Independent online COMSOL feedback. The complete nodal beam is also solved with the CHNN feedback inside COMSOL. The first two modal coordinates and velocities are projected from its current beam state; two global algebraic equations enforce the learned Legendre inverse. The physical tip force and magnetic moment are recomputed online. This experiment is distinct from the prescribed-field S-path replay. Generalized-alpha integration uses $\rho_\infty = 0$, a nonlinear relative tolerance of 10^{-8} and fixed steps. Table S10 reports four 4 s refinements and compares them at common sampled times with the independent full-state Python run. At the finest step, the maximum relative change in the learned energy is 0.0002958. These numerical errors are reported separately from the analytic KAM and all-time amplitude certificates. Adaptive stepping stalled at very small steps in two preliminary runs; fixed stepping completed each refinement.

The two certified physical angular frequencies are approximately 11.41719351 and 70.24034116 rad/s. Their

Table S10. Online COMSOL feedback versus independent Python integration. Errors are in μm; no step-size tolerance is interpreted as a rigorous error bound.

Step (ms)	Tip RMSE	Maximum tip difference
1	147.852	355.845
0.5	41.356	99.176
0.25	12.006	28.161
0.125	3.848	8.658

exact specification and all-integer Diophantine bound follow from the KAM certificate under the stated time normalization. Magnetic feedback therefore preserves a certified nonresonant frequency pair while compensating damping. The complete nonlinear command contains the harmonics determined by the torus; it is not approximated by two sinusoidal inputs when invoking the lifting theorem. The rigorous state and command envelopes provide a simultaneous amplitude check. The operating-domain fraction is examined separately in Section S5.6.

S7 Clearance guarantees and magnetic safety control

S7.1 A whole-beam clearance certificate for the same magnetic model

The conservative normal form and the safety calculation use the same clamped-free beam, static field, magnetic dipole, and energy. Safety is a statement about the driven physical equations, with their nonnegative drag; it does not require a driven trajectory to lie on a conservative KAM torus. For the planar, small-slope model, write $w(s,t)$ for transverse displacement, $m=\rho A$ for line density, $c_d \ge 0$ for distributed viscous drag, and $\vartheta = w_s(L,t)$ for the tip slope used in the dipole potential. The equations and natural end conditions are

$$mw_{tt} + c_d w_t + EIw_{ssss} = 0, \qquad 0 < s < L, \tag{S116}$$

$$w(0,t) = w_s(0,t) = 0, \qquad w_{sss}(L,t) = 0, \tag{S117}$$

$$EIw_{ss}(L,t) = \mu\{B_y(t)\cos\vartheta - B_0\sin\vartheta\}. \tag{S118}$$

The end moment is the derivative of the same cosine potential used to generate the training trajectories. In particular, the varying transverse field is an input port, not an additional autonomous Hamiltonian perturbation. For a sufficiently regular solution satisfying the mechanical power identity, the stored energy relative to the aligned static-field equilibrium is

$$E(t) = \frac{1}{2}\int_0^L \{mw_t^2 + EIw_{ss}^2\}\,\mathrm{d}s + \mu B_0(1-\cos\vartheta), \tag{S119}$$

$$\dot{E} = -c_d\int_0^L w_t^2\,\mathrm{d}s + \mu B_y\cos\vartheta\,\dot{\vartheta}. \tag{S120}$$

All inequalities below also hold for the conforming cubic-Hermite finite element (FE) system and any of its Galerkin subspaces, because their energies are the restrictions of this expression and retain the same input power. In the balanced coordinates of the learning experiment, $E=\mu B_0 h_*$ for the physical target energy, $\vartheta = d^{\mathsf{T}}X$, and the port force in the Y equation is $d(B_y/B_0)\cos(d^{\mathsf{T}}X)$.

Proposition S11. Clearance and field-variation budget. *Assume $B_0 > 0$, $c_d \ge 0$, and an absolutely continuous applied field B_y on $[0,T]$. Define*

$$S_T = \sup_{0\le t\le T}|B_y(t)| + |B_y(0)| + \int_0^T |\dot{B}_y(t)|\,\mathrm{d}t, \tag{S121}$$

$$C_T = \mu\sqrt{\frac{2L}{EI}}S_T, \tag{S122}$$

$$R_T = \frac{C_T + \sqrt{C_T^2 + 4E(0)}}{2}. \tag{S123}$$

Then, throughout $[0,T]$,

$$E(t) \le R_T^2, \qquad |w(s,t)| \le \sqrt{\frac{2s^3}{3EI}}R_T, \qquad |w_s(s,t)| \le \sqrt{\frac{2s}{EI}}R_T. \tag{S124}$$

For a straight corridor with wall half-width a and rod radius r, let $c = a - r > 0$. A sufficient condition for positive clearance of the whole rod is $R_T < \sqrt{E_c}$, where

$$E_c = \frac{3EIc^2}{2L^3}. \tag{S125}$$

For $E(0) < E_c$, the following directly checkable operating condition is equivalent to this sufficient energy condition:

$$S_T < \frac{E_c - E(0)}{\mu\sqrt{2LE_c/EI}}. \tag{S126}$$

These conclusions hold for every nonnegative drag coefficient, including zero.

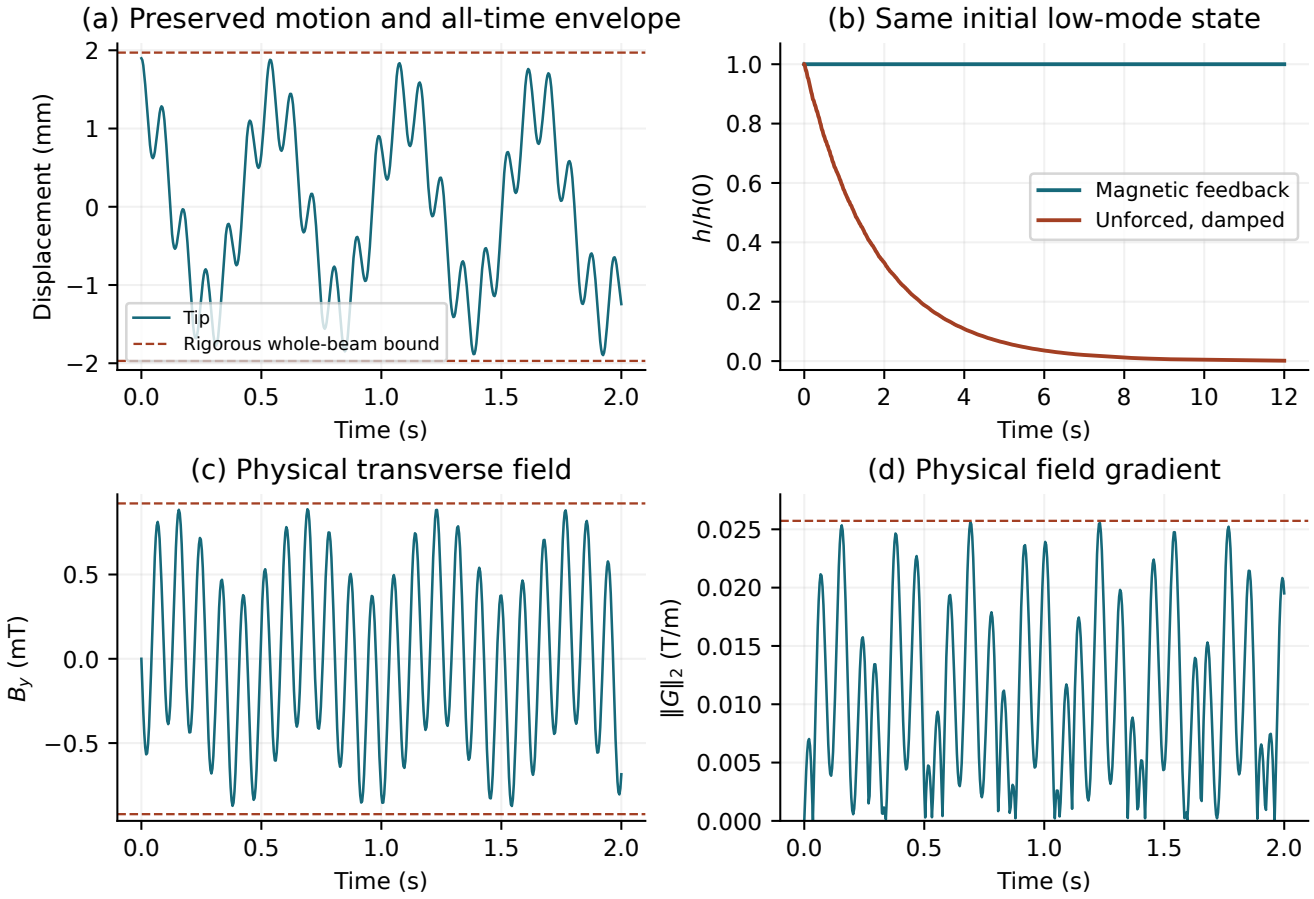


Figure S11. Magnetic preservation of the COMSOL-integral learned Hamiltonian on the complete exported modal FE plant. The higher modes start at zero and undergo a decaying transient toward their unique quasiperiodic response graph. Dashed bounds in (a), (c) and (d) are validated all-phase bounds, including that startup. The field and gradient in (c,d) jointly realize physical tip force and torque. Panels (a,c,d) show the first 2 s; panel (b) shows the complete 12 s run.

Proof. Since $B_0 > 0$, the static magnetic energy is nonnegative and $\|w_{ss}\|_{L^2}^2 \le 2E/EI$. The clamped conditions imply $w(s) = \int_0^s (s-\xi) w_{ss}(\xi)\,\mathrm{d}\xi$ and $w_s(s) = \int_0^s w_{ss}(\xi)\,\mathrm{d}\xi$. Cauchy–Schwarz gives the two spatial inequalities in (S124) with $\sqrt{E}$ in place of R_T. Write $D_c(t) = \int_0^t c_d \|w_t\|_{L^2}^2\,\mathrm{d}\tau$. Integrating (S120) by parts in the input term gives

$$\begin{aligned} E(t) + D_c(t) - E(0) &= \mu B_y(t)\sin\vartheta(t) \\ &\quad - \mu B_y(0)\sin\vartheta(0) \\ &\quad - \mu \int_0^t \dot{B}_y(\tau)\sin\vartheta(\tau)\,\mathrm{d}\tau. \end{aligned}$$

Set $M_T = \sup_{[0,T]} \sqrt{E(t)}$. Using $|\sin\vartheta| \le |\vartheta| \le \sqrt{2L/EI}\, M_T$ and discarding the nonnegative dissipation yields $M_T^2 \le E(0) + C_T M_T$. The positive root is R_T. The entire radius-r tube about the centerline remains between the walls when $\sup_s |w(s,t)| < a - r$. Solving $R_T < \sqrt{E_c}$ gives (S126).

For a nonuniform planar corridor, let $c(s) > 0$ be the Euclidean distance from the undeformed point $(s, 0)$ to the wall, less the rod radius. The distance-to-wall function is one-Lipschitz, so $|w(s)| < c(s)$ keeps the radius-r tube clear. The same argument replaces E_c by $\inf_{0<s\le L} 3EIc(s)^2/(2s^3)$. For an inclined or curved wall, a local vertical half-width is not a substitute for this Euclidean clearance. This sufficient condition applies to the specified corridor geometry; vessel-specific use requires its own wall data. The continuum inequality applies directly to solutions with the stated initial-energy and applied-field bounds, and holds separately for conforming FE solutions. It therefore requires neither a two-mode reconstruction error estimate nor an identification of an FE trajectory with a continuum solution.

For the shared SI parameters, choose $a = 4.6$ mm and $r = 0.6$ mm, so that $c = 4$ mm. The resulting E_c is 23.8565 nJ. For a monotone field ramp from zero to A followed by a hold, $S_\infty = 2A$. With zero initial energy, a sufficient

condition is

$$A < \frac{\sqrt{3}\,EIc}{4\mu L^2} = 6.8868\ \mathrm{mT}. \tag{S127}$$

For $B_y = A\rho(t)\sin(\Omega t)$ with $\rho(t) = 1 - \exp[-(t/0.25\,\mathrm{s})^2]$, $S_T \le A(2 + \Omega T)$, because ρ increases from zero and is at most one. Thus amplitude, frequency and operating duration enter an explicit joint condition by substituting this bound in (S126). For $T = 8$ s, $\Omega = 9.444949$ rad/s and $E(0) = 0$, its sufficient amplitude limit is approximately 0.1776 mT. This conservative open-loop test does not certify the default 6 mT sinusoid, even when a simulated trajectory has positive clearance.

S7.2 A magnetic-input governor with a full-model energy guarantee

The field-variation test can be strengthened for feedback operation by measuring the actual mechanical input port. Let $\overline{E}_0 \ge E(0)$ be a valid initial-energy bound and integrate

$$W(t) = \overline{E}_0 + \int_0^t \mu B_y \cos\vartheta\,\dot{\vartheta}\,\mathrm{d}\tau. \tag{S128}$$

Equation (S120) gives $E(t) \le W(t)$ without measuring or estimating the dissipated energy. Given $E(0) \le \overline{E}_0 < E_{\max} < E_c$, a bounded requested field $|B_{\mathrm{req}}| \le B_{\max}$, and $\kappa > 0$, choose the applied field as the closest value to B_{req} satisfying

$$\begin{aligned} |B_y| &\le B_{\max}, \\ a_p B_y &\le \kappa(E_{\max} - W), \\ a_p &= \mu\cos\vartheta\,\dot{\vartheta}. \end{aligned} \tag{S129}$$

This is a scalar projection onto an interval. Whenever the requested input violates the work constraint and $a_p \neq 0$, the applied field is $B_y = \kappa(E_{\max} - W)/a_p$; otherwise it is unchanged. On $W \le E_{\max}$, zero field is feasible, and this projection never increases the requested field amplitude.

Proposition S12. Port-work safety invariant. *Suppose the actual applied field and actual tip slope and slope rate are available to* (S128)–(S129)*, and the field command is applied without unbudgeted error or delay. Every closed-loop solution satisfying the mechanical power identity obeys*

$$\begin{aligned} E(t) \le W(t) \le E_{\max}, \\ \sup_s |w(s,t)| \le \sqrt{\frac{2L^3 E_{\max}}{3EI}} < c. \end{aligned} \tag{S130}$$

Consequently the entire rod retains positive wall clearance for all times on which the stated beam model and port measurements apply, for any $c_d \ge 0$.

Proof. The power identity gives $W - E = \overline{E}_0 - E(0) + \int_0^t c_d\|w_t\|_{L^2}^2\,\mathrm{d}\tau \ge 0$. The input constraint yields $\dot{W} \le \kappa(E_{\max} - W)$, hence $E_{\max} - W(t) \ge (E_{\max} - \overline{E}_0)\exp(-\kappa t) \ge 0$. The spatial estimate proved above completes the result.

The guarantee uses the full physical input power, including the work associated with unobserved high modes. A learned model supplies the requested field. Accurate actual-port measurements and ideal command application are assumed; sensor, actuation and delay errors require additional work-error budgets. The projection constrains field amplitude and accumulated work, with no imposed slew-rate bound. Because dissipated energy is not credited back, the observer can restrict the request even when stored energy has decreased. Tracking performance is therefore evaluated separately under this conservative safety constraint.

A bounded port-measurement error can be included without assuming an exact learned state. If $|a_p - \widehat{a}_p| \le \delta_a(t)$, replace the observer derivative and its input constraint by

$$\begin{aligned} \dot{W} &= \widehat{a}_p B_y + \delta_a|B_y| \\ &\le \kappa(E_{\max} - W). \end{aligned} \tag{S131}$$

Actual input power is at most this observer derivative, so the same proof applies. Zero field remains feasible and the scalar feasible set is convex. Here B_y must be the actual applied field; an uncertain command-to-field map requires a further bound on actual input power. The numerical tests below use $\delta_a = 0$ and do not experimentally validate a sensor-error bound.

S7.3 Full-FE safety, violation and truncation checks

All tests use the shared 32-element consistent-mass FE beam, retain all 64 unconstrained displacement/slope degrees of freedom, and integrate the exact cosine tip potential. The coordinates are the complete 64-mode static-field basis; this is a change of coordinates, not a modal truncation. The default drag is $C = \alpha M$ with $\alpha = 0.55558524\ \mathrm{s}^{-1}$. Unless stated otherwise, trajectories start from rest at the straight equilibrium. The sinusoidal tests use the frequency $\Omega = 9.444949$ rad/s and the 0.25 s ramp above; the monotone tests omit the sine factor. The integration interval is 8 s. The continuous-time claims come from Propositions S11 and S12; cubic-Hermite spatial extrema are evaluated exactly at each saved time for the numerical diagnostics.

Table S11 separates numerical maxima from continuous-time certificates. The 6 mT monotone ramp satisfies an all-time displacement bound of 3.48494 mm, including zero drag. The default 6 mT sinusoid has a numerical peak of 3.09031 mm, but fails the conservative field-variation test; that peak alone is not a safety proof. Increasing its amplitude to 10 mT produces a 5.14478 mm free-motion peak and crosses the 4 mm boundary. The continuation after crossing is a diagnostic of a violated constraint, not a simulation of wall contact.

With $E_{\max} = 0.95E_c = 22.66365$ nJ and $\kappa = 5\ \mathrm{s}^{-1}$, the work governor applied to the same 10 mT request has a rigorous whole-centerline bound of 3.89872 mm and thus a minimum guaranteed clearance of 0.10128 mm. Its numerical peak is 3.14362 mm with the default drag and 3.55853 mm with zero drag. The same energy cap also implies $\sup_s |w_s| \le 0.08441$, consistent with the declared small-slope regime for the certified trajectories. Tightening both relative and absolute integration tolerances tenfold for the full-FE governed case changes the tip trajectory by at most 8.19×10^{-11} m; the refined power-balance residual is 3.86×10^{-18} J (1.74×10^{-10} relative to the work scale). The governor changes the requested field at 55.37% of saved times in this oscillatory case.

For the 6 mT sinusoid, the maximum tip discrepancies of the 2-, 4-, and 8-mode reconstructions against the full 64-mode FE trajectory are 22.7168, 5.9661, and 1.5156 μm, respectively (0.73510%, 0.19306%, and 0.04904% of the full-FE peak). These are measured truncation errors for this input, not rigorous bounds for all neglected modes or all trajectories. The full-continuum energy certificate does not substitute any of them for an unknown error constant. An independent COMSOL HermitianBeam integration of the same 32-element consistent-mass model gives a default-sinusoid peak of 3.09032 mm; its largest tip difference from the present integration is 0.11835 μm. Tightening the COMSOL relative tolerance to 10^{-8} reduces the largest tip difference to 0.02160 μm. Independent COMSOL runs also reproduce the 6 mT monotone-ramp safe case and 10 mT sinusoidal violation: their peaks are 0.924046 and 5.144787 mm, with maximum cross-solver tip discrepancies of 0.00663 and 0.03489 μm, respectively. This cross-check concerns the same mathematical model and does not calibrate material or actuation parameters against hardware.

The closed-loop governor was also implemented inside COMSOL, coupling a global ODE for $W/E_{\max}$ to the nonlinear magnetic end moment. This run used the generalized-α time integrator with $\rho_\infty = 0$, relative tolerance 10^{-5}, and a maximum internal step of 0.002 s; the 2001 saved outputs are spaced by 0.004 s. Its full-FE centerline peak is 3.143643 mm, leaving a sampled net clearance of 0.856357 mm. The maximum tip discrepancy from the refined Python trajectory is 0.582073 μm, or 0.018516% of its peak. The applied field remains below 9.816986 mT. The COMSOL-integrated work observer peaks at 22.253032 nJ, or $0.9818823E_{\max}$; $E \le W \le E_{\max}$ and the magnetic-power projection hold at every saved time. Its maximum work-observer discrepancy from Python is 2.263×10^{-12} J.

For an independent energy ledger, reconstruct E from all displacement and velocity DOFs and compute

$$D_h(t) = \int_0^t \alpha \dot{q}^{\mathsf{T}} M \dot{q}\, \mathrm{d}\tau,$$
$$R_E(t) = E(t) - E(0) - [W(t) - W(0)] + D_h(t).$$

Simpson quadrature on the saved outputs gives $\max_t |R_E(t)| = 5.796 \times 10^{-12}$ J after rounding up, or 2.605×10^{-4} of the work scale. The Simpson–trapezoid difference for that dissipation integral is 2.424×10^{-13} J. This COMSOL run uses high-frequency numerical damping and a looser tolerance than the refined Python integration; its ledger residual includes time-integration and output-quadrature errors. The separately integrated COMSOL work state is used for the observer comparison, rather than replacing it by a coarse-output input power integral. These numerical checks corroborate the implementation; the continuous-time clearance statement remains the analytic invariant in Proposition S12.

S7.4 The learned CHNN generates a feasible magnetic tracking request

To evaluate learned magnetic tracking, the fitted GC model with seed 17 and its identified dissipation coefficient generate a nominal magnetic field for the full FE plant. Let $z = (X_1, X_2, Y_1, Y_2)$ denote the observed projection onto the first two static-field modes, $\ell^{\mathsf{T}} X$ its reconstructed tip displacement, and h_θ the fitted energy, including all learned mixed and momentum-dependent quartic coefficients. Define, in the dimensionless time $\tau = \omega_0 t$,

$$y = \ell^{\mathsf{T}} X, \qquad y' = \ell^{\mathsf{T}} h_{\theta,Y}, \tag{S132}$$
$$f_0 = \ell^{\mathsf{T}} h_{\theta,YX} h_{\theta,Y} - \ell^{\mathsf{T}} h_{\theta,YY}(h_{\theta,X} + D_\theta h_{\theta,Y}), \tag{S133}$$
$$g_0 = \ell^{\mathsf{T}} h_{\theta,YY} d \cos(d^{\mathsf{T}} X), \tag{S134}$$
$$D_\theta = \widehat{\gamma}\ \mathrm{diag}(\nu)^{-1}. \tag{S135}$$

Table S11. Safety tests on the same full 32-element, 64-DOF FE beam. Numerical maxima are cubic-Hermite spatial extrema at 2001 saved times over 8 s. Bounds are continuous-time sufficient bounds from the stated propositions. "No" means the sufficient test does not certify the case. All fields use a 0.25 s ramp; sine frequency is 9.444949 rad/s. Except the stated $E_c/4$ case, initial energy is zero.

Applied protocol	Peak (mm)	Bound (mm)	Certified
6 mT ramp/hold	0.92405	3.48494	Yes
3 mT ramp, $E(0) = E_c/4$	2.06222	3.05276	Yes
0.15 mT sine	0.07731	3.37863	Yes
6 mT sine	3.09031	135.145	No
10 mT sine	5.14478	225.242	No
Governed 10 mT sine	3.14362	3.89872	Yes
Governed 10 mT, zero drag	3.55853	3.89872	Yes
6 mT ramp, zero drag	0.95405	3.48494	Yes

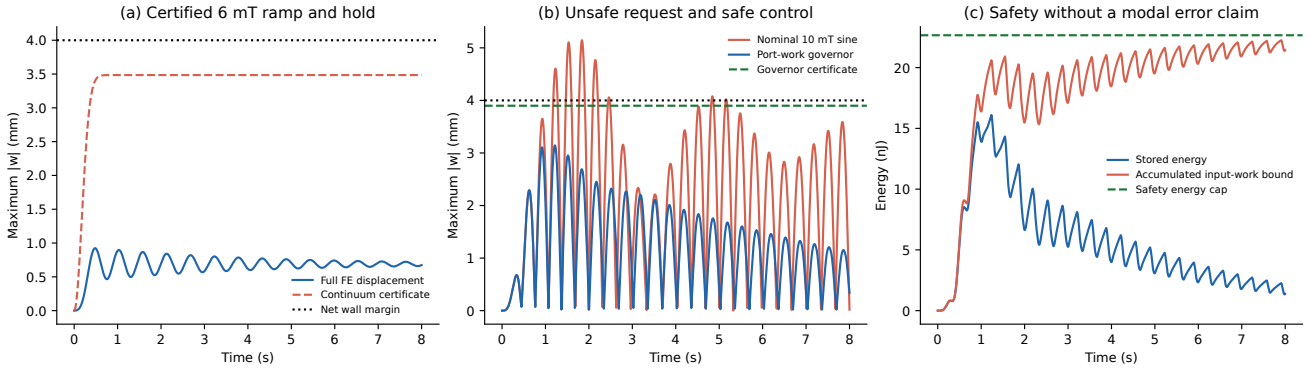


Figure S12. Whole-beam safety for the shared full FE plant, with zero initial energy and default drag. (a) A 6 mT monotone ramp and its all-time continuum energy certificate. (b) The 10 mT ramped sinusoidal request at 9.444949 rad/s violates the 4 mm net corridor margin; the actual port-work governor preserves it with an analytically guaranteed bound. (c) Stored energy and accumulated input-work bound for the governed case. The wall half-width is 4.6 mm and rod radius 0.6 mm. These geometric constraints define the simulated corridor; patient-specific clearance requires a corresponding geometry and parameter model.

Where $g_0 \neq 0$, the nominal request is

$$a_r = r'' - k_p(y - r) - k_d(y' - r'), \tag{S136}$$

$$u_* = (a_r - f_0)/g_0, \tag{S137}$$

$$B_{\mathrm{req}} = B_0 \,\mathrm{clip}_{[-b,b]}(u_*), \tag{S138}$$

$$b = B_{\max}/B_0. \tag{S139}$$

The primes on r denote derivatives in τ. We use $k_p = 16/\omega_0^2$, $k_d = 8/\omega_0$, and $\widehat{\gamma} = 0.0486617964$. Both Hessian blocks are evaluated from the fitted model; they are not replaced by their quadratic-prior values. The plant receives only the scalar B_y through its actual tip moment, after the work-governor projection. It receives no independently assigned modal forces.

The requested SI tip trajectory is $y_{\mathrm{ref}}(t) = 2\,\mathrm{mm}\,(10s^3 - 15s^4 + 6s^5)$ for $s = t/(4\,\mathrm{s}) \in [0, 1]$, followed by a hold through 8 s, and $r(\tau) = y_{\mathrm{ref}}(\tau/\omega_0)$. The field limit is $B_{\max} = 20$ mT for this separate tracking protocol. This limit follows from the static input authority: a 2 mm equilibrium requires $\vartheta = 2y_{\mathrm{tip}}/L = 0.05$ and

$$B_y^{\mathrm{eq}} = \frac{EI\vartheta/(\mu L) + B_0 \sin\vartheta}{\cos\vartheta} \simeq 17.43\ \mathrm{mT}. \tag{S140}$$

The 10 mT sinusoidal safety tests and the 2 mm tracking experiment thus use different input amplitudes, with the tracking limit set by the required equilibrium field.

On the full 64-DOF plant, the actual tip has an RMS tracking error of 33.734 μm and a maximum error of 47.069 μm. The final tip position is 2.03515 mm and the largest centerline displacement is 2.03838 mm. The applied field peaks at 17.75695 mT; saturation and governor intervention occur at 0% of saved times. Thus the feasible learned request passes through the filter unchanged, and the separate unfiltered run is identical. The input-work bound peaks at 9.05293 nJ, below the 22.66365 nJ cap, and the same continuous-time safety invariant applies. The minimum absolute input gain along this run is 0.0138678 m, so the division in (S139) does not approach a singularity. The physical power-balance residual is 3.53×10^{-20} J. This tracking result assumes access to the two-mode projection of the full state. Extending it to incomplete sensing requires a state observer and a corresponding tracking analysis.

A second run uses the CHNN identified by integral fitting of full-FE COMSOL displacement and velocity exports, with cosine multiplier 1.0223248822. Its complete gradient

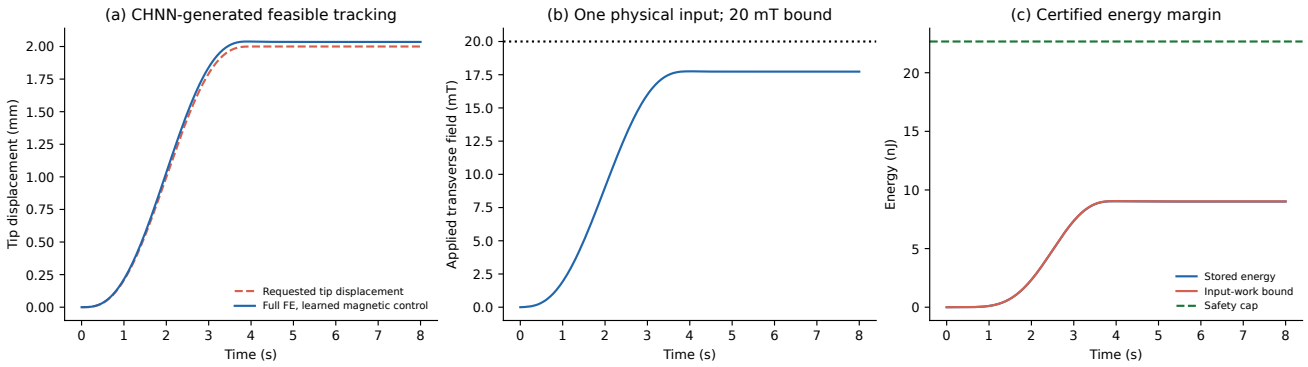


Figure S13. The fitted GC17 CHNN and identified drag generate a single bounded transverse magnetic input for the same full 32-element FE beam. (a) A feasible 2 mm tip request over 4 s and the actual full-FE tip motion. (b) The applied field remains below the declared 20 mT limit. (c) Stored energy and the actual-port work observer stay below the safety cap. The controller uses the exact first-two-mode projection available in this simulation; the safety observer uses full-model tip power. No modal actuation authority or unmeasured-state observer is assumed. The governor is inactive for this feasible tracking request; its active effect is tested separately in Figure S12.

and Hessian, including learned momentum quartics, generate the same 2 mm request on the same full-FE Python plant. For this conservative fit, a separate PSD drag identification on the existing Python-generated damped beam trajectories gives $\widehat{\gamma} = 0.0486617702$; this dissipative training source is distinct from the COMSOL conservative data. The tip RMSE is $33.726\ \mu$m, its maximum error is $47.065\ \mu$m, and the largest centerline displacement is 2.038365 mm. The applied field peaks at 17.756854 mT and the work observer at 9.052842 nJ. Saturation and governor intervention are again zero at the saved times, and the physical power-balance residual is 3.53×10^{-20} J. This COMSOL-trained model and the Python-trained GC17 model use the same physical plant. The full-state projection and actual-port measurement assumptions are unchanged, and the same continuum clearance guarantee applies.

Finally, the single uniform-field dipole supplies one boundary moment. In static equilibrium with no distributed transverse loading, $w_{ssss} = 0$ and the free-tip shear condition imply $w(s) = \vartheta s^2/(2L)$. This one-parameter family cannot realize an arbitrary S-shaped static centerline. The moving-base endpoint task in Section S7.6 instead follows an S-shaped tip history using axial base feed and magnetic bending. The governor preserves the corridor for feasible magnetic requests. Prescribing more general static shapes requires additional actuation authority, such as distributed magnetic moments or independently controlled loads.

S7.5 Finite initialization and port-error tolerances

The transverse action half-width 10^{-220} in Appendix S11 quantifies local invariant-torus measure. For operation, we instead construct a finite energy set and a robust input-work constraint on the same full beam. This construction accepts an ordinary finite-precision initial state, explicit sensing errors and bounded command errors. It preserves clearance when the safety filter changes the KAM-realizing request; exact quasiperiodic motion is then no longer asserted.

Physical object and a measurable initialization set. For this calculation the mathematical plant is the frozen, symmetric nodal 64×64 matrices M, K supplied with the independent FE comparison. All of their exported binary entries are treated as exact. This retains the complete physical energy

$$\begin{aligned} E(q,v) &= \tfrac{1}{2} v^{\mathsf{T}} M v + \tfrac{1}{2} q^{\mathsf{T}} K q + E_0(1-\cos\vartheta), \\ \dot{E} &= -\alpha v^{\mathsf{T}} M v + F v_w + \mu B \cos\vartheta\, v_\vartheta, \end{aligned} \tag{S141}$$

where $v = \dot{q}$, $\vartheta = e_\vartheta^{\mathsf{T}} q$, $v_w = e_w^{\mathsf{T}} v$, $v_\vartheta = e_\vartheta^{\mathsf{T}} v$, $E_0 = \mu B_0$, and F, B are the actual transverse tip force and field. The same two physical ports appear in (S109). The use of nodal matrices here includes their small diagonalization discrepancy relative to the frozen modal plant in Theorem S10; no exact conjugacy between these two finite-precision definitions is required for the work bound.

Let b_r list the four cubic Bernstein control rows on every Hermite element, including the fixed zero base coordinates, and define

$$C_w = \max_r b_r K^{-1} b_r^{\mathsf{T}}, \qquad E_c = \frac{c^2}{2C_w}, \qquad c = 4\,\mathrm{mm}. \tag{S142}$$

The Bernstein convex-hull property and Cauchy–Schwarz in the K metric give $\sup_s |w(s)| \le \sqrt{2C_w E}$ for every FE

state, including all higher modes. Thus $E < E_c$ is a whole-centreline clearance condition. Using exact Hermite coefficients $\ell/3 = 1/1200$ m, 256-bit ball matrix inverses and positive LDL pivots gives $C_w < 335.338810137\,\mathrm{m\,N^{-1}}$ and $E_c > 23.85646921\,\mathrm{nJ}$. This discrete certificate agrees with the continuum compliance while avoiding an unbudgeted matrix-rounding step.

Set $K_b = K + E_0 e_\vartheta e_\vartheta^{\mathsf{T}}$ and

$$Q(q,v) = \tfrac{1}{2} v^{\mathsf{T}} M v + \tfrac{1}{2} q^{\mathsf{T}} K_b q. \tag{S143}$$

For full-state estimates $(\widehat{q}, \widehat{v})$, require the initialization error to satisfy $Q(q-\widehat{q}, v-\widehat{v}) \le \mathcal{E}_{\mathrm{init}}$. Since $1-\cos\vartheta \le \vartheta^2/2$ and $\sqrt{Q}$ is a norm,

$$E(q,v) \le Q(q,v) \le \overline{E}_0 := \left[\sqrt{Q(\widehat{q},\widehat{v})} + \sqrt{\mathcal{E}_{\mathrm{init}}}\right]^2. \tag{S144}$$

This specifies a nonzero finite-dimensional admissible initialization region with nonempty interior, without deciding membership in an exact invariant torus. The energy-error bound is a full-state condition; tip sensing alone does not bound unobserved bending or kinetic energy.

Sensing and execution error envelopes. Let $u = (u_F, u_B)$ be the commanded ports and suppose that, throughout the specified interval,

$$\begin{aligned} F &= (1+g_F)u_F + e_F, \\ B &= (1+g_B)u_B + e_B, \\ |g_i| &\le \eta_i, \qquad |e_i| \le \epsilon_i. \end{aligned} \tag{S145}$$

Assume available estimates obey $|v_w - \widehat{v}_w| \le \delta_v$, $|\vartheta - \widehat{\vartheta}| \le \delta_\vartheta$ and $|v_\vartheta - \widehat{v}_\vartheta| \le \delta_\omega$. Define the estimated power coefficients and their error bounds by

$$\begin{aligned} \widehat{a}_F &= \widehat{v}_w, & \Delta_F &= \delta_v, \\ \widehat{a}_B &= \mu\cos\widehat{\vartheta}\,\widehat{v}_\vartheta, & \Delta_B &= \mu(\delta_\omega + |\widehat{v}_\vartheta|\delta_\vartheta). \end{aligned} \tag{S146}$$

The cosine is one-Lipschitz on the real line, hence $|a_i - \widehat{a}_i| \le \Delta_i$. Both sensing error and relative execution error are covered by the convex, positively homogeneous function

$$P^+(u) = \sum_{i\in\{F,B\}} \left\{\widehat{a}_i u_i + [\Delta_i + \eta_i(|\widehat{a}_i| + \Delta_i)]|u_i|\right\}. \tag{S147}$$

In particular, $P^+(0) = 0$. There is no division by a measured velocity, so a zero or incorrectly signed noisy velocity creates no singularity.

The independent additive errors need a separate energy allowance. For $\alpha > 0$, write $m_F = e_w^{\mathsf{T}} M^{-1} e_w$ and $m_\vartheta = e_\vartheta^{\mathsf{T}} M^{-1} e_\vartheta$, and put

$$p_e = \frac{(\epsilon_F\sqrt{m_F} + \mu\epsilon_B\sqrt{m_\vartheta})^2}{4\alpha}, \qquad R_T = T p_e. \tag{S148}$$

For time-dependent bounds one may instead integrate their corresponding $p_e(t)$. With $r = e_w e_F + \mu e_\vartheta e_B \cos\vartheta$, completion of the square gives

$$r^{\mathsf{T}} v - \alpha v^{\mathsf{T}} M v \le \frac{r^{\mathsf{T}} M^{-1} r}{4\alpha} \le p_e. \tag{S149}$$

All quantities in these equations use SI units: P^+, p_e are powers, and Q, E, R_T are energies. No dissipated energy needs to be measured.

Proposition S13. Robust finite-horizon work and clearance. *Fix $T > 0$ and $\overline{E}_0 < C := E_{\max} - R_T < E_c - R_T$. Initialize $W(0) = \overline{E}_0$ and integrate $\dot{W} = P^+(u)$. Clip a requested force and field to symmetric command limits, then apply $u = \sigma u_{\mathrm{req}}$, with $\sigma = 1$ if $P^+(u_{\mathrm{req}}) \le \kappa(C-W)$, and otherwise*

$$\sigma = \frac{\kappa(C-W)}{P^+(u_{\mathrm{req}})}, \qquad \kappa > 0. \tag{S150}$$

For solutions satisfying the mechanical power identity and the stated error envelopes, throughout $0 \le t \le T$,

$$\begin{aligned} E(t) &\le W(t) + t p_e \le C + R_T = E_{\max}, \\ \sup_s |w(s,t)| &\le \sqrt{2 C_w E_{\max}} < c. \end{aligned} \tag{S151}$$

Zero command remains feasible, even with nonzero sensing errors. To enforce actual actuator limits $F_{\lim}, B_{\lim}$, use command limits $(F_{\lim} - \epsilon_F)/(1+\eta_F)$ and $(B_{\lim} - \epsilon_B)/(1+\eta_B)$, respectively.

Proof. The two non-additive port powers are at most $P^+(u)$, by (S146) and (S145). Equation (S149) bounds the remaining power and damping together. Thus $\dot{E} \le \dot{W} + p_e$, and integration with (S144) proves $E \le W + t p_e$. Positive homogeneity makes the radial choice satisfy $\dot{W} \le \kappa(C-W)$; consequently $C - W(t) \ge (C - W(0))e^{-\kappa t} > 0$. The spatial compliance bound completes the proof. The amplitude limits follow directly from $|F| \le (1+\eta_F)|u_F| + \epsilon_F$ and its field counterpart. If additive errors vanish, $R_T = 0$ and the same argument holds for all time and for any $\alpha \ge 0$. A constant nonzero additive error allowance has the stated finite horizon; it is not an all-time KAM or clearance assertion.

Sampled sensing and delayed execution. The sensing bounds must cover the age of the data, not only sensor quantization. For a sample at t_k used for an interval of length h_k, let the maximum age including sensing latency be ℓ_k. If

valid bounds $|\dot{v}_w| \le A_w$, $|v_\vartheta| \le V_\vartheta$ and $|\dot{v}_\vartheta| \le A_\vartheta$ hold on that interval, replace

$$\delta_v \mapsto \delta_v + A_w \ell_k, \quad \delta_\vartheta \mapsto \delta_\vartheta + V_\vartheta \ell_k, \quad \delta_\omega \mapsto \delta_\omega + A_\vartheta \ell_k. \tag{S152}$$

Compute P_k^+ with this interval envelope. A fully discrete sufficient rule is to hold u_k such that $h_k P_k^+(u_k) \le C - W_k$ and set $W_{k+1} = W_k + h_k P_k^+(u_k)$. Linear interpolation of W proves the same within-interval energy bound; $u_k = 0$ remains feasible. This rule does not treat a value at one sample as an interval guarantee.

Similarly, if a continuously specified command has a verified slew bound $|\dot{u}_i| \le S_i$ and the applied copy is delayed by at most d_i, including a zero-order-hold age when applicable, it suffices to use

$$\epsilon_i \ge \epsilon_{i,0} + (1 + \eta_i) S_i d_i \tag{S153}$$

in the additive reserve. This is an explicit joint delay/slew budget. The radial governor alone does not impose a slew limit. A controller that also enforces slew must verify its feasibility, or use the discrete interval-power rule above. The numerical experiment below uses continuous bounded port estimates and the declared direct execution-error envelope; it does not identify a hardware latency or an unmeasured acceleration bound.

A nonzero SI budget next to the finite-amplitude reference. The reference $(\widehat{q}, \widehat{v})$ is the frozen nodal reconstruction of phase $(0, 0)$ of the computed Fourier candidate for the integral-COMSOL fit, with the learned velocity h_Y, rather than an assumed exact torus point. Its quadratic energy is below 21.45855219 nJ. For $\mathcal{E}_{\mathrm{init}} = 5$ pJ, ball evaluation of (S144) gives $\overline{E}_0 < 22.11866366$ nJ. This full-state ellipsoid implies, among other coordinate bounds, an initial tip-displacement error below 56.006 μm and tip-angle error below 1.19851 mrad. It also implies mass-weighted RMS velocity error below 0.316981 mm/s, where the squared RMS is $\delta v^{\mathsf{T}} M \delta v / (\rho A_c L)$. These are projections of one energy ellipsoid; their endpoints cannot be independently combined as a Cartesian error box.

The declared actual limits are 20 mT and 20 μN; the latter corresponds to a gradient norm 1 T/m in the ideal local magnetic realization. The command bounds reserve the relative and additive errors before clipping. Matrix ball arithmetic, rather than a two-mode estimate of unobserved velocity, gives $m_F < 4.452635925\,10^6$ and $m_\vartheta < 3.940922387\,10^{13}$ in their respective SI inverse-mass units, which establish the work reserve in the table.

Table S12. One sufficient operational budget for the full nodal FE plant. Bounds are deterministic hypotheses; they are not measured sensor specifications.

Quantity	Value or outward bound
Horizon T	4 s
Energy cap $E_{\max}$	22.66 nJ
Initialization error $\mathcal{E}_{\mathrm{init}}$	5 pJ
Angle / angular-rate error	1 mrad / 0.01 rad/s
Tip-velocity error	0.1 mm/s
Relative force and field execution error	1% each
Additive force / field error	0.5 nN / 0.05 μT
Additive work reserve R_T	< 0.09677917 nJ
Available commanded-work cap C	> 22.56322083 nJ
Initial commanded-work margin	> 0.44455717 nJ
Whole-centreline displacement bound	< 3.898405 mm
Minimum net wall clearance	> 0.101595 mm

Full-model numerical check. The reproducibility directory `reviewer_revision_20261008/robustness` contains the complete input matrices, learned coefficients, finite initial reference, original-file hashes, code and trajectories. The physical integrator retains $\Phi^{\mathsf{T}} M \Phi$ and $\Phi^{\mathsf{T}} K_b \Phi$ in full, so it integrates the frozen nodal plant in a well-scaled modal basis instead of discarding diagonalization residuals. The requesting controller uses the same fitted h_Y, h_{YX}, h_{YY} and velocity-to-momentum inverse as (S108), with an additional bounded low-state perturbation. Independent deterministic sinusoids supply the stated port-sensor, gain and additive execution errors. The full-state initialization error is verified by direct Arb matrix products, including the modal-to-nodal transformation.

The initialized energy is 21.743724 nJ, with a 46.354 μm tip position error and a 0.347654 mm/s tip velocity error; its combined initial error energy is below 4.951 pJ. Over 4 s the saved-time Bernstein centreline bound is below 1.944145 mm and the physical energy is below 21.744899 nJ. The work filter is active at 97.8011% of the 2001 saved times, illustrating its conservative use of work without crediting dissipated energy. Its activation changes the desired Hamiltonian trajectory. Clearance follows from Proposition S13, while invariant-torus persistence remains the separate conservative and ideal-feedback result.

The Radau run uses relative tolerance $2\,10^{-8}$ and componentwise absolute tolerances $2\,10^{-13}$ for modal positions, $2\,10^{-11}$ for modal velocities and $2\,10^{-13}$ for normalized energy states. Simultaneously reducing them tenfold changes the saved tip trajectory by less than 0.010 μm and the energy by less than $3.2\,10^{-15}$ J; the refined energy-ledger residual is below $3\,10^{-16}$ J. These are numerical implementation checks, not a validated enclosure of the integrated trajectory. The continuous-time clearance statement uses the analytic work proof and its independently validated initialization and parameter bounds.

S7.6 A physically actuated S-shaped endpoint path

The moving clamped base permitted by the distributed formulation supplies an axial feed coordinate $b(t)$, while the same single transverse magnetic field bends the catheter. In the Cartesian small-slope model, the centreline is

$$\begin{aligned} r(s,t) &= \big(b(t)+s,\, w(s,t)\big), \\ r_{\mathrm{tip}}(t) &= \big(b(t)+L,\, w(L,t)\big). \end{aligned} \tag{S154}$$

The axial feed and magnetic bending together prescribe the endpoint history. The material length remains 80 mm throughout; advancing a varying length through a guide is a different boundary-value problem.

The base starts at rest. Over the first second its speed is $\dot b = v_b(10\xi^3 - 15\xi^4 + 6\xi^5)$, where $\xi = t/(1\,\mathrm{s})$ and $v_b = 4\,\mathrm{mm\,s^{-1}}$; afterwards $\dot b = v_b$. The transverse request is zero until 2 s. During the subsequent 12 s stroke, put $\sigma = (t - 2\,\mathrm{s})/(12\,\mathrm{s})$ and prescribe

$$y_r(t) = \frac{8A}{3\sqrt{3}}\left(\frac{1}{2}\sin 2\pi\sigma - \frac{1}{4}\sin 4\pi\sigma\right), \quad A = 1.5\,\mathrm{mm}. \tag{S155}$$

Outside this interval $y_r = 0$. Its position, velocity and acceleration join continuously to zero, and its two lateral extrema are $\pm A$. The resulting S-shaped endpoint stroke advances 48 mm axially. The base acceleration is at most $7.5\,\mathrm{mm\,s^{-2}}$ and ends before any bending command is applied. Consequently the beam is exactly straight during axial acceleration and the base has zero acceleration throughout the bending stroke. There is no axial fluid load, guide friction, or imposed axial prestress in this test.

Base work and clearance. Let $m_b = \rho A_c L$ denote the rod mass, with A_c its cross-sectional area. Orthogonality of the axial and transverse Cartesian velocities gives

$$E_{\mathrm{total}} = E_\perp + \frac{1}{2}m_b\dot b^2, \qquad P_b = F_b\dot b = m_b\ddot b\dot b. \tag{S156}$$

Here $E_\perp$ is the same transverse kinetic, bending and static-field energy used in the clearance theorem. The chosen timing also avoids transverse geometric-stiffness effects from an accelerating bent beam: the axial inertial load acts only while $w = 0$. During bending, axial translation is force-free in the stated model. Thus

$$\dot E_{\mathrm{total}} = \mu B_y \cos\vartheta\,\dot\vartheta - \alpha\dot q^{\mathsf T} M\dot q + P_b. \tag{S157}$$

The base work is $W_b = m_b\dot b^2/2$ and is accounted for separately from the magnetic input budget. The straight, open-ended corridor is invariant under axial translation. Its lateral noncollision proof therefore applies to $E_\perp$ and the actual magnetic port, while the separate base-energy balance is retained. For the same 4 mm net clearance and $0.95E_c$ transverse work cap, the continuous-time centreline bound remains 3.898718 mm. This statement would require additional work and geometry terms for curved walls, axial contact, variable insertion length or axial drag.

Learned control and comparisons. The conservative readout is the integral fit to full-FE COMSOL position–velocity data in Section S5.5. It has cosine multiplier 1.0223248822 and includes all fitted quartic terms; its invariant torus is certified independently. The conservative coefficients and the nonnegative drag readout are held fixed throughout this experiment. The controller evaluates its drift and magnetic input gain from the complete learned gradient and Hessian, as in the preceding tracking construction. For this experiment the feedback error uses the actual full-beam tip position and velocity; the first two modal projections supply the learned drift and gain. The request is clipped to $|B_y| \le 20$ mT and passed through the same physical input-work governor. Every plant run integrates all 64 FE degrees of freedom. The imposed base motion has the same force and speed budget in every comparison.

The known-physics static inverse has the smallest tracking error in Table S13 for this slow, weakly nonlinear nominal trajectory. The learned controller also follows the S-shaped endpoint path, with accuracy depending on the measured output: full-tip feedback reduces the CHNN error relative to its two-mode reconstruction while using the same fitted energy and physical input.

The learned controller uses a peak field of 12.9177 mT and a sampled peak field slew of $10.4268\,\mathrm{mT\,s^{-1}}$. Its largest centreline displacement is 1.48892 mm, leaving 2.51108 mm of net lateral clearance at the saved times; the largest slope is 0.03718 rad. The recorded work-based energy estimate peaks at 4.84997 nJ, below the 22.66365 nJ cap, and neither field saturation nor governor intervention occurs. The base's maximum kinetic energy is 0.796205 nJ, its peak ideal inertial force is $0.746442\,\mu$N, and its peak power is 1.90256 nW. Numerical total-energy balance includes this base work and has maximum residual 2.44×10^{-18} J. Tightening integration tolerances by a factor of ten and halving the maximum step changes the tip by at most 3.14×10^{-12} m and the commanded field by at most 1.47×10^{-10} T at saved times.

Independent COMSOL field replay. An additional COMSOL solve integrates the same full beam for 16 s using the saved learned-controller field, with piecewise-cubic

Table S13. S-path tracking on the full FE plant. RMS errors use the 12 s stroke; maxima use the full 16 s simulation. All baselines share the same trajectory, physical input and field bound.

Controller	RMS (μm)	Max. (μm)
CHNN, measured tip feedback	6.419	12.907
Quadratic model, measured tip feedback	6.207	12.476
Known-physics static inverse feedforward	4.124	8.005
CHNN, modal-output feedback	25.308	41.544

field interpolation, generalized-alpha integration, relative tolerance 10^{-8} and maximum step 0.002 s. The prescribed field is the controller output from the preceding simulation. Its tip differs from the Python result by 1.312×10^{-10} m RMS and at most 4.038×10^{-10} m, and its S-stroke tracking RMSE is $6.419\,\mu$m. COMSOL independently integrates the mechanical response to this prescribed command; the controller is not recomputed online in this replay. Axial translation is reconstructed using the same analytic base protocol. Power balance reconstructed by trapezoidal integration of the 0.004 s output samples has a 1.244×10^{-14} J residual; this diagnostic is distinguished from the much tighter internal Python work-state integration and is not used as a rigorous solver-error bound.

S8 Structure-preserving time integration

For a real smooth Hamiltonian, implicit midpoint is

$$x_{k+1} = x_k + hJ\nabla H\big((x_k + x_{k+1})/2\big). \tag{S158}$$

Let $A = JH''(\bar{x})$. Differentiation gives $D\Phi_h = (I - hA/2)^{-1}(I + hA/2)$. Since $A^{\mathsf{T}}J + JA = 0$, this Cayley transform satisfies $D\Phi_h^{\mathsf{T}} J D\Phi_h = J$ whenever the implicit equation defines a differentiable local map. The proof assumes the nonlinear step is solved exactly; residual tolerances add a numerical defect. Explicit RK4 is not symplectic in general. For the separable physical Hamiltonian, the kick–drift–kick Störmer–Verlet map is also symplectic, because each substep is the exact flow of the kinetic or potential Hamiltonian. A general learned momentum-dependent quartic no longer admits that separable split, so the same formula cannot be reused without justification.

Energy error and symplecticity are distinct diagnostics. Finite-time energy plots are evaluated against the specified reference Hamiltonian and accompanied by a step-size check. A small integration error does not certify a Diophantine inequality, nor does a positive finite-time Lyapunov estimate alone establish asymptotic chaos. These facts determine the labels used in the numerical figures.

S9 The fourth-order central generating-function scheme

Let $f = J\nabla H$, $A = H''$ and define a central generating function

$$S_h(x) = hH(x) + \frac{h^3}{24} f(x)^{\mathsf{T}} A(x) f(x). \tag{S159}$$

The map is specified implicitly by

$$x_+ - x_- = J\nabla S_h\big((x_- + x_+)/2\big). \tag{S160}$$

The central generating-function construction and fourth-order formula are classical (Feng et al. 1989, Theorem 24, p. 90; formula on p. 95); geometric integration is developed systematically by Hairer et al. (2006). Here every derivative is computed from the same learned scalar Hamiltonian. The identity

$$J\nabla(f^{\mathsf{T}} A f) = f''[f,f] - 2(f')^2 f \tag{S161}$$

follows by differentiating both occurrences of f, using $f' = JA$ and $J^{\mathsf{T}} = -J$. It gives the displacement implemented in the reproducibility code:

$$g_h = hf + \frac{h^3}{24}\big(f''[f,f] - 2(f')^2 f\big). \tag{S162}$$

The Hessian S_h'' is symmetric, so the midpoint Cayley-transform argument in Supplement S8 proves exact symplecticity of the locally defined map. Since $S_{-h} = -S_h$, interchanging its endpoints proves reversibility.

To verify order, expand an exact orbit about its temporal midpoint x_0. Its endpoint midpoint m and endpoint difference satisfy

$$m = x_0 + \tfrac{1}{8}h^2 f'(x_0) f(x_0) + O(h^4), \tag{S163}$$

$$x_+ - x_- = hf(x_0) + \tfrac{1}{24}h^3\big(f''[f,f] + (f')^2 f\big)(x_0) + O(h^5). \tag{S164}$$

Substituting $x_0 = m - \tfrac{1}{8}h^2 f'(m) f(m) + O(h^4)$ gives $x_+ - x_- = g_h(m) + O(h^5)$, proving fourth order under the required smoothness and local solvability assumptions.

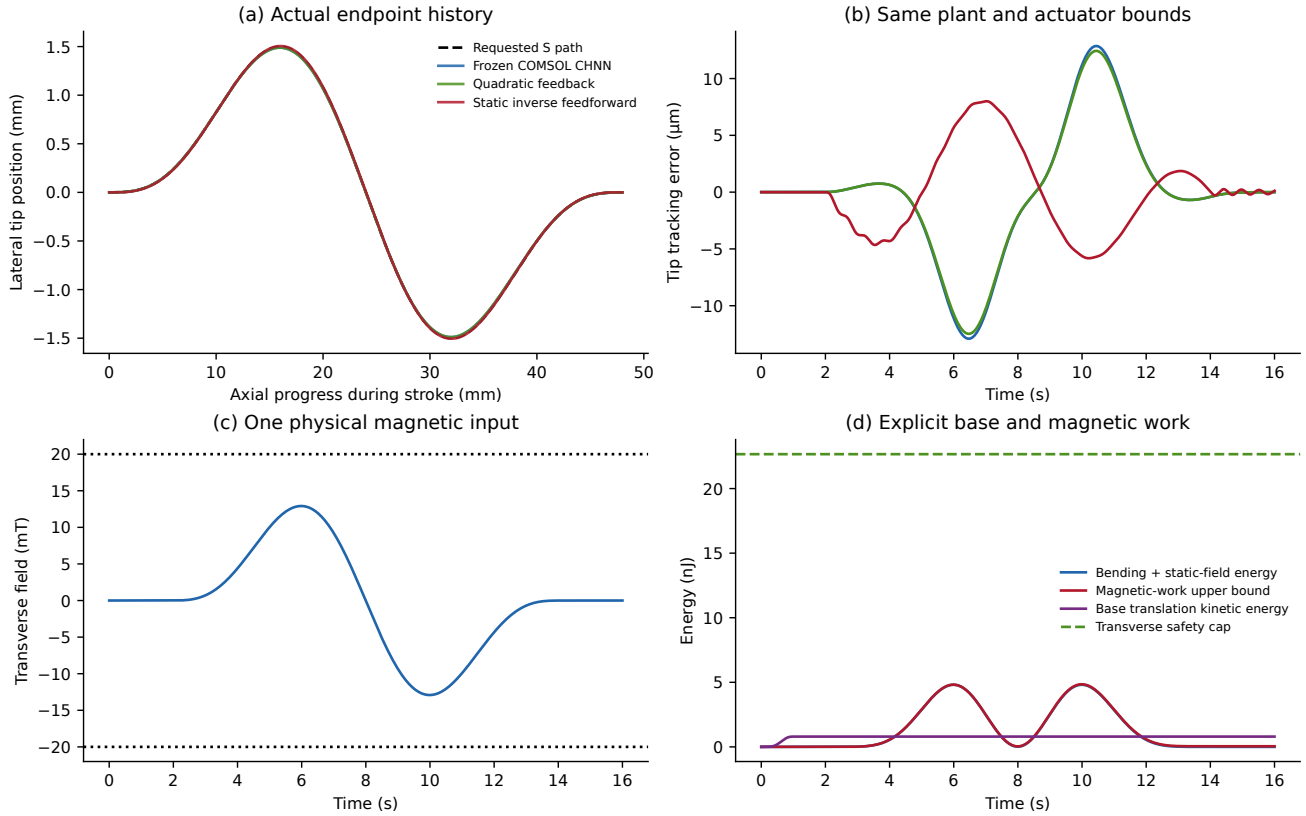


Figure S14. A realized S-shaped endpoint path generated by axial base feed and one transverse magnetic input. The controller uses the fitted COMSOL-trained CHNN on the full FE plant. Axial translation work is explicitly included in the energy balance. The matched controls reveal that known-physics static inverse feedforward is more accurate on this particular slow nominal path.

S9.1 Integration accuracy for the learned magnetic energy

We implement (S160) for the Python-trained GC17 Hamiltonian, retaining its complete cosine and all mixed and momentum-dependent quartics. Polynomial and trigonometric derivatives through order four are evaluated analytically from the fitted energy; they supply both the central displacement and its Newton Jacobian. This test closes all forcing and damping ports. Starting at $(X_1, X_2, Y_1, Y_2) = (0.1, 0.06, 0, 0.02)$, we integrate to $\tau = 10$ and compare against DOP853 with relative and absolute tolerances 3×10^{-14} and 3×10^{-16}. Tightening the reference tolerances from 2×10^{-13} and 2×10^{-15} changes the sampled reference by at most 8.32×10^{-14} per state component. The endpoint errors in Table S14 approach fourth order for Central4 and second order for implicit midpoint.

Table S14. Conservative integration of the GC17 energy. Errors are Euclidean endpoint errors in the balanced dimensionless state; orders compare successive step halvings.

h	Midpoint error	Order	Central4 error	Order
0.08	7.1351×10^{-2}	—	1.8346×10^{-3}	—
0.04	1.9383×10^{-2}	1.8802	1.1784×10^{-4}	3.9606
0.02	4.8966×10^{-3}	1.9849	7.4149×10^{-6}	3.9902
0.01	1.2265×10^{-3}	1.9972	4.6422×10^{-7}	3.9976

The largest implicit-step residual in the infinity norm is 2.00×10^{-15}, with at most three Newton evaluations per step. For each tested step size, differentiating the converged implicit equation gives its one-step tangent map $D\Phi_h$. The largest measured defect $\|D\Phi_h^{\mathsf{T}} J D\Phi_h - J\|_\infty$ is 3.34×10^{-16} for Central4 and 2.24×10^{-16} for midpoint. An independent fourth-order central-difference evaluation of the solved step map agrees with the implicit tangent to 1.39×10^{-12}; its largest symplectic defect is 2.75×10^{-12}. Thus both methods preserve the symplectic structure to the measured implicit-solve and arithmetic accuracy, while their convergence orders differ. The test assesses numerical integration of the complete momentum-dependent Hamiltonian; the KAM statement follows separately from its analytic certificate.

S10 Complex-gradient descent with a strict descent margin

Theorem 3 in the published version of Zhang et al. (2022) motivates complex-gradient optimization, but a stationary-point conclusion requires a strict descent margin. The following convention-explicit statement also applies to a smooth CHNN regression objective.

Proposition S14. Complex descent. *Let $\mathcal{L} : \mathbb{C}^P \simeq \mathbb{R}^{2P} \to \mathbb{R}$ be bounded below and have an L-Lipschitz real gradient on every segment traversed by the iterates. For $w = a + ib$, define $G(w) = \nabla_a \mathcal{L} + i\nabla_b \mathcal{L} = 2\partial\mathcal{L}/\partial\bar{w}$. Choose a constant $\rho \in \mathbb{C}$ with $\kappa = \operatorname{Re}\rho - (L/2)|\rho|^2 > 0$, and update $w_{t+1} = w_t - \rho G(w_t)$. Then*

$$\mathcal{L}(w_{t+1}) \le \mathcal{L}(w_t) - \kappa\|G(w_t)\|_2^2, \qquad \lim_{t\to\infty} \|G(w_t)\|_2 = 0,$$

and

$$\min_{0\le t<T} \|G(w_t)\|_2^2 \le \frac{\mathcal{L}(w_0) - \inf \mathcal{L}}{\kappa T}.$$

Proof. Under realification, multiplication by $\rho = u + iv$ is $uI + vJ_c$, with $J_c^\top = -J_c$ and $J_c^\top J_c = I$. Hence the update has inner product $-u\|G\|^2$ with the real gradient and squared norm $|\rho|^2\|G\|^2$. The real descent lemma gives the first inequality. Summing it and using the lower bound for $\mathcal{L}$ gives $\kappa\sum_{t\ge0}\|G(w_t)\|^2 \le \mathcal{L}(w_0) - \inf\mathcal{L}$. Square summability proves convergence of the gradient norm and the minimum bound. The result asserts convergence to stationarity in gradient norm, not convergence to a global minimum or to a particular parameter vector.

For a real step this condition is $0 < \rho < 2/L$ under the stated convention. The case $\rho = 0$, allowed by a non-strict printed step-size interval, cannot imply gradient convergence. Nonsmooth complex activations also need separate treatment; a smooth polynomial CHNN satisfies the differentiability requirements, with a Lipschitz gradient bound on any bounded iterate region. This proposition concerns smooth gradient descent; the experimental readouts are fitted by linear least squares.

S11 An explicit local KAM measure estimate

S11.1 A fixed-domain bound and its scope

This appendix quantifies a sufficient local measure bound for the Hamiltonian identified from COMSOL integral observations. The construction uses the exact finite-amplitude torus and bounds the complete transformed Hamiltonian, including the action-domain remainder. The resulting width illustrates the conservatism of the explicit constants in Chierchia and Koudjinan (2021).

Table S15. Outward bounds for the explicit finite-amplitude tube. The smallness quotient is the tested perturbation divided by the sufficient threshold; failure of that test does not imply chaos.

Quantity	Bound or value
Initial chart action radius	10^{-3}
Quadratic-normalization radius	10^{-17}
Certified tube half-width w	10^{-220}
$\|\overline{A}\|_\infty$, $\|\overline{A}^{-1}\|_\infty$	3, 5100
Smallness quotient at $w = 10^{-220}$	< 0.407586
Same quotient at $w = 10^{-219}$	> 4.0758
Certified fraction in U_w	$> 96.6938\%$

Theorem S15. Explicit finite-amplitude tubular measure. *Let h be the fitted Hamiltonian certified in Section S4.5, and let K_∞ be its exact invariant torus. There is an explicitly constructed analytic symplectic chart $\Xi = \Psi \circ \Phi_\chi$, specified in Sections S11.2–S11.3, that fixes this torus at zero action. For the fixed domain*

$$U_w = \Xi\big(\mathbb{T}^2_{2\pi} \times (-w, w)^2\big), \qquad w = 10^{-220}, \tag{S165}$$

the KAM fraction satisfies

$$f_{\mathrm{KAM}}(U_w) \ge 0.9669383. \tag{S166}$$

The centre torus is the previously certified motion with the 1.8977904 *mm tip witness and the* 1.8977962 *mm all-phase centreline bound. The width w is a dimensionless action half-width, not a displacement amplitude.*

Proof. The explicit chart and a convergent quadratic normalization give

$$h \circ \Xi = e + \omega \cdot I + \tfrac{1}{2} I^\top \overline{A} I + R_3, \qquad \omega = 2\pi\varpi.$$

The outward bounds are $\|\overline{A}\|_\infty \le 3$, $\|\overline{A}^{-1}\|_\infty \le 5100$ and $\|R_3\|_{2w,1/128} \le 4 \times 10^{51} w^3$. Use the global KAM theorem with real action radius w, complex radius $2w$, final angle strip $1/256$, frequency exponent $\tau = 2$, boundary allowance $\rho = w/500$ and $\alpha = w/(1000 \times 5100)$. The dimensionless smallness quantity is at most 3.1212×10^{-155}, below the theorem threshold 7.6577×10^{-155}. The geometric estimate in Section S11.4 gives

$$\begin{aligned} 1 - f_{\mathrm{KAM}}(U_w) &\le \frac{25}{16}\left(\frac{4\rho}{w} + \frac{8\alpha L\zeta(2)}{w}\right) \\ &= \frac{25}{16}\,(0.008 + 0.008\zeta(2)) < 0.0330617, \end{aligned} \tag{S167}$$

where $L = 5100$. The two terms account for the boundary layer and all integer resonance strips, respectively. Symplecticity transfers this relative measure to U_w without a density correction.

The width in Table S15 quantifies the conservatism of this general theorem. The domain is centred on the exact millimetre-amplitude torus, but its thickness is far below any operational resolution. In particular, the computed Fourier approximation has a certified canonical-embedding error upper bound below 2.1×10^{-41}. This available guarantee cannot resolve membership in a tube whose transverse distance is bounded by $41w = 4.1 \times 10^{-219}$; it is not an estimate of the actual embedding error. The domain is mathematically specified through K_∞; this certificate does not permit an experimental initial state to be assigned to that tube from the finite-precision approximation. It supplies a concrete fixed-domain measure result while leaving the much wider operating-domain question to a separate calculation. Section S5.6 reports that wider-domain numerical diagnostic, with its own error checks and interpretation.

S11.2 An explicit symplectic tube at the certified physical amplitude

Let K_∞ be the exact torus supplied by the integral-COMSOL certificate, with its phase normalization fixed by that construction. Write $t = 2\pi\theta$, $k(t) = K_\infty(t/(2\pi))$, and $\omega = 2\pi\varpi$. Thus $D_\omega k = X_{h_\eta}(k)$, and the already verified Diophantine inequality becomes

$$|\omega \cdot \ell| \geq \gamma_0 |\ell|_1^{-3}, \qquad \gamma_0 = 2\pi \frac{9}{1750}, \quad 0 \neq \ell \in \mathbb{Z}^2. \tag{S168}$$

The exponent 3 here concerns this fixed central torus. The family measure estimate below uses the separate exponent $\tau = 2$.

For each coordinate plane, define

$$\begin{aligned} j_a(t) &= \tfrac{1}{2}(k_{X_a}^2 + k_{Y_a}^2), \\ F_a(t) &= t_a + \frac{\log(1+u_{a,+}) - \log(1+u_{a,-})}{2i}, \\ u_{a,\pm}(t) &= 10(k_{X_a} \mp i k_{Y_a}) e^{\mp i t_a} - 1. \end{aligned} \tag{S169}$$

The logarithms use the branches at 1. On the real torus, F_a is the phase of $k_{X_a} - ik_{Y_a}$ and $j_a > 0$. Set $B(t) = DF(t)^{-\mathsf{T}}$ and define

$$\begin{aligned} J(t,I) &= j(t) + B(t)I, \\ \Psi_{X_a}(t,I) &= \sqrt{2J_a(t,I)}\cos F_a(t), \\ \Psi_{Y_a}(t,I) &= -\sqrt{2J_a(t,I)}\sin F_a(t). \end{aligned} \tag{S170}$$

Equivalently, each pair (k_{X_a}, k_{Y_a}) is multiplied by $\sqrt{1 + (BI)_a/j_a}$; this expression is convenient for interval evaluation and fixes the square-root branches.

This is an exact symplectic chart. Indeed, polar coordinates have canonical one-form $J \cdot dF$, and

$$\Psi^*(J \cdot dF) = j(t) \cdot DF(t)dt + I \cdot dt.$$

The first term is closed because k is Lagrangian. Taking the exterior derivative gives $\sum_a dI_a \wedge dt_a$. This argument requires closedness, not exactness, of the first term. Since DF is uniformly close to the identity, its real lift is a global diffeomorphism of $\mathbb{T}^2$; positive J then makes Ψ injective on its real tube. Moreover $\Psi(t,0) = k(t)$.

The invariant-torus estimates and outward-rounded interval arithmetic give the following upper bounds:

$$\begin{gathered} \|k - k_0\|_{1/16} < 7\,10^{-6}, \\ k_0 = 0.1(\cos t_1, \cos t_2, -\sin t_1, -\sin t_2), \\ \|u_\pm\|_{1/16} < 1/5000, \qquad \|j - 0.005\mathbf{1}\|_{1/16} < 3\,10^{-6}, \\ \|F - t\|_{1/16} < 0.000201, \\ \max\{\|DF - \mathrm{Id}\|_\infty, \|DF - \mathrm{Id}\|_1\}_{1/32} < 0.013, \\ \|B\|_{1/32,\infty} < 1.014. \end{gathered} \tag{S171}$$

Here a subscript is the complex angular strip width in radians; vector norms are maximum norms and matrix norms are induced norms. The first bound includes the a posteriori correction to the numerical embedding. For example, Cauchy's estimate bounds each entry of $D(F-t)$ by $0.000201/(1/32)$, so both row and column sums are below 0.013. For $R_0 = 10^{-3}$, $|I|_\infty < R_0$ therefore implies

$$\begin{gathered} |(BI)_a/j_a| < \frac{1.014\,10^{-3}}{0.005 - 3\,10^{-6}} < \tfrac{1}{4}, \\ \|\Psi\|_{1/32,R_0} < 0.12. \end{gathered}$$

All logarithms, square roots and inverses above are thus holomorphic on the declared domain.

Write $h_\eta = h_2 + h_{\mathrm{nl}}$. The full cosine and the exact exported quartic coefficients give

$$\|h_{\mathrm{nl}} \circ \Psi\|_{1/32,R_0} < 10^{-6}, \qquad \|h_\eta \circ \Psi\|_{1/32,R_0} < 0.05. \tag{S172}$$

The second estimate uses the identity $h_2 \circ \Psi = \nu \cdot (j + BI)$, whose norm is less than $7.153(0.005003 + 0.001014) < 0.044$. Invariance and symplecticity give the exact expansion

$$h_\eta \circ \Psi = e + \omega \cdot I + \tfrac{1}{2} I^{\mathsf{T}} A(t) I + O(|I|^3). \tag{S173}$$

Only $h_{\mathrm{nl}} \circ \Psi$ contributes to A: Cauchy bounds on its diagonal and off-diagonal second derivatives give $\|A\|_{1/32,\infty} \leq 3$.

The averaged matrix $\overline{A}$ is nonsingular. To check the normalization, the tangent frame with radian angles is the certified cycle-angle frame divided by 2π; a symplectically dual normal frame is multiplied by 2π. Replacing that normal frame by $\partial_I \Psi(t,0)$ adds a symmetric tangential shear. Its contribution to the torsion is a directional derivative with zero angular average. Consequently

$$\overline{A} = (2\pi)^2 \langle T_\infty \rangle, \qquad \|\overline{A}^{-1}\|_\infty \le \frac{2\,10^5}{(2\pi)^2} < 5100. \tag{S174}$$

The inverse-torsion bound applies to the exact final torus through the uniform a posteriori iteration.

S11.3 A complete analytic remainder on a fixed tube

Let S be the symmetric, zero-average solution of $D_\omega S = A - \overline{A}$. Its Fourier coefficients for $\ell \neq 0$ are $S_\ell = A_\ell/(i\omega\cdot\ell)$. Equation (S168) and the angular strip loss $\delta = 1/64$ imply

$$\begin{aligned} \|S\|_{1/64,\infty} &\le \frac{3}{\gamma_0} 4 \sum_{m\ge 1} m^4 e^{-m/64} \\ &\le \frac{288 e^{1/64} 64^5}{\gamma_0} < 10^{13} =: T. \end{aligned} \tag{S175}$$

There are $4m$ integer vectors of one-norm m. The second inequality uses $\sum_{m\ge1} m^4 e^{-\delta m} \le e^\delta \Gamma(5)\delta^{-5}$, obtained by integrating on each interval $[m, m+1]$.

Use the exact time-one flow Φ_χ of $\chi(t,I) = \frac{1}{2} I^{\mathsf{T}} S(t) I$, not a truncated Lie series. Fix $R_1 = 10^{-17}$. On the intermediate strip $3/256$, Cauchy's estimate from the strip $1/64$ gives $\|\partial_{t_a} S\|_\infty \le 256 T$. As long as $|I|_\infty < 2R_1$, the flow satisfies

$$\begin{aligned} |\dot t|_\infty &\le 2TR_1 = 2\,10^{-4} < 1/256, \\ |\dot I|_\infty &\le 1024 T R_1^2 = 0.1024 R_1 < R_1. \end{aligned}$$

These strict bounds close the continuation argument from the initial strip $1/128$ and action polydisc R_1: throughout the unit time interval, the angle stays inside $3/256$ and the action inside $2R_1 < R_0$. The flow is holomorphic on the initial domain and symplectic on its real part.

With the pullback convention $L_\chi H = \{H,\chi\}$, $\{\omega\cdot I,\chi\} = -D_\omega \chi$. Hence the exact transformed Hamiltonian has the form

$$h_\eta \circ \Psi \circ \Phi_\chi = e + \omega\cdot I + \tfrac{1}{2} I^{\mathsf{T}} \overline{A} I + \mathcal{R}_3, \tag{S176}$$

where $\mathcal{R}_3$ starts at total action degree three. The complete left-hand side remains bounded by 0.05 on the strip $1/128$ and radius R_1. Cauchy's coefficient estimate in the two action variables therefore yields, for $0 < r \le R_1/2$,

$$\begin{aligned} \|\mathcal{R}_3\|_{1/128,r} &\le 0.05 \sum_{m\ge3} (m+1)(r/R_1)^m \\ &\le \frac{0.5}{R_1^3} r^3 = 5\,10^{50} r^3. \end{aligned} \tag{S177}$$

This bound includes the full magnetic cosine, all fitted quartics, the exact chart and the exact normalization flow.

S11.4 A quantitative measure bound with all resonances included

The action-domain dependence of KAM measure estimates is central in Biasco and Chierchia (2018). We use the fully quantified formulation of Chierchia and Koudjinan (2021, Theorems 1 and 3, Eqs. (4), (9), (13)–(14)). Consider a Hamiltonian $K(I) + \mathcal{R}(I,t)$ on $B_R(I_c)\times\mathbb{T}^2_s$, with $K = e + \omega\cdot I + \frac{1}{2} I^{\mathsf{T}} A I$, real symmetric invertible A, and $\|\mathcal{R}\|_{R,s} \le E$. Let $M \ge \|A\|_\infty$, $L \ge \|A^{-1}\|_\infty$, $\vartheta = ML$, and $\epsilon = ME/\alpha^2$. For $0<\rho<r\le R$, $0<s_*<s\le 1$, and $\tau > 1$, the sufficient conditions are

$$\alpha \le c_0 \rho/L, \qquad \epsilon \le \epsilon_* := \frac{(s-s_*)^{7(\tau+1)+10}}{c_* \vartheta^6}. \tag{S178}$$

Here c_0, c_* are exactly the constants in their Appendix A, Remark A.4, Eq. (A.38), evaluated for $d=2$ and the chosen τ. Upper bounds on M, L, E, c_*, and a lower bound on c_0, are used in the appropriate directions. The normalization is $H = K + \varepsilon P$ with $\varepsilon\|P\| \le E$.

Take the centre set in that theorem to be the singleton $\{I_c\}$. The real domain is precisely $D = B_r(I_c)$, the inner domain is $B_{r-\rho}(I_c)$, and the boundary layer in Theorem 3 is $B_{r+\rho}(I_c)\setminus B_{r-\rho}(I_c)$, of area $16 r\rho$. Let $\mathcal{K} \subset D\times\mathbb{T}^2$ be the certified KAM set. The Lipschitz estimate in their Eq. (9) is at most $1/8$, so the factor multiplying the bad action measure in Eq. (14) is bounded by $(25/16)(2\pi)^2$.

For completeness the remaining estimate includes every integer resonance. For $m = |\ell|_1$, the strip

$$|\ell\cdot(\omega + AI)| < \alpha/m^\tau$$

has intersection with D of area at most $4r\alpha/(m^\tau \|A\ell\|_\infty)$. Since $\|A\ell\|_\infty \ge m/(2L)$ and there are $4m$ vectors on the shell, summing over all shells gives

$$\operatorname{meas}(R_\alpha) \le 32 r\alpha L \zeta(\tau).$$

Counting both signs of each resonance gives a conservative upper bound. Dividing Theorem 3's bound by $\operatorname{meas}(D\times$

$\mathbb{T}^2) = 4r^2(2\pi)^2$ proves

$$\frac{\mathrm{meas}(\mathcal{K})}{4r^2(2\pi)^2} \geq 1 - \frac{25}{16}\left(\frac{4\rho}{r} + \frac{8\alpha L\zeta(\tau)}{r}\right). \tag{S179}$$

The estimate therefore includes resonances of every integer order.

Apply this result to (S176), with

$$\begin{gathered} I_c = 0, \quad r = w, \quad R = 2w, \\ s = 1/128, \quad s_* = 1/256, \quad \tau = 2, \\ M = 3, \quad L = 5100, \quad \vartheta = 15300, \\ E = 4\,10^{51} w^3, \\ \rho = w/500, \qquad \alpha = w/(1000L). \end{gathered} \tag{S180}$$

Any specified positive $w \leq R_1/4$ satisfying

$$4\,10^{57} ML^2 w \leq \frac{256^{-31}}{c_* \vartheta^6} \tag{S181}$$

passes the smallness condition. The first condition in (S178) follows already from $c_0 > 1$. For every such fixed w, define the physical canonical tube

$$U_w = (\Psi \circ \Phi_\chi)(\mathbb{T}^2 \times B_w(0)). \tag{S182}$$

It is an injective symplectic image of a declared domain, so its Liouville volume is $4w^2(2\pi)^2$. Equations (S179) and (S180) give the explicit bound

$$\begin{aligned} \frac{\mu_L(U_w \cap \mathcal{K}_{h_\eta})}{\mu_L(U_w)} &\geq 1 - \frac{25}{16}\left(\frac{4}{500} + \frac{8}{1000}\frac{\pi^2}{6}\right) \\ &> 0.9669383. \end{aligned} \tag{S183}$$

The numerical certificate takes the exact value $w = 10^{-220}$. It encloses ϵ_* above $7.6577\,10^{-155}$ and the smallness ratio ϵ/ϵ_* below 0.40759, verifying (S181) with outward rounding.

The transverse size of this particular tube is also bounded. For any real point of U_w, write $(t', I') = \Phi_\chi(t, I)$. The same flow estimate on the smaller action domain gives $|I'|_\infty < 2w$. Comparison with the central point $k(t')$, rather than with a fixed phase label before the flow, eliminates the irrelevant phase shift. For real $q > -1$, $|\sqrt{1+q} - 1| = |q|/(1 + \sqrt{1+q}) \leq |q|$; hence

$$\begin{aligned} \|\Psi(t', I') - k(t')\|_\infty &\leq \frac{0.100007\,(1.014)\,2w}{0.005 - 3\,10^{-6}} \\ &< 41w = 4.1\,10^{-219}. \end{aligned} \tag{S184}$$

This is a bound in the dimensionless Cartesian canonical variables; the SI displacement bound follows by applying the fixed modal reconstruction matrix. The millimetre value describes the amplitude of the central motion, not the thickness in (S184).

The fraction concerns the conservative Hamiltonian on a four-dimensional symplectic tube. The constant SI scaling cancels from the ratio. As quantified in Section S11.1, the transverse width is far below operational resolution and the available embedding-error guarantee. The exact set is mathematically defined, but membership cannot be resolved from the computed torus approximation. The lift to the damped closed loop carries only the low-mode parameter measure specified in Section S4.6.

References

Abbott JJ, Diller E and Petruska AJ (2020) Magnetic methods in robotics. *Annual Review of Control, Robotics, and Autonomous Systems* 3: 57–90. doi: 10.1146/annurev-control-081219-082713.

Antman SS (2005) *Nonlinear Problems of Elasticity*. 2nd ed. Springer. doi: 10.1007/0-387-27649-1.

Biasco L and Chierchia L (2018) Explicit estimates on the measure of primary KAM tori. *Annali di Matematica Pura ed Applicata* 197(1): 261–281. doi: 10.1007/s10231-017-0678-8. https://arxiv.org/abs/1612.01903.

Bounemoura A (2016) Non-degenerate Liouville tori are KAM stable. *Advances in Mathematics* 292: 42–51. doi: 10.1016/j.aim.2016.01.012. Preprint version used: https://arxiv.org/abs/1412.0509v1.

Bounemoura A (2020) Positive measure of KAM tori for finitely differentiable Hamiltonians. *Journal de l'École polytechnique – Mathématiques* 7: 1113–1132. doi: 10.5802/jep.137.

Chen Y, Matsubara T and Yaguchi T (2022) KAM theory meets statistical learning theory: Hamiltonian neural networks with non-zero training loss. *Proceedings of the AAAI Conference on Artificial Intelligence* 36(6): 6322–6332. doi: 10.1609/aaai.v36i6.20582.

Chierchia L and Koudjinan CE (2021) V. I. Arnold's "Global" KAM theorem and geometric measure estimates. *Regular and Chaotic Dynamics* 26(1): 61–88. doi: 10.1134/S1560354721010044. Preprint version used: https://arxiv.org/abs/2010.13243v1.

Das S, Saiki Y, Sander E and Yorke JA (2017) Quantitative quasiperiodicity. *Nonlinearity* 30(11): 4111–4140. doi: 10.1088/1361-6544/aa84c2.

Desai SA, Mattheakis M, Sondak D, Protopapas P and Roberts SJ (2021) Port-Hamiltonian neural networks for learning explicit time-dependent dynamical systems. *Physical Review E* 104: 034312. doi: 10.1103/PhysRevE.104.034312.

Duignan N and Meiss JD (2023) Distinguishing between regular and chaotic orbits of flows by the weighted Birkhoff average. *Physica D: Nonlinear Phenomena* 449: 133749. doi: 10.1016/j.physd.2023.133749.

Feng K, Wu HM, Qin MZ and Wang DL (1989) Construction of canonical difference schemes for Hamiltonian formalism via generating functions. *Journal of Computational Mathematics* 7(1): 71–96. https://computmath.cjoe.ac.cn/jcm/EN/Y1989/V7/I1/71.

Francescon V, Murasovs N, Lloyd P, Onaizah O, Chathuranga DS and Valdastri P (2025) Closed-loop shape-forming control of a magnetic soft continuum robot. *IEEE Robotics and Automation Letters* 10(6): 6071–6078. doi: 10.1109/LRA.2025.3565124.

Greydanus S, Dzamba M and Yosinski J (2019) Hamiltonian neural networks. In: *Advances in Neural Information Processing Systems*, vol. 32. https://arxiv.org/abs/1906.01563.

Hairer E, Lubich C and Wanner G (2006) *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*. 2nd ed. Springer. doi: 10.1007/3-540-30666-8.

Haro À and Luque A (2019) A-posteriori KAM theory with optimal estimates for partially integrable systems. *Journal of Differential Equations* 266(2–3): 1605–1674. doi: 10.1016/j.jde.2018.08.003. Preprint version used: https://arxiv.org/abs/1711.05515v1.

Hornik K (1991) Approximation capabilities of multilayer feedforward networks. *Neural Networks* 4(2): 251–257. doi: 10.1016/0893-6080(91)90009-T.

Kim Y, Parada GA, Liu S and Zhao X (2019) Ferromagnetic soft continuum robots. *Science Robotics* 4(33): eaax7329. doi: 10.1126/scirobotics.aax7329.

NASA (2025) Risk of spaceflight associated neuro-ocular syndrome (SANS). NASA Human Research Program risk page (updated 16 May 2025; accessed 7 October 2026).

Pöschel J (2001) A lecture on the classical KAM theorem. In: *Proceedings of Symposia in Pure Mathematics*, vol. 69, pp. 707–732. doi: 10.1090/pspum/069/1858551. Preprint version used: https://arxiv.org/abs/0908.2234v1 (Version 1.3, 2009).

Simo JC (1985) A finite strain beam formulation. The three-dimensional dynamic problem. Part I. *Computer Methods in Applied Mechanics and Engineering* 49(1): 55–70. doi: 10.1016/0045-7825(85)90050-7.

Tong D, Hao Z, Li J, Sun B, Liu M, Wang L and Huang W (2025) Real-time simulation enabled navigation control of magnetic soft continuum robots in confined lumens. *Journal of the Mechanics and Physics of Solids* 203: 106198. doi: 10.1016/j.jmps.2025.106198.

Trabelsi C, Bilaniuk O, Zhang Y et al. (2018) Deep complex networks. In: *International Conference on Learning Representations*. https://arxiv.org/abs/1705.09792.

Zhang SQ, Gao W and Zhou ZH (2022) Towards understanding theoretical advantages of complex-reaction networks. *Neural Networks* 151: 80–93. doi: 10.1016/j.neunet.2022.03.024. Published version used for theorem numbering.

Zhong YD, Dey B and Chakraborty A (2020) Dissipative SymODEN: Encoding Hamiltonian dynamics with dissipation and control into deep learning. In: *ICLR Workshop on Integration of Deep Neural Models and Differential Equations*. https://arxiv.org/abs/2002.08860.